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MEASUREMENT OF THE CHARM-YUKAWA COUPLING VIA CHARM-QUARK ASSOCIATED HIGGS BOSON PRODUCTION USING CMS RUN-2 DATA

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Abstract

A primary goal of the Large Hadron Collider, located in Geneva, Switzerland, is to measure the properties of the Higgs boson. Specifically, a measurement of the interaction strength between the Higgs boson and the charm quark, know as the charm quark Yukawa coupling, is a quantity of considerable interest in contemporary particle physics. The measurement of this coupling represents a key consistency check of the Standard Model of particle physics, where any deviation may provide exciting hints about the presence of new physics. In this thesis, such a measurement is presented using LHC Run-2 proton-proton collision data recorded by the CMS detector. Specifically, the the charm- quark associated Higgs boson production process (cH) where the Higgs boson decays to four muons via a pair of Z boson is targeted, which has previously not been explored at the LHC. Key challenges such as the low transverse momentum of the charm quark produced in the cH process are adressed using likelihood based jet selection methods. Additionally, an interpretation of the analysis in the context of Effective Field Theory is presented with a specific focus on the previously unconstrained chromomagnetic dipole operator of the charm quark, which may be uniquely constrained with the cH process.

Samenvatting

Een belangrijk doel van de Large Hadron Collider, gelegen in Genève, Zwitserland, is het meten van de eigenschappen van het Higgs boson. In het bijzonder is het bepalen van de interactiesterkte tussen het Higgs-boson en het charm quark, ook wel bekend als de charm-quark Yukawa-koppeling, een groot aandachtspunt in de hedendaagse deeltjesfysica. De meting van deze koppeling vertegenwoordigt een belangrijke test van de consistentie van het Standaardmodel van de deeltjesfysica, waarbij eventuele afwijkingen kunnen duiden op de aanwezigheid van nieuwe fysica. In dit proefschrift wordt een dergelijke meting voorgesteld met behulp van LHC Run-2 proton- protonbotsingsgegevens die zijn verzameld met de CMS-detector. Er wordt specifiek gekeken naar het proces van Higgs bosonproductie in associatie met een charm-quark (cH), een proces dat nog niet eerder onderzocht werd met de CMS data, waarbij het Higgs boson vervalt via $\mathrm{H} \to \mathrm{ZZ} \to 4$ muonen. Belangrijke uitdagingen, zoals het lage transversale momentum van het charm quark dat in het cH- proces wordt geproduceerd, worden aangepakt met behulp van op waarschijnlijkheid gebaseerde jetselectiemethoden. Daarnaast wordt een interpretatie van de analyse in de context van Effective Field Theory voorgesteld, met specifieke aandacht voor de chromomagnetische dipooloperator van het charm quark, waar we nu voor het eerst op een unieke wijze informatie over kunnen geven via het cH proces.

Contents

Chapter 1

Introduction

The Standard Model (SM) of particle physics is the theory that best characterises our current understanding of fundamental particles and their interactions. It describes a broad range of phenomena and makes a plethora of predictions, many of which have been confirmed via measurement to great degrees of accuracy [1]. A noteable feature of the SM is the Brout-Englert-Higgs (BEH) mechanism [2][3], which predicts the existence of a Higgs boson. The BEH mechanism is considered a central part of the SM as it provides a unique mechanism by which SM particles may acquire mass through their interaction with the Higgs field. As such, the experimental discovery of a Higgs-like scalar boson at the CERN Large Hadron Collider (LHC) in 2012 [4][5] was a major milestone in particle physics. Since this discovery, measuring the exact properties of this particle (henceforth simply referred to as the Higgs boson) has been a major goal of LHC experiments such as the CMS collaboration [6] as ultimately, physicists seek to determine whether the particle is indeed the Higgs boson described by the SM. A particular subset of these properties are the so-called Yukawa interactions between the Higgs boson and massive fermions, which are proportional to the mass of the fermions. As can be seen in Figure 1.1, a number of these have previously been measured and indeed align with the values predicted by the SM. However, the measurement of the Yukawa couplings of several of the lighter fermions still remain an open challenge, as these these couplings decrease in strength with smaller fermion masses.

The next lightest fermion candidate for such a measurement is the charm quark. Consequently, the study of the Yukawa coupling between the Higgs boson and the charm quark is of significant interest [7]. This is as, experimentally speaking, relatively little is known about the interaction of the Higgs boson and lighter SM particles, such as the first and second-generation quarks. Studies of the Yukawa coupling of the charm quark thus provide an essential glimpse into previously unexplored territory, allowing us to better characterise the nature of the Higgs boson in this regime. Most importantly, potential deviations from the SM could be present in the couplings of the Higgs boson to lighter quarks. Since Higgs production rates at an experiment such as the LHC largely depend on the couplings of the Higgs boson to weak gauge bosons and third-generation quarks, such deviations may not be visible in our current experimental picture. As a result, the motivation to study the Yukawa coupling of the charm quark (henceforth referred to as the charm-Yukawa coupling) becomes clear.

Experimental approaches to measuring this coupling fall into two categories. The first involves targeting processes in which the Higgs boson decays to a charm and anti-charm quark pair. With this strategy, the CMS collaboration recently reported that it is able to exclude deviations of

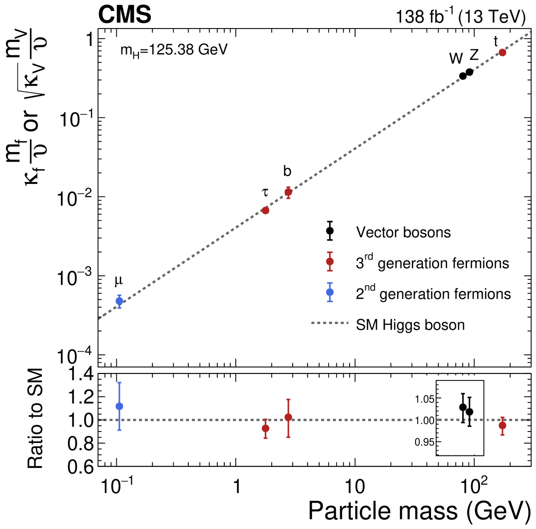


Figure 1.1: The measured coupling modifiers κ_f and κ_V of the coupling between the Higgs boson and fermions as well as heavy gauge bosons as functions of fermion or gauge boson mass m_f and m_V, where ν is the vacuum expectation value of the Higgs field. [6]

the charm-Yukawa coupling that are greater than 3.5 times of what is predicted by the SM, the most stringent constraint to date [8]. The second strategy is to target charm quark-associated Higgs production modes (henceforth referred to as cH production), where an interaction between a charm quark and Higgs boson occurs in the production of the Higgs boson. While analyses based on this strategy have thus far yielded comparatively weaker constraints on the charm-Yukawa coupling [9][10], they constitute important, complimentary measurements. This is due to the strong statistical limitations that both types of strategies face, given that the cross section of processes involving the charm-Yukawa coupling are extremely small in comparison to the plehora of processes that occur at the LHC. As such, a final measurement of this coupling at the 5σ significance level may very well rely on a statistical combination that includes both strategies. While such a feat is still currently out of reach with the current datasets gathered at the LHC, it may come within reach with the significantly larger dataset expected at the High Luminosity LHC [11]. As a result, the preparation of well-understood analysis strategies to reach this important goal is clearly of importance.

Using data from the Run-2 dataset of the CMS detector, this work seeks to contribute to this endeavour by presenting such a complimentary analysis of cH production in a previously unconsidered Higgs decay channel. In Chapter 2, an introduction to the SM and the charm-Yukawa coupling is given, together with a discussion of cH process. Additionally, an argument is presented indicating why the cH process may be uniquely suited to probing certain types of processes not described by the SM, which are here parametrised through so-called effective field theory. Subsequently in Chapter 3, the CMS detector is presented alongside the algorithms used to reconstruct proton-proton collions produced at the LHC. In Chapter 4, the analysis strategy that is employed in this work is outlined, along with the results that are dervied by applying this strategy and a discussion thereof. Finally, a demonstration of the ability of this analysis strategy to constrain the presence of the aforementioned non-SM processes is presented in Chapter 5.

Chapter 2

The Standard Model and the charm-Yukawa coupling

In this chapter, the theoretical underpinnings of the analysis presented in this work are discussed. This includes a brief discussion of the SM in Section 2.1, after which the charm-Yukawa coupling is introduced in Section 2.2. Following this, the cH process is discussed along with how it may be targeted by experiments such as the CMS detector Subsection 2.3.1. Finally, an effective field theory (EFT) interpretation of the cH process is discussed in Section 2.4.

2.1 The Standard Model of particle physics

The SM is formulated through the formalism of Quantum Field Theory (QFT). This is a formalism that combines concepts of classical field theory, quantum mechanics as well as special relativity into a single, coherent description of fundamental particles as excitations of underlying fields that pervade space-time. In this description, SM particles fall into two categories: fermions and bosons. The former are the massive particles which make up the matter of the universe while the latter are the force-carrying particles of the strong and electroweak forces. The distinction between these categories is made based on the spin of the particle, which may be of either half-integer or integer, respectively.

The fermion content of the SM consists of 12 unique particles. These include six leptons, namely the electron, muon and tau as well as their respective neutrinos alongside six different quarks that are distinguished by their so-called flavour. The different quark flavours include up, down, charm, strange, bottom and top and specifies a quark's mass eigenstate as well as electric charge. These fermions are typically arranged into three generations depicted as

$$\begin{pmatrix} e \\ \nu_e \end{pmatrix} \begin{pmatrix} \mu \\ \nu_\mu \end{pmatrix} \begin{pmatrix} \tau \\ \nu_\tau \end{pmatrix}, \begin{pmatrix} u \\ d \end{pmatrix} \begin{pmatrix} c \\ s \end{pmatrix} \begin{pmatrix} t \\ b \end{pmatrix}. \tag{2.1}$$

There are distinct differences between the leptons and quarks. Leptons carry integer (or no) electrical charge while quarks carry fractional electrical charges. More importantly, while both quarks and leptons may interact via the electroweak force, only the quarks interact via the strong force. A special feature of the strong force is that the potential energy associated with it grows, as the distance between strongly interacting particles increases. This leads to a phenomenon referred to as *confinement*, where quarks almost exlusively form compositive states called hadrons. As a result, the emission of individual quarks or gluons in scattering processes typically produce complex cascades of particles referred to as *jets* and overall gives rise to a very complex phenomenology of proton-proton (pp) collisions such as those at the LHC. At very high energies these processes that involve the strong force, typically referred to as quantum chromodynamics (QCD) processes, can be understood through perturbative descriptions. This refers to an expansion in the strong coupling constant α_s in the descriptions of QCD processes. However, at lower energies the perturbative approach loses its validity and phenomenological descriptions are relied on instead.

Lastly, the existence of anti-fermions must be mentioned. These carry the exact opposite quantum numbers (e.g. charge) as their fermion counterparts, though otherwise behave similarly (take the electron and positron for instance). For simplicity, references to a fermion in this work may be understood as referencing both the fermion and anti-fermion counterpart, unless otherwise indicated explicitly. Examples of the latter are e.g. referring explicitly to electrons e^- and positrons e^+ or charm quark c and anti-charm quark $\bar{\mathrm{c}}$ pairs.

There exist 13 unique bosons in the SM. These include the photon γ, $\mathrm{W}^\pm$ and Z which mediate the electroweak force as well as 8 gluons g that mediate the strong force. The final piece is the Higgs boson. Contrary to the force carriers, which all are spin 1, the Higgs boson is spin 0. The massive particles of the SM acquire their mass by interacting with the BEH field, excitations of which are identified as the Higgs boson, and is therefore a central element of the SM.

Considering the introduced particles and forces, the SM has a rich and detailed phenomeology. A great example of a mathematically rigorous delineation of this can be found for example in [12]. Given the focus of this work on the Yukawa coupling between the Higgs boson and charm quark, only this aspect of the SM is discussed in further detail.

2.2 The charm-Yukawa coupling

The quantity that defines the strength of the interaction between massive fermions and the Higgs boson is the so-called Yukawa coupling. To better understand this and associated concepts, some knowledge of the electroweak sector of the SM is required. These are discussed in this section while a comprehensive overview may be found in [13].

To understand the origin of the Yukawa couplings, a brief discussion of Lagrangian densities, gauge transformations and the role of symmetries in the SM is useful. The Lagrangian density $\mathcal{L}(\phi_i; a_i)$ is a quantity dependent on a set of fields ϕ_i and constants a_i from which the equations of motions for the particles associated with these fields may be derived. Commonly, theories of particles and their behaviour in a QFT are therefore defined through the formulation of a Lagrangian density. The form of this expression determines the nature of the particles that are included as well as their interactions.

A central component of the way in which particle interactions are introduced in the SM is the concept of gauge symmetries. These originate from the fact that the quantum fields in a QFT carry phase information, which may depend on the space-time coordinate of the field. This phase information describes (local) degrees of freedom of the field and should have no effect on the physical observables of the system. Thus, $\mathcal{L}$ should remain invariant under arbitary phase transformations. These transformations are typically referred to as a choice of gauge and the corresponding invariance is accordingly referred to as a *local gauge symmetry.*

In the Lagrangian of the SM, invariance in the presence of local gauge symmetries is insured through the addition of additional fields. These gauge fields couple to the previously existing fields and effectively serve as mediators of phase information between space-time points of the original fields. It is exactly these gauge fields that we identify as the fields of the previously introduced force-mediating bosons, and which are required to maintain local gauge symmetry. A very interesting conclusion from this is that the dynamics of the bosons and the corresponding force are determined entirely by the structure of the local gauge symmetry that must be preserved. For the electroweak force, the corresponding symmetry is referred to as $SU(2)_L \,\mathrm{x}\, U(1)_Y$. Here, the L denotes that the associated force only acts on left-handed chiral particles while the Y denotes the electroweak charge that is carried by the corresponding bosons and is referred to as the weak hypercharge. There are a total of four boson associated with the electroweak force. These are the photon γ, which mediates the electromagnetic force, as well as the electromagnetically charged $\mathrm{W}^{\pm}$ and electromagnetically neutral Z boson that mediate the weak force.

With these concepts in mind the nature of the electroweak sector's Lagrangian in the SM may be discussed. Naively, the form of this would be given by

$$\mathcal{L}_{\mathrm{EW}} = i\overline{\psi}_L \gamma^\mu D^L_\mu \psi_L + i\overline{\psi}_R \gamma^\mu D^R_\mu \psi_R - \frac{1}{2}\mathrm{Tr}\left(W^a_{\mu\nu} W^{a\mu\nu}\right) - \frac{1}{4} B_{\mu\nu} B^{\mu\nu} \,. \tag{2.2}$$

for a generic combination of a left-handed isospin doublet ψ_L and and right-handed isospin singlet ψ_L that represent (groupings of) particles and anti-particles respectively. The individual elements of $\mathcal{L}_{\mathrm{EW}}$ are briefly summarised below:

g' :	coupling constant of $\mathrm{U}(1)_Y$
g :	coupling constant of $\mathrm{SU}(2)_L$
ψ_L,	left-handed isospin doublet
ψ_R,	right-handed isospin singlet
B_μ :	gauge field of $\mathrm{U}(1)_Y$
W^a_μ :	gauge fields of $\mathrm{SU}(2)_L$, $a = 1,2,3$
$W_{\mu\nu}$:	field strength tensor
$B_{\mu\nu}$:	field strength tensor
$t^a = \dfrac{\sigma^a}{2}$,	SU(2) generators
$Y_L = -1$,	left chiral hypercharge
$Y_R = -2$,	right chiral hypercharge
$D^L_\mu = \partial_\mu + ig'\dfrac{Y_L}{2}B_\mu + igt^a W^a_\mu$	
$D^R_\mu = \partial_\mu + ig'\dfrac{Y_R}{2}B_\mu$	

The terms $D_\mu^{L/R}$ are so-called covariant derivates that ensure the local $\mathrm{SU}(2)_L \,\mathrm{x}\, \mathrm{U}(1)_Y$ gauge symmetry is uphheld for $\mathcal{L}_{EW}$. These contain the coupling constants g‘ and g, which determine the coupling strength of fermions to the B_μ and W_μ^a gauge fields, respectively. The generators of the SU(2) group are the pauli matrices, with which abritrary SU(2) gauge transformations may be described. Further details can be found in [13].

In this formulation, the observed charged gauge bosons $\mathrm{W}^\pm$ arise from linear combinations of the W_1 and W_2 gauge fields

$$W^\pm = \frac{1}{\sqrt{2}}(W_1 \mp iW_2), \tag{2.3}$$

while the Z boson and photon γ arise from linear combinations of the W_3 and B gauge fields achieved via a rotation

$$\begin{pmatrix} \gamma \\ Z \end{pmatrix} = \begin{pmatrix} \cos\theta_\mathrm{W} & \sin\theta_\mathrm{W} \\ -\sin\theta_\mathrm{W} & \cos\theta_\mathrm{W} \end{pmatrix} \begin{pmatrix} B \\ W_3 \end{pmatrix} . \tag{2.4}$$

with the weak mixing angle θ_W.

The massive natures of of the $\mathrm{W}^\pm$ and Z bosons, as first reported in [14], are however incompatible with such a formulation. The reason is that naive mass term such as

$$m_W^2 W_\mu^+ W^{-,\mu} + \frac{1}{2} m_Z^2 Z_\mu Z^\mu. \tag{2.5}$$

do not remain invariant under arbitrary $\mathrm{SU(2)_L}$ gauge transformations. This is as gauge fields A_μ generically transform as

$$A_\mu \rightarrow A_\mu^{'} = A_\mu - \frac{1}{g}\partial_\mu \mathcal{V}(x), \tag{2.6}$$

where $\mathcal{V}(x)$ is some arbitrary phase depending on the spacetime coordinate x. Substituting Eq. 2.6 into Eq. 2.5 thus introduces additional terms that do not cancel. The same is true for fermion mass terms in the form of

$$m_f \overline{\psi}\psi. \tag{2.7}$$

There is, however, a subtle distinction in this case, as the invariance breaking terms in Eq. 2.7 arise from the different transformation behaviour of the ψ_L and ψ_R components of ψ under $\mathrm{SU}(2)_L \,\mathrm{x}\, \mathrm{U}(1)_Y$ gauge transformations.

2.2.1 The Brout-Englert-Higgs mechanism

The BEH mechanism provides a way to circumvent the gauge symmetry breaking nature of the aforementioned generic mass terms. This is achieved through a process referred to as spontaneous symmetry breaking. A spontaneously broken symmetry refers to a symmetry that is upheld in a global view of the system (i.e. the overall Lagrangian density $\mathcal{L}_\mathrm{EW}$ remains invariant under a relevant gauge transformation) while the energetic ground state of the system explicitly breaks this symmetry. This is a process formally described by the Goldstone theorem [15], which states that each broken symmetry in a relativistic QFT generates an additional massless boson. These introduce additional degrees of freedom into the theory and are coined Goldstone bosons. The BEH mechanism exploits this by adding an additional term

$$\mathcal{L}_{\text{Higgs}} = D_\mu \phi^\dagger D^\mu \phi - V(\phi) \tag{2.8}$$

$$V(\phi) = -\mu^2 \phi^\dagger \phi + \lambda(\phi^\dagger \phi)^2. \tag{2.9}$$

to $\mathcal{L}_{\text{EW}}$, that describes an additional complex field ϕ. This is a $\text{SU}(2)_L$ doublet

$$\phi = \begin{pmatrix} \phi^+ \\ \phi^0 \end{pmatrix} \tag{2.10}$$

with the scalar components ϕ^+ and ϕ^0. Here, $V(\phi)$ corresponds to the potential energy term of the field. Again, the covariant derivative

$$D_\mu = \partial_\mu + ig' \frac{Y_\phi}{2} B_\mu + igt^a W_\mu^a \tag{2.11}$$

ensures $\mathcal{L}_{\text{Higgs}}$ remains locally gauge invariant under $\text{SU}(2)_L \,\text{x}\, \text{U}(1)_Y$ transformations and therefore introduces couplings between the ϕ field and the B_μ and W_μ fields. The constants of the potential term Eq. 2.9 are chosen in such a way that the ground state of $V(\phi)$ is non-zero. This can be achieved by choosing them such that $\lambda > 0$ and $\mu^2 > 0$. The result is a ground state of V that is identified as the vacuum expectation value

$$v = \sqrt{\frac{\mu^2}{2\lambda}}\,. \tag{2.12}$$

The center of the potential is now an unstable local maximum and the only stable configuration can be found in the non-zero ground state. This is demonstrated visually in Figure 2.1. Through this, the symmetry of the potential is effectively broken. A popular choice of gauge for ϕ is

$$\phi = \begin{pmatrix} 0 \\ v + \frac{h}{\sqrt{2}} \end{pmatrix}, \tag{2.13}$$

where h is a new scalar field that is used to parametrise radial perturbations of the potential's ground state. This choice is referred to as the unitary gauge and h is identified as the field corresponding to the physical Higgs boson. By expanding Eq. 2.8 with this choice of ϕ, a range of terms are introduced to $\mathcal{L}_{\text{EW}}$. These contain a variety of interaction terms between the gauge fields and the Higgs field, as well as newly generated mass terms for the Z and W bosons

$$\left(\frac{g}{2}\right)^2 v^2 W_\mu^+ W^{\mu-} = m_{\text{W}}^2 W_\mu^+ W^{\mu-} \tag{2.14}$$

$$\left(\frac{\sqrt{g^2 + g'}}{2}\right)^2 v^2 Z_\mu Z^\mu = m_{\text{Z}}^2 Z_\mu Z^\mu\,. \tag{2.15}$$

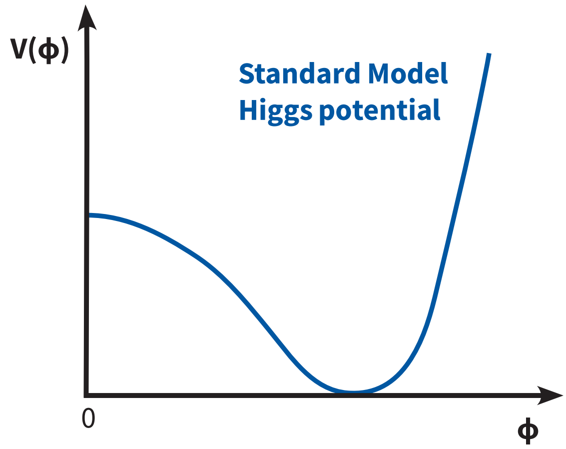


Figure 2.1: A sketch of the shape of the Higgs potential in the Standard Model, which is simplified by showing a cross section of the two dimensional potential. A non-stable maximum at the center of a potential can be seen alongside a stable configuration in a non-zero ground state.

This can be understood to mean that the electroweak coupling constants g and g' along with v effectively determine the mass of the Z and $W^{\pm}$ bosons. A full description and compilation of all the terms of the electroweak Lagrangian density of the SM can be found in [13].

2.2.2 The Yukawa couplings

By including this additional contribution in our theory, mass terms for fermions may now be generated by including a term of the form

$$\begin{aligned} \mathcal{L}_{\text{Yukawa}} &= -y_f \overline{\psi} \phi \psi, && (2.16) \\ &= -y_f v \overline{\psi} \psi (1 + \frac{1}{v} \frac{h}{\sqrt{2}}), && (2.17) \end{aligned}$$

which is now invariant under$\mathrm{SU}(2)_L \, \mathrm{x} \, \mathrm{U}(1)_Y$ gauge transformations due to the addition of ϕ. Similarly to the W and Z mass terms, the relation

$$m_f = y_f v. \tag{2.18}$$

is obtained. A curious feature of the SM is that the Yukawa couplings y_f are free parameters of the theory. As a result these must be measured experimentally, with the measurement of the charm-Yukawa coupling y_{c} being the goal of this work. Since the charm quark mass has previously been determined from experiment to be $m_{\mathrm{c}} = 1.27$ GeV [1], a measurement of y_{c} thus represents an important consistency test of the SM. To this end, one can exploit that an interaction between fermions and the Higgs field is introduced as can be seen in Eq. 2.17, with an interaction strength proportional to y_{c}. It is this feature that may be utilised by experiments at the LHC to measure y_c. The presence of such interactions is also how other Yukawa couplings, such as of the bottom and top quarks, are determined experimentally and have been found to be consistent with the SM expectation [16][17].

2.3 Measuring the charm-Yukawa coupling

By measuring the frequency of occurence of physics processes in which the coupling between the Higgs boson and charm quark appears, y_c may be determined. As such, a suitable process must be found that can be detected by an experiment such as the CMS experiment, which records pp collisions at the LHC. These fall into two categories. The first consists of processes in which a Higgs boson decays into a charm and anti-charm quark pair ($H \to c\bar{c}$). This typically involved targetting processes in which the Higgs boson is produced in association with e.g. a top quark pair ($t\bar{t}H$) or weak gauge bosons (VH), to better identify the presence of the Higgs boson. The second category consists of processes in which a Higgs boson is produced in association with a charm quark. This latter category of processes is the focus of this work and is henceforth referred to as the cH process, which is described in Subsection 2.3.1. Subsequently, in Subsection 2.3.2, the κ-framework is introduced. This framework serves as a tool for analyses of different charm-Yukawa sensitive processes to interpret their results in terms of a single, common parameter denoted as κ_c.

2.3.1 The cH process

The cH process encompasses processes in proton-proton collisions in which a charm quark is produced alongside a Higgs boson. At leading order in the perturbative description of the strong force, this consists of two processes sensitive to y_c, represented by the Feyman diagrams shown in Figure 2.2. The two diagrams, namely the s and t-channel diagrams, constitute the y_c sensitive contribution due to the presence of an interaction between the Higgs boson and the charm quark. There exist also additional cH processes, mediated through the effective Higgs boson to gluon coupling, which are not sensitive to y_c. These account for approximately 80% of the inclusive cH cross section and thus represents a significant background to the cH process sensitive to the charm-Yukawa coupling.

Targeting the cH process to measure y_c is a relatively novel strategy in comparison to targeting $H \to c\bar{c}$ decays. A key advantage of the former approach is that contributions from the abundant QCD multijet background at the LHC are greatly reduced due to only needing to identify the flavour of single jet resulting from a charm quark, as opposed to two. This is especiialy relevant, given that the task of jet flavour identification remains to be an experimentally challenging one. Additionally, since the sensivity to y_c does not originate from the decay of the Higgs boson, the Higgs boson decay mode to target can be chosen freely. Especially signatures such as $H \to ZZ \to 4\mu$, which may be resolved cleanly by an experiment such as the CMS experiment, can be targeted. Motivated by this feature, this work focusses on the $H \to ZZ \to 4\mu$ decay mode in cH processes.

However, an analysis of the cH process also comes with notable drawbacks. A significant experimental difficulty results from the fact that the associated charm flavour jets are typically produced at lower transverse momenta p_T, as seen in Figure 2.3. These are experimentally difficult to reconstruct and thus a significant portion of the cH signal is expected to be lost due to such detector acceptance effects. Another drawback is that though Higgs boson decay channels such as $H \to ZZ \to 4\mu$ may be very cleanly resolved in a detector, they come with very small branching ratios (e.g. $\mathcal{B}(H \to ZZ \to 4\mu) = 0.3\%$ [1]) and thus the already small cross section of the

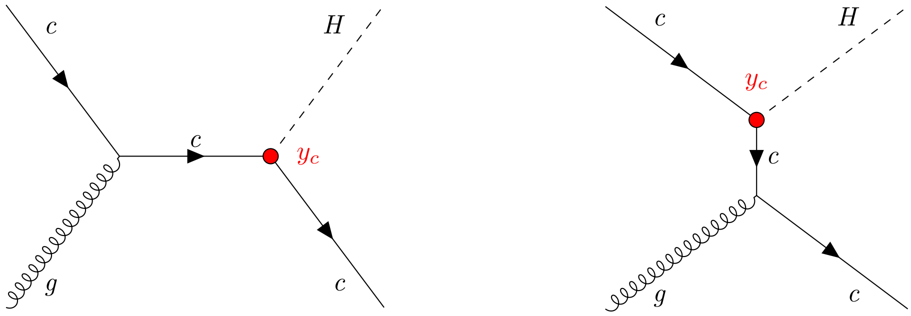


Figure 2.2: The leading-order cH processes through which y_c may be probed as each diagramm contains a vertex with a charm quark and Higgs boson, here denoted in red. The corresponding diagrams with an anti-charm quark $\bar{c}$ are implied.

cH process is further reduced. As a result of these effects, a key challenge in targeting the cH process is expected to arise from the significant statistical uncertainties associated of the analysis.

2.3.2 The κ-framework

The κ-framework [20] is a commonly used framework to parametrise modifications to the couplings between the Higgs boson and other particles with respect to the expected SM values. It introduces the coupling modifier for the charm-Yukawa coupling as

$$\kappa_c = \frac{y_c}{y_c^{\mathrm{SM}}}, \tag{2.19}$$

where y_c is the measured Yukawa coupling and y_c^{SM} is the expected Yukawa coupling of the SM, calculated from the known charm quark mass. Thus modifications to the Yukawa coupling of the charm quark are parametrised in this way as deviations from $\kappa_c = 1$. However, y_c is not a quantity that can be measured directly. Instead a signal strength measurement μ_{if} relative to the SM expectation, where i represents the production process and f represents the decay process, is made. Thus a measurement of μ_{if} must be converted into an interpretation of κ_c. This is a step that contains some finer subtleties.

As described in [7], the rate of a Higgs production and decay process in relation to the expected SM signal (i.e. a signal strength) may be written as

$$\mu_{if} = \frac{\sigma_i \mathcal{B}_f}{(\sigma_i \mathcal{B}_f)^{\mathrm{SM}}}, \tag{2.20}$$

where σ_i is the production cross section in a given channel i and $\mathcal{B}_f$ is the decay branching ratio in a given channel f. This can alternatively be expressed as

$$\sigma_i \mathcal{B}_f = \kappa_{r,i}^2 \sigma_i^{\mathrm{SM}} \frac{\kappa_f^2 \Gamma_f^{\mathrm{SM}}}{\Gamma_{\mathrm{H}}}, \tag{2.21}$$

where modifications with respect to the SM expectation are introduced via the modifiers $\kappa_{r,i}$ and κ_f. The former modifies the SM production cross section σ_i^{SM}, while the latter modifies the

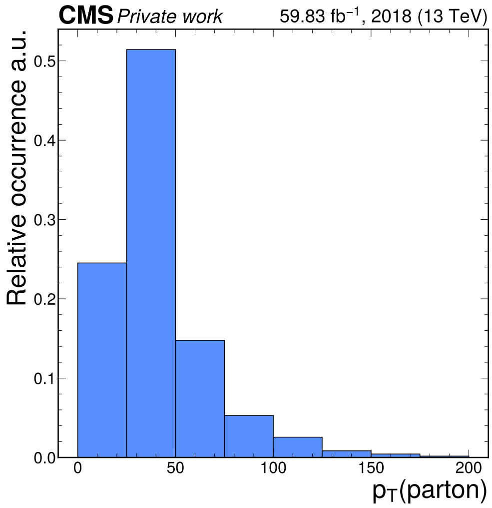


Figure 2.3: Transverse momentum of the parton produced alongside a Higgs boson in a Madgraph5_aMC@NLO [19] simulation of the cH process within the CMS detector, which typically takes on very small values. However, since jets at very low transverse momentum cannot be well reconstructed in experiments such as the CMS detector, a minimum p_T requirement is introduced in the simulation to reduce the reconstruction efficiency loss associated with this effect. This results in the appearance of a peak in the [25, 50] GeV bin of the distribution.

partial SM decay width Γ_f^{SF} in the given decay channel. The denominator Γ_{H}, represesents the total decay width and can be written as

$$\begin{aligned}\Gamma_{\mathrm{H}} =&\Gamma_{\mathrm{H}}^{\mathrm{SM}}(\kappa_{\mathrm{b}}^2\mathcal{B}_{\mathrm{bb}}^{\mathrm{SM}} + \kappa_{\mathrm{W}}^2\mathcal{B}_{\mathrm{WW}}^{\mathrm{SM}} + \kappa_{\mathrm{g}}^2\mathcal{B}_{\mathrm{gg}}^{\mathrm{SM}} + \kappa_{\tau}^2\mathcal{B}_{\tau\tau}^{\mathrm{SM}} + \kappa_{\mathrm{Z}}^2\mathcal{B}_{\mathrm{ZZ}}^{\mathrm{SM}} + \kappa_{\mathrm{c}}^2\mathcal{B}_{\mathrm{cc}}^{\mathrm{SM}} \\ &+ \kappa_{\gamma}^2\mathcal{B}_{\gamma\gamma}^{\mathrm{SM}} + \kappa_{\mathrm{Z}\gamma}^2\mathcal{B}_{\mathrm{Z}\gamma}^{\mathrm{SM}} + \kappa_{\mathrm{s}}^2\mathcal{B}_{\mathrm{ss}}^{\mathrm{SM}} + \kappa_{\mu}^2\mathcal{B}_{\mu\mu}^{\mathrm{SM}}) \qquad (2.22)\\ :=&\Gamma_{\mathrm{H}}^{\mathrm{SM}}\kappa_{\mathrm{H}}^2 \qquad (2.23)\end{aligned}$$

Here, $\Gamma_{\mathrm{H}}^{\mathrm{SM}}$ is the total decay width of the Higgs boson in the SM and the $\mathcal{B}_f^{\mathrm{SM}}$ are the branching ratios of the possible decay modes. Note that the loop induced coupling of the Higgs boson to gluons and photons are included as independent quantities. Again, each partial width contribution to the total width comes with a modifier κ_i. Substituting these expressions into Eq. 2.20, the general rate modifier may be rewritten as

$$\mu_{if} = \frac{\kappa_{r,i}^2\kappa_f^2}{\kappa_{\mathrm{H}}^2}. \qquad (2.24)$$

Now, assuming that only modifications to the charm-Yukawa coupling play a role in Higgs boson production and no modifications are made to the decay mode (e.g. $\mathrm{H} \rightarrow \mathrm{ZZ} \rightarrow 4\mu$), Eq. 2.24 becomes

$$\mu_{if} = \frac{\kappa_{\mathrm{c}}^2}{\kappa_{\mathrm{H}}^2} \qquad (2.25)$$

for a signal strength measurement of the cH(ZZ$\rightarrow 4\mu$) process.

Using the flat direction approach discussed in [7] and [21], a simplification of κ_H can be introduced. This approach is based on the finding that, when performing fits to existing Higgs boson production and decay rates, increases in the Yukawa couplings of light quarks (including the charm quark) can be compensated by increases in the couplings of the gauge bosons and heavy fermions. This is referred to as a “flat direction” in the fit, where observed Higgs boson production and decay rates can be modeled equally well for any value of κ_c by a respective scaling of all other processes. The individual modifiers in the sum of Eq. 2.22, with the exception of κ_c, as well as the modifiers in the nominator of Eq. 2.24, can thus be replaced with a single modifier κ. This allows Eq. 2.24 to be expressed as

$$\mu_{if} = \frac{\kappa^4}{\kappa^2(1-\mathcal{B}_\mathrm{cc}^\mathrm{SM}) + \kappa_\mathrm{c}^2\mathcal{B}_\mathrm{cc}^\mathrm{SM}}, \tag{2.26}$$

for the described scaling of Higgs production and decay with κ. This expression, for some measured μ, has a solution for κ given by

$$\kappa^2 = \frac{(1-\mathcal{B}_\mathrm{cc}^\mathrm{SM})\mu}{2} + \frac{\sqrt{(1-\mathcal{B}_\mathrm{cc}^\mathrm{SM})^2\mu^2 + 4\mu\mathcal{B}_\mathrm{cc}^\mathrm{SM}\kappa_\mathrm{c}^2.}}{2}. \tag{2.27}$$

Here, the expected SM decay width $\mathcal{B}_\mathrm{cc}^\mathrm{SM} = 0.03$ can be substituted. Additionally, the fact that observed Higgs boson rates have been well measured to be close to their expected values (see e.g. [22]) can be reflected by setting $\mu \approx 1$. As a result, only a dependence on κ_c remains in the expression. Thus by replacing κ_H in Eq. 2.25 with Eq. 2.27, a final expression relating a measured signal strength of the cH process to κ_c is obtained, given by

$$\mu_{\sigma_\mathrm{cH}\mathcal{B}(\mathrm{H}\to\mathrm{ZZ})} = \frac{2\kappa_\mathrm{c}^2}{0.97 + \sqrt{(0.97)^2 + 4\cdot 0.03\kappa_\mathrm{c}^2}}. \tag{2.28}$$

Rearranging for κ_c gives

$$\kappa_\mathrm{c} = \pm\sqrt{0.97\mu_{\sigma_\mathrm{cH}\mathcal{B}(\mathrm{H}\to\mathrm{ZZ})} + 0.3\mu^2_{\sigma_\mathrm{cH}\mathcal{B}(\mathrm{H}\to\mathrm{ZZ})}}. \tag{2.29}$$

Effectively, this approach to interpreting κ_c from a signal strength measurement $\mu_{\sigma_\mathrm{cH}\mathcal{B}(\mathrm{H}\to\mathrm{ZZ})}$ ensures compatibility with existing Higgs boson rate measurements by connecting the two different scaling behaviours. This is relevant as a non-unity value of κ_c leads to modifications of the Higgs boson partial decay widths, leading to a slight modification of the quadratic relationship one might naively expect between κ_c and μ_cH. It should be noted that constraints from such a compatibility requirement already indirectly implies upper limits on κ_c, as discussed in [7].

2.4 The cH process in an EFT context

The cH process may also be interpreted in the context of Standard Model Effective Field Theory (SMEFT). In SMEFT, potential effects from physics processes not described by the SM (commonly referred to as beyond-the-SM or BSM physics) are parametrised in a mostly model-independent way. Specifically, the SMEFT framework can be used at colliders with a characteristic energy scale E to describe the effects of processes with a characteristic energy scale above E. This concept is illustrated in Figure 2.4.

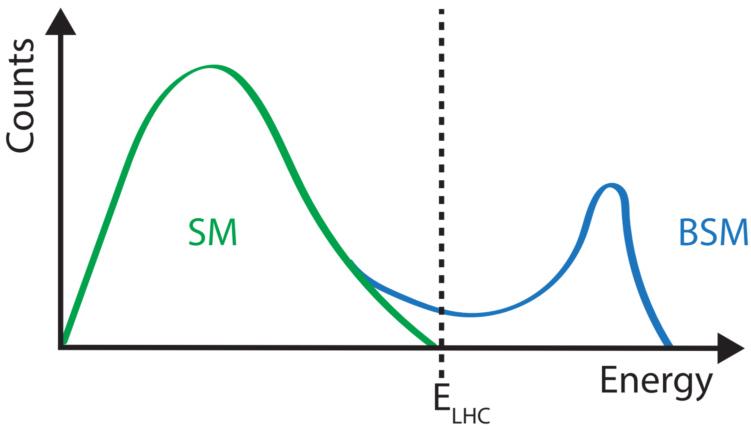


Figure 2.4: Illustration of how the presence of BSM physics, which is primarily visible beyond the reach of current collider energies (e.g. $\mathrm{E}_{\mathrm{LHC}}$), can lead to subtle modifications of SM observables. These effects can be parametrised by SMEFT.

Formally, SMEFT is a collection of all possible combinations of field interactions that obey the gauge invariance conditions of the SM. Generically, this can be expressed as an expansion in the energy scale of the new physics Λ

$$\mathcal{L}_{\mathrm{SMEFT}} = \mathcal{L}_{\mathrm{SM}} + \sum_{d>4} \sum_{i} \frac{C_i}{\Lambda^{d-4}} \hat{O}_i^d \tag{2.30}$$

where $\mathcal{L}_{\mathrm{SM}}$ is the SM lagrangian density, O_i^d denotes a particular operator (i.e. a particular combination of fields) with a dimensionless coupling coefficient C_i (typically referred to as Wilson coefficients) and d denotes the dimension of the operator. The dimensionality is derived through a dimensional analysis of a lagrangian and its fields, where energy dimensions of terms may be deduced from the requirement that the action

$$\mathrm{S} = \int \mathcal{L}\, d^4x \tag{2.31}$$

remains dimensionless. Accordingly, $\mathcal{L}_{\mathrm{SM}}$ is of energy dimension four. Since the SMEFT operators O_i^d all have energy dimensions higher than four and Λ comes with energy dimension one, the terms in the sum of Eq. 2.30 are scaled with $1/\Lambda^{d-4}$ to ensure the combination also has an energy dimension of four.

Typically, operators in SMEFT are grouped by their energy dimension. In $d = 5$, only one possible operator exists that violates lepton number [23] and is not relevant in this work. In $d = 6$ however, a plethora of valid operators exist. These are commonly described in the Warsaw basis [24], which is representation of $d = 5$ and $d = 6$ operators. In this basis, a total of 59 independent operators are found. Since $d = 7$ operators again violate lepton number and each additional dimension adds a suppresive factor of Λ^{-1}, a simplified SMEFT schema is commonly used in which only the contribution of $d = 6$ operators is considered in the expansion. Thus Eq. 2.30 simplifies to

$$\mathcal{L}_{\text{SMEFT}} = \mathcal{L}_{\text{SM}} + \sum_i \frac{C_i}{\Lambda^2} \hat{O}_i^{(6)} \tag{2.32}$$

A good overview of SMEFT can be found in [25].

2.4.1 The chromomagnetic dipole operator

A particular operator relevant to this work is referred to as the chromomagnetic dipole (CMD) operator $\hat{O}_{qG}$, which introduces a four-particle interaction involving a Higgs boson, a gluon and a pair of quarks. For the charm quark, the CMD operator is written as

$$\hat{O}_{cG} = (\overline{q}_{2,L}\sigma^{\mu\nu}T^a c)\tilde{\phi}G^a_{\mu\nu}. \tag{2.33}$$

Here, $\overline{q}_{2,L}$ is the second generation, left-handed quark doublet, $\sigma^{\mu\nu} = i[\gamma_\mu, \gamma_\nu]/2$ with the Dirac matrices γ_μ , $T^a c$ are the generators of the SU(3) symmetry group, $\tilde{\phi}$ is the adjoint Higgs doublet and $G^a_{\mu\nu}$ is the field strength tensor of the strong interaction. This operator may be uniquely bounded with the cH process due its unique chiral structure, which mixes left and right-handed spinors, and which is otherwise only found in the Yukawa and quark-Higgs boson interaction terms of the SM.

To better understand this, it is worth considering other processes such as inclusive Higgs boson production, which have been successfully leveraged to set strong contraints on the top quark CMD operator $\hat{O}_{tG}$ [26]. Typically, the strategy that is used to probe even small Wilson coefficients, such as C_{tG}, is to exploit interference of the relevant (small) SMEFT contribution with a larger SM contribution. Though the pure SMEFT contribution itself may be small and experimentally neglible due to limited analysis sensitivity, the much larger contribution of the SM process it interfers with can result in a non-neglible interference effect with respect to the SM process. However, the chiral structure of the CMD operator influences the effectiveness of this strategy. Since the $\hat{O}_{qG}$ operator effectively flips the chirality of the incoming and outgoing quarks, an additional *chirality flip* must be inserted for the SMEFT contribution to interfere with the SM process. This is visualised in Figure 2.5. Such a chirality flip is proportional to the mass m_q of the respective quark. As a result, the interference contribution for a much lighter quark is significantly supressed in comparison to the top quark, as also argued for the bottom quark in [27]. Effectively, this means that the processes that prove effective in targeting $\hat{O}_{tG}$ due to the large mass of the top quark, are thus much less sensitive to $\hat{O}_{cG}$. However, since the cH process itself contains the chirality flipping quark-Higgs boson vertex, interference terms between the EFT and SM contributions do not suffer from the above described effect. Furthermore, due to the very low expected cross section of the cH process, quadratic contributions from $\hat{O}_{cG}$ may be comparatively large, even at small values of C_{cG}. Accordingly, the cH process may be an excellent target to constrain $\hat{O}_{cG}$.

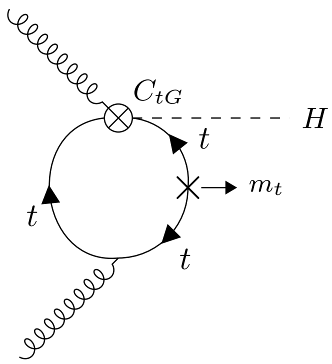


Figure 2.5: A modification of the gluon fusion process with a top quark loop by including the vertex introduced by the top quark CMD operator. Note that the arrows indicate chirality and not momentum flow. A chriality flip, denoted by the cross, proportional to the top quark mass m_t is required for the inclusion of the top quark CMD vertex.

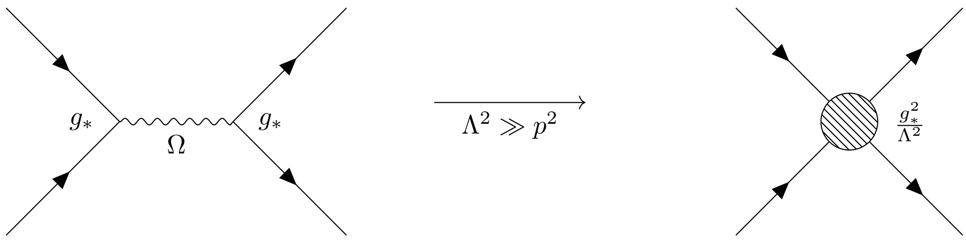


Figure 2.6: Feynman diagrams depicting a resonant process in which the new mediator particle Ω is created (left) and the approximate description of this in an EFT, where the diagram is reduced to a four-point interaction.

2.4.2 Validity of an EFT

In addition to EFT terms needing to satisfy the gauge invariance conditions of the SM, two additional key validity conditions are typically required of an EFT. The first is related to the fact that in an EFT, the particle nature of e.g. new, heavy mediator particles is simplified into the introduction of a new effective vertex. For example, a 2→2 particle resonant scattering via a new heavy mediator particle Ω associated with a newly introduced coupling constant g_*, is simplified via the introduction of a four-point interaction, as visualised in Figure 2.6. This corresponds to a first-order approximation of the new particle's mediator as

$$\frac{g_*}{p^2 - m_\Omega} \xrightarrow[m_\Omega^2 \gg p^2]{} -\frac{g_*}{m_\Omega}\left(1 + \frac{p^2}{m_\Omega^2} + \frac{p^4}{m_\Omega^4} + ...\right) \approx -\frac{g_*}{m_\Omega} \tag{2.34}$$

For the EFT description of this simplification to be valid, the energy involved in processes containing the effective vertex introduced by the relevant operator must thus lie well below m_Ω, which represents the previously introduced new physics scale Λ. Practically, this can be achieved by placing an upper limit M_{cut} on the total energy that is considered in measurements of such processes. The requirement can be expressed as

$$M_{cut} < \Lambda. \tag{2.35}$$

A good estimator of M_{cut} is the inviariant mass of the final state particles of a process. In case of the cH process, the invariant mass of the Higgs boson and jet system is a natural choice.

The second condition that must be met is related to the perturbativity of the theory. Concretely, this means that higher dimensional operators should consitute increasingly smaller corrections, so that the sum of operator contributions converges. In the case of this work where only d=6 operators are considered, this means ensuring contributions from d=8 operators and higher are sufficiently small. While this cannot be determined with certainty without explicit knowledge of the underlying theory the EFT is estimating, a popular choice is to require that at most $g_* \sim 4\pi$ [28].

These two conditions may be combined into a single, simultaneous requirement. In [28], an effective lagrangian (ignoring relevant indices for simplicity) of the general form

$$\mathcal{L}_{\mathrm{eff}} = \frac{\Lambda^4}{g_*^2}\mathcal{L}\left(\frac{D_\mu}{\Lambda}, \frac{g_h\phi}{\Lambda}, \frac{g_{\psi_{L,R}}\psi_{L,R}}{\Lambda^{3/2}}, \frac{gF_{\mu\nu}}{\Lambda^2}\right) \tag{2.36}$$

is obtained when a BSM coupling g_* is introduced. This provides a prescription for the powers of the couplings and Λ that are associated with the SM fields ϕ, ψ and $F_{\mu\nu}$, and the covariant derivate D_μ. Here, g represents the unaltered gauge field couplings of the SM, while $g_{\psi_{L,R}}$ and g_h represent the coupling of SM fermion and the Higgs doublet to the BSM theory. In a single BSM coupling scenario, this simplifies to $g_{\psi_{L,R}} = g_h = g_*$. Applying this prescription to the CMD operator gives

$$\hat{O}_{cG} \longrightarrow \frac{\Lambda^4}{g_\star^2}\left[\left(\frac{g_\star\psi_{L,R}}{\Lambda^{3/2}}\right)\cdot\left(\frac{g_\star\psi_{L,R}}{\Lambda^{3/2}}\right)\cdot\left(\frac{g_\star\phi}{\Lambda}\right)\cdot\left(\frac{g_sG}{\Lambda^2}\right)\right] \tag{2.37}$$

$$= \frac{g_\star g_s}{\Lambda^2}\left(\psi_{L,R}\cdot\psi_{L,R}\cdot\phi\cdot G\right)\,. \tag{2.38}$$

Reading off from Eq. 2.38, one can see that the coupling of the CMD operator is given by $g_\star g_s/\Lambda^2$. Comparing to Eq. 2.30 thus reveals that the CMD Wilson coefficient is given by $C_{cG} = g_* g_s$. By requiring the first discussed validity condition, the relation

$$\frac{C_{cG}}{\Lambda^2} < \frac{g_* g_s}{M_{cut}^2} \tag{2.39}$$

is obtained. Since both C_{cG} and Λ are a priori unknown, we can redefine $\tilde{C}_{cG} = \frac{C_{cG}}{\Lambda^2}$. With this redefinition, and by setting $g_* \sim 4\pi$, the expression

$$\frac{\left|\tilde{C}_{cG}\right| M_{cut^2}}{4\pi g_s} < 1\,. \tag{2.40}$$

can be used to define a plane in $\tilde{C}_{cG}$ and M_{cut} that satisfies both of the previously discussed conditions. This plane is visualised in Figure 2.7, showing regions in which the validty condition is (loosely) statisfied and in which it is not satisfied. Constraints placed on the CMD operator for a given M_{cut} value can be cross-checked against this to ensure the validity of the constraint.

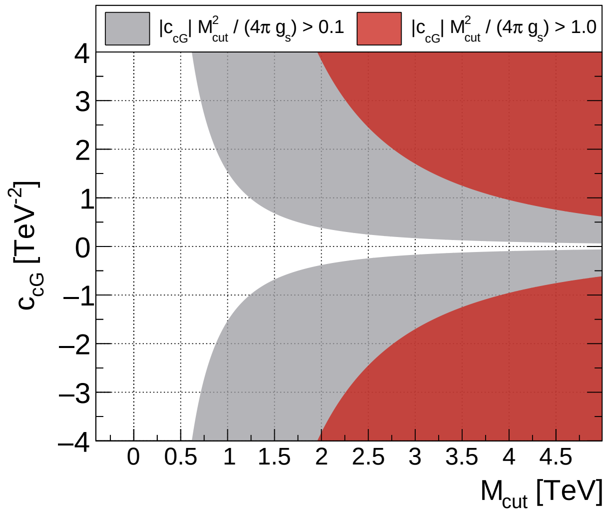


Figure 2.7: Diagram showing the plane spanned by $\tilde{C}_{cG}$ and $M_{cut}$for the CMD operator [**PhenoPaper**]. The white region shows the region in which the perturbativity requirement is well satisfied. The grey region represents the transition from the white region to the red region, in which the perturbativity requirement is no longer valid.

2.5 State of the art on κ_c

Given the previously described status of the charm-Yukawa coupling in our current, experimental picture of the Higgs boson, analyses targetting the charm-Yukawa coupling are becoming of increased interest. This pertains particularly to the cH process, since it is considered a relatively novel process. In reflection of this, recent results in the cH($\gamma\gamma$) and cH(WW) channels using Run-2 data of the CMS experiment have been published. These results are interpreted in terms of the charm-Yukawa coupling modifier κ_c and upper limits on κ_c at 95% CL are reported at $|\kappa_c| < 38.1$ for cH($\gamma\gamma$) [9] and $|\kappa_c| < 47$ for cH(WW) [10], respectively.

While not as sensitive as the 95% CL upper limit observed in a recent combination of VH($c\bar{c}$) and $t\bar{t}$H($c\bar{c}$) channels, which at $\kappa_c < |3.5|$ [8] is the most stringent to date, these analyses of the cH process nonetheless provide important complementary results and may contribute significantly in a statistical combination. This is especially important given that even at the High-Luminosity LHC (HL-LHC), the projected sensitivity to the charm-Yukawa coupling in individual channels is only starting to approach one [18]. Accordingly, the development of well-understood analysis methods, together with a combination of analysis channels, are key facets to exceeding the expecations set by such a projection. By performing a first-time analysis of the previously unexplored cH(ZZ$\rightarrow 4\mu$) channel, this work intends to contribute to making such a future measurement of the charm-Yukawa coupling possible.

Chapter 3

The CMS experiment at the LHC

The Compact Muon Solenoid (CMS) detector [29] is a large, general purpose particle detector located at the Large Hadron Collider (LHC)[30] accelerator in Geneva, Switzerland. Run by the European Organisation for Nuclear Research (CERN), the LHC's largest ring spans a circumference of 27 km, making it the largest particle accelerator in the world. In their circular trajectory through the beam pipe, collimated bunches of $\sim 10^{11}$ protons are accelerated in both directions of the ring. At four different collision points, the trajectories of these proton bunches are crossed such that highly energetic proton-proton collisions are produced. The CMS detector is located at one of these collision points. A sketch of the LHC accelerator complex can be seen in Figure 3.1. A detector such as CMS effectively acts as a camera taking very complex snapshot of each collision. During Run-2 (2016 - 2019) of the LHC, approximately 30 protons collide on average per bunch crossing with a centre-of-mass energy of $\sqrt{s} = 13$ TeV. These collisions produce a plethora of particles, many of which decay to sets of particles of varying multiplicities themselves. As such, these collision produce a complex and varied phenomenology that require a complex machine such at the CMS detector to fully capture. By recording the information from many millions of collisions, a multitude of different statistical analyses may be performed. This includes analyses of the Higgs boson and its properties, such as the Yukawa coupling of the charm quark. To this end, this chapter gives an overview of the CMS detector and its subsystems as well as the techniques used to reconstruct individual proton-proton collisions.

3.1 The CMS detector

The CMS detector is designed to be able to detect a wide range of signatures and is built from a set of complementary subdetectors. An overview of the detector may be seen in Figure 3.2. By combining data from these subdetectors, a comprehensive reconstruction of individual proton-proton collisions, commonly referred to as an *event*, may be made. The role and functioning of the individual subdetectors is covered in this section. While several of the detector components have undergone changes for the current Run-3 (2022 - 2026) of the LHC[33], the configuration relevant to this work is that of Run-2.

3.1.1 The CMS coordinate system

Due to the cylindrical nature of the CMS detector, using cylindrical coordinates to describe positions within the detector is a natural choice. Thus, the z coordinate describes the position along the beam pipe, r the radius, and ϕ the azimuthal angle, where the proton-proton collision

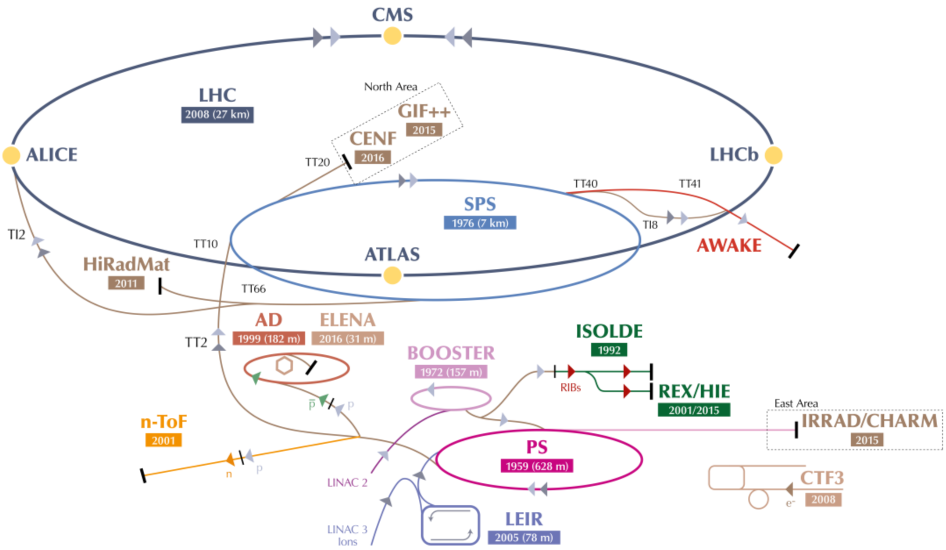


Figure 3.1: An overview of the LHC accelerator complex [31]. Before enetering the large LHC ring, particles must pass through a number of increasingly powerful set of accelerators.

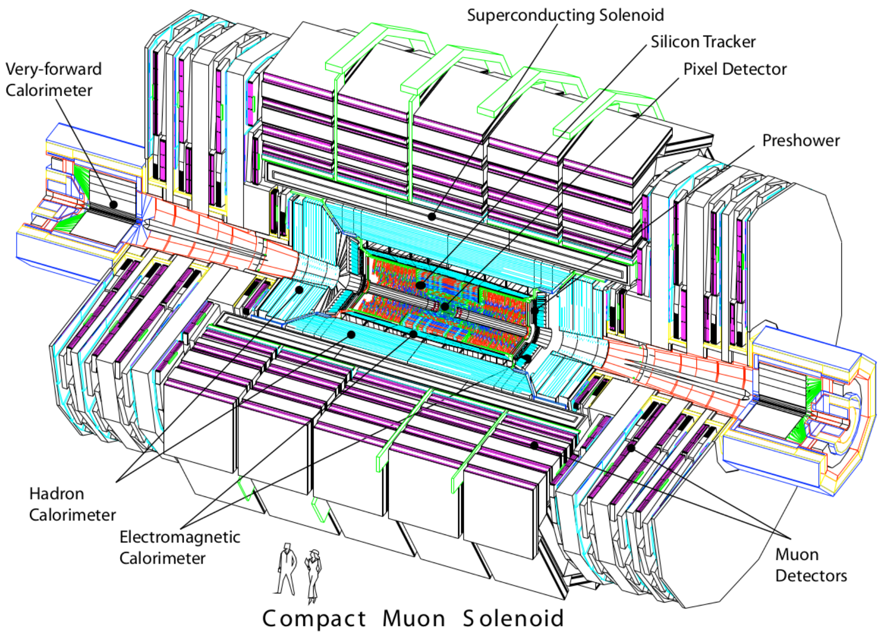


Figure 3.2: An overview of the CMS detector [32].

point is taken as a the coordinate system's centre. Trajectories of particles with energy E within the detector into the plane perpendicular to z may be described by the rapidity

$$y = \ln\sqrt{\frac{E + p_z c}{E - p_z c}}. \tag{3.1}$$

Small momenta in the z-direction p_z give a rapidity of zero, while the rapidity tends to $\pm\infty$ for large p_z. However, this requires knowledge of E and p_z, which can be difficult to measure. By assuming the particle is ultra-relativistic, as is typically the case at the LHC, it is possible to simply this description and introduce the pseudorapidity

$$\eta = \ln\left(\tan\left(\frac{\theta}{2}\right)\right) \tag{3.2}$$

which is dependent solely on θ, the polar angle. A convenient feature of the (pseudo)rapidity is that differences of (pseudo)rapidity are Lorentz invariant and thus not dependant on the initial longitudinal boost of the proton-proton system, which is a priori not known due to the varying momenta fractions of its constitutents. Together with the particle's transverse (to the beam axis) momentum p_T and mass m, a particles four-vector may be described by

$$p = \begin{pmatrix} m \\ p_T \\ \eta \\ \phi \end{pmatrix}. \tag{3.3}$$

The CMS detector may be broadly split into two distinct regions inward and outward of the boundary $|\eta|= 1.479$. The inner region or *barrel* consists of concentric layers around the beam pipe. The outer *endcap* region consists of two caps that close off the detector at either end. In this way, the CMS detector is designed for the best possible hermetic coverage around the collision point.

3.1.2 The silicon tracker

The silicon tracker [34] is the innermost system of the CMS detector, situated closest to the beampipe. It is designed to track the trajectories of charged particles as they emerge from the collision point while producing minimal energy losses of the particles themselves. This subdetector is split into two main components, the pixel detector and silicon strip detector. A sketch of these components may be seen in Figure 3.3.

The pixel detector is situated right around the beampipe and as of 2017 consists of four circular layers of individual silicon pxiels in the barrel region and three disk layers in the endcap region. These consist of rectangular silicon chips with a size of 100 x 150 μm^2. When a charged particle traverses through the active material of these chips, an electrical signal is induced that is recorded. This is typically referred to as a *hit*. The small pixel size allows for position measurements with a very high resolution, namely $\sim 10\mu$m in the $r\phi$ direction and $\sim 20\mu$m in the z direction [35]. An important feature of the pixel detector its high radiation tolerance due to the close proximity of these modules to the beam pipe.

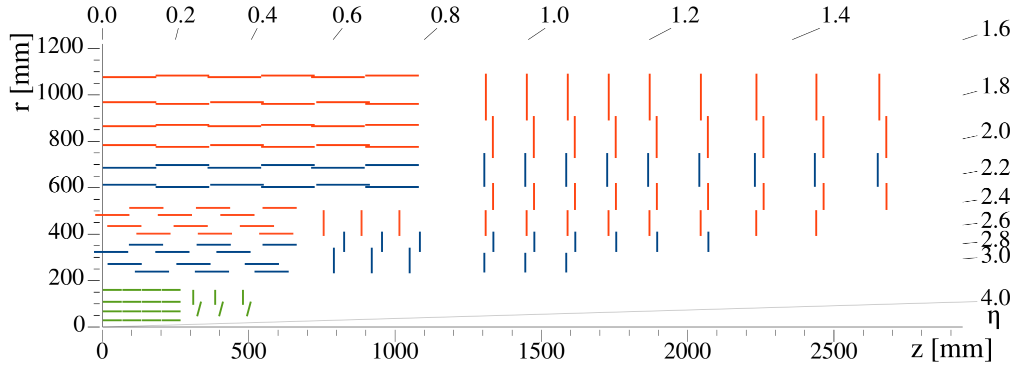


Figure 3.3: An overview of the CMS silicon tracker [34], shown in the r-z plane after its upgrade during Run-2. The pixel detector is denoted in green while the silicon strip detector is denoted in blue and orange.

Following the pixel detector is the silicon strip detector. It is composed of silicon strips of varying sizes, with increases in size at greater distances to the beam pipe due to the reduced overall particle flux they must contend with. In the barrel region, this consists of 10 layers of silicon strips, while in the endcap regions this consists of nine layers. The latter extend the coverage of the detector to $|\eta|= 2.5$.

The tracking system provides key information that is essential to the reconstruction of events. As charged particle fly through the CMS detector, their trajectories are curved due to the magnetic field generated by the solenoid magnet (see Subsection 3.1.5). By measuring the curvature of these trajectories with this system, the transverse momentum p_T of particles can be constructed. Additionally, the tracker plays a key role in methods used to determine the nature of hadronic particle cascades and the progenitor particles (quarks or gluons) from which these originate.

3.1.3 The electromagnetic calorimeter

The second innermost subsystem is the electromagnetic calorimeter (ECAL) [36][37], a sktech of which may be seen in Figure 3.4. It is designed to measure the energies of electromagnetic showers initiated by photons and electrons. The ECAL is a homogenous calorimeter, consisting of over 75,000 lead tungstate crystals. These crystals scintillate as charged particles pass through them and the produced photons can be collected via photodiodes, producing an electrical signal. This signal may be evaluated to infer the energy that is deposited. Not only do the crystals scintillate but they are also extremely dense and thus are very effective in absorbing the energy of incoming electrons and photons. This allows a very compact thickness of 23cm (22cm) in the barrel (endcap) region, which corresponds to $\sim$26 ($\sim$25) radiation lengths. An additional component of the ECAL is the preshower detector. This consists of lead absorbers interlaced with scintillating layers and help to distinguish high energy photons from neutral pions. The latter decays into photon pairs which may mimic high energy photons in this part of the detector with an increased likelihood. The increased granularity of the preshower detector helps mitigate this effect. The energy resolution of the ECAL is $\sim$ 1–4%.

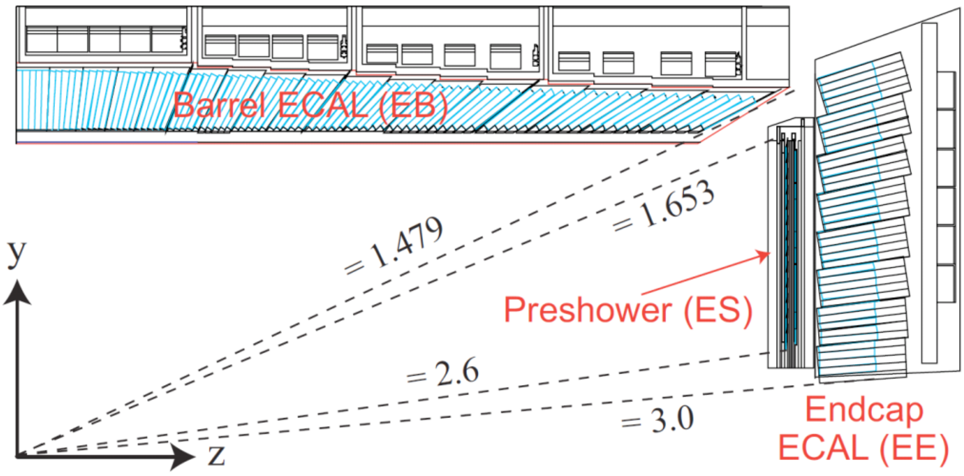


Figure 3.4: An overview of the CMS ECAL [38], shown in the $r(y)$-z plane. The dashed lines denote the coverage of the barrel and endcap ECAL region as well as the preshower detector.

3.1.4 The hadronic calorimeter

Following the ECAL is the hadronic calorimeter (HCAL) [39]. It is designed to measure the presence and energy of hadrons, which typically traverse the ECAL with minor energy losses. It is the most hermetic part of the CMS detector, with a coverage out to $|\eta| = 5.0$, in order to absorb almost all particle produced in the proton-proton collision. The only exceptions to this are muons which are particles that minimally deposit their energy, and neutrinos, which have an interaction probability that is so low that they cannot be measured with the CMS detector at all.

In contrast to the ECAL, the HCAL is a sampling calorimeter. This means layers of absorber are interleaved with layers of a scintillator. Different materials are used in different parts of the calorimeter, which is split into the barrel ($|\eta| < 1.5$), endcap ($1.5 < |\eta| < 3.0$) and forward ($3.0 < |\eta| < 5.0$) regions. Since the HCAL component inside the magnet system does not sufficiently absorb all hadronic showers, the system also extends past the magnet. Due to the sampling nature of the calorimeter, a lower number of respective interaction lengths and larger energy fluctuations in hadronic particle showers, the energy resolution of the HCAL is significantly worse than the ECAL. It lies in the order of 10–30% and with a strong dependence on the energy and pseudorapidity of the initiating particles.

3.1.5 The superconducting solenoid magnet

A key component of the CMS detector is the superconducting solenoid magnet [40]. It is responsible for maintaining a strong 3.8 T magnetic field that homogenously permeates the barrel of the detector. A measurement of the field strength can be seen in Figure 3.5. With its toroidal shape, the field is orientated along the z-axis and covers the 12.9 m long barrel region of the detector, curving the the trajectories of charged particles emerging from the interaction point in the ϕ-direction. This allows for a measurement of the particles transverse momentum p_T, which together with the ϕ and η directions fully characterise the particle's momentum vector. The magnet itself is composed of superconducting noibium-titanum coils that are cooled to a

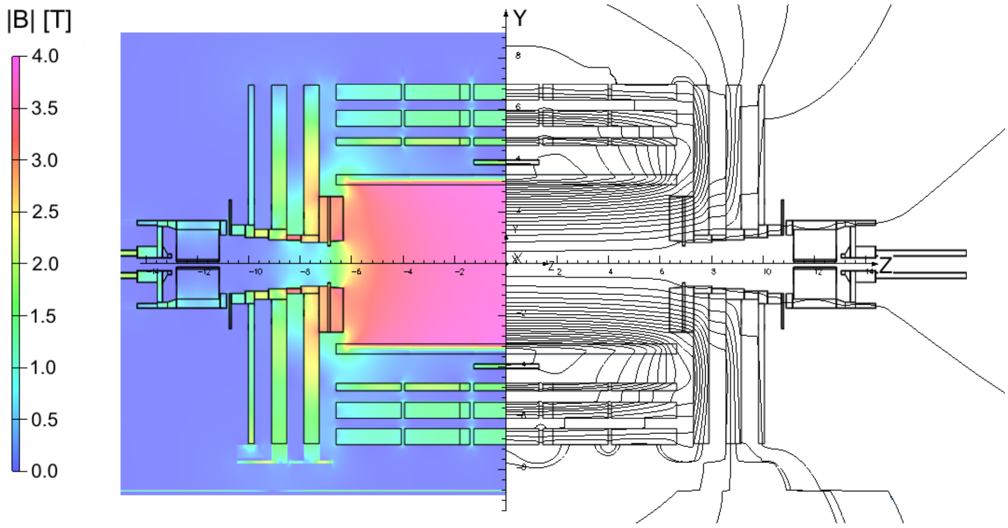


Figure 3.5: An overview of the magnetic flux (left) and magnetic field lines(right) inside the CMS detector, shown in the $r - z$ plane [41].

temperature of 4.65 K, at which these are superconducting. The magnet is encased by a 12,000 t steel yoke that captures the magentic field that is produced outside of the solenoid.

3.1.6 The muon chambers

The muon subdetector consists of a dedicated system of gaseous detectors [42][43], which are placed outside of the solenoid magnet. As suggested by the CMS name, a strong focus is placed on the performance of this subdetector. The reasons is that muons may often be produced in collisions that are of physics interest (such as in this work) and thus an emphasis is laid on detecting these with great efficiency. Due to muons being minimally ionising particles, they easily pass through the inner subdetector layers to reach the muon chambers and information from the moun chambers as well as the tracker and calorimeters may be used to identify and reconstruct them.

Like the other subdetectors, the muon chambers are separated in a barrel ($|\eta| < 1.2$) and endcap ($1.2 < |\eta| < 2.4$) region, which are composed of drift tubes (DT) and cathode strip chambers (CSC), respectively. The drift tubes each consists of a gas volume containing a mixture of Argon and CO_2 in which a posititively charged wire is stretched through the center. When charged particles such as muons traverse these tubes, the gas is ionised. Due to the positive charge of the wire, the resulting electrons drift towards the wire producing an electrical signal. Thus the presence of muons may be determined by activation of the drift tubes. The cathode strip chambers on the other hand consist of layers of positively charged (anode) wires, which are arranged in a perpendicular fashion to a set of negatively charged (cathode) strips. Combining signals from both the wires and strips allows for a position measurement in both the r and ϕ direction. Both types of detector are supplemented by resistive plate chambers (RPC), which act as a trigger providing a precise timing resolution of $\sim$1ns. This makes it possible to unambiguously assign muons to individual collisions. These consist of parallel, oppositely charged plastic plates that are coated with a conductive graphite layer and are contained in a gas volume. Ionisation of the

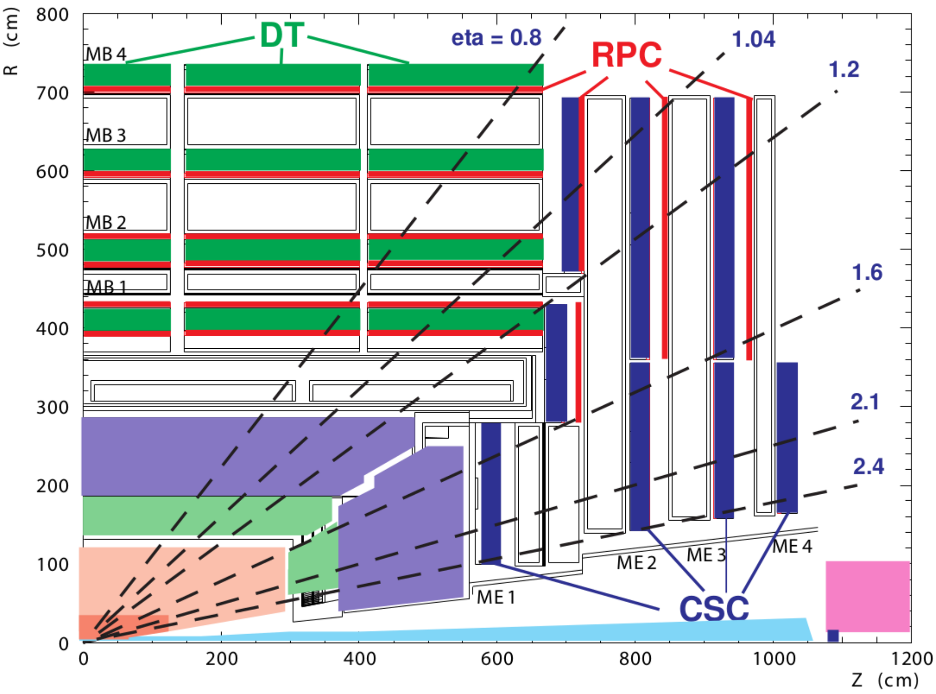


Figure 3.6: An overview of the CMS muon system, shown in the $r-z$ plane [32]. Shown are the drift tube (DT), the cathode strip chambers (CSC) and resistive plate chambers (RPC).

gas due to the traversal of a charged particle thus leads to an electrical signal. An overview of the spatial arrangement of these systems can be see in Figure 3.6. With this system, the bulk of muons may be measured with a precise momentum resolution of ∼1-2%.

3.1.7 The triggering system

The triggering system is an essential component in managing the data output of the CMS detector [44]. With a nominal collision rate of ∼40 MHz, the data rate the CMS detector provides is close to 40 TB/s. Not only is the storage of such a quantity of data unfeasible but a significant portion consists of low-energy scattering events which are not of interest to analysts. As such, the triggering system is implemented to extract a subset of events that are of physics interest.

The trigger systems is composed of two subsystems. The first the so-called level one (L1) trigger. This is a very fast hardware-based system which reduces the event rate to ∼100 kHz by evaluating the presence of e.g. energetic muons or other interesting signatures such as large energy deposits in the calorimeters in an event. The total time allocated to decide whether an event should be kept is 3.2μs. Subtracting for signal propagation in the detector, the L1 system must make a decision within 1μs. From the L1, the events are passed to a software based high-level trigger (HLT) system. This is composed of several thousand CPU cores, performing a simple reconstruction of the event signatures to make a decision whether an event should be stored. Since different analyses are interested in different signatures, a set of trigger paths are defined so that only one such path must be satisfied for an event to pass the HLT. Since the HLT is software-based, the trigger paths may be continuously updated. After the HLT, the event rate is reduced to ∼100 Hz and the passing events are permanently stored.

3.1.8 The 2018 dataset of the CMS detector

The dataset used in this work is comprised of the 2018 dataset, recorded by the CMS experiment during Run-2 of the LHC. This dataset is comprised of an integrated luminosity of 58.83fb^{-1} worth of data at a center-of-mass energy of 13 TeV. It is used in the estimation of reducible backgrounds described in Subsection 4.2.3 as well as in Section 4.3, where it is used to verify the background estimation in sidebands of the primary signal region.

3.2 Event reconstruction with the CMS detector

Events that pass the triggering system are stored and reconstructed using a more complicated set of reconstruction algorithms. An overview of the reconstruction techniques for the objects relevant to this work, namely muons and jets, is given in this section.

3.2.1 Track and vertex reconstruction

Particle tracks, describing the trajectories of particles through the detector, can be obtained by leveraging information from the pixel and strip detectors of the tracker [45]. By determining the track of a charged particle and thus the curvature of its trajectory in the detectors magnetic field, the particle's transverse momentum p_T may be implicitly determined. Since track reconstruction is a computationally intensive procedure given the large number of permutations in which individual pixel or strip hits may be combined, this procedure is performed iteratively. Inititally, tracks which are easily identifiable due to e.g. their relatively high p_T or proximity to the interaction point are identified by matching hits in the pixel and silicon strip subdetectors and performing a fitting procedure. The hits associated with these tracks are then removed from the collection of unassociated hits. This procedure is repeated anew with looser fitting criteria so that hits that may originate from low p_T tracks or those with an origin displaced from the collision point, may also be associated to tracks.

From the reconstructed tracks, common track origins or *vertices* may be identified. Since several proton-proton collisions may occur in a single bunch crossing, this amounts to identifying the location of the individual collisions in an event. Tracks with a low perpendicular distance or low *impact parameter* to the center of the bunch crossing and that satisfy requirements on the number of pixel and strip detector hits as well as the quality of the track fit are chosen for this purpose. These tracks are clustered using a deterministic annealing algorithm [46], thus producing a set of candidate vertices with some location along the z-axis. The vertex candidate which is associated with the highest $\sum {p_T}^2$ is assigned as the primary vertex of the collision. The remaining vertex candidates are referred to as pileup vertices since they typically result from other, lower energy interactions produced in pp collisions.

3.2.2 The Particle Flow algorithm

The Particle Flow (PF) algorithm [47] is used to combine information from many of the different CMS subsystems to give an improved and holistic description of an event. This includes reconstructed tracks, the energy deposits in the ECAL and HCAL as well as hits in the muon chamber system. Since different types of particles will interact with the CMS subdetector systems in unique ways, the properties of individual particles can be extrapolated from this information. These are briefly summarised in Table 3.1.

Table 3.1: Overview of particle signatures in the CMS detector

Particle	**Signature**
Muons	Muons produce tracks in the tracker as well as the muon system with minimal energy deposits in the calorimeters.
Electrons	Electrons produce tracks in the tracker as well as energy deposits in the ECAL with minimal deposits in the HCAL.
Photons	Photons do not produce tracks in the tracker due to being uncharged and deposit their energy in the ECAL.
Charged hadrons	Charged hadrons produces tracks in the tracker, primarily depositing their energy in the HCAL.
Neutral hadrons	Neutral hadrons produce no tracks in the tracker, primarily depositing their energy in the HCAL.

A visual overview of these signatures and the particle type they correspond to can be found in Figure 3.7. The PF algorithm leverages exactly these properties. Initially, matched tracks in the tracker and muon systems are identified as muons and the corresponding components are removed from the event. Subsequently, matched tracks and energy deposit clusters in the ECAL are identified as electrons and the corresponding components are removed. An isolated cluster in the ECAL with no associated track is reconstructed as a photon candidate and the corrsponding cluster is removed. This is expected to leave only charged and neutral hadrons. Clusters of energy deposits in the HCAL associated with a track are thus identified as charged hadrons. However, it frequently occurs that photons are produced in the decay of neutral hadrons. Thus, if the energy estimated from a track is considerably less than the associated cluster in the HCAL and there is a corresponding energy deposit in the ECAL, an additional photon candidate is reconstructed that is associated with the hadron. Finally, HCAL clusters with no associated track are reconstructed as neutral hadrons.

This of course is a greatly simplified description, a more comprehensive version of which can be found in [47]. The following section describe in greater detail the reconstruction of objects relevant to this work. This includes muons, *jets*, which are collimated particle showers that typically consist of a collection of reconstructed objects, and missing transverse momentum.

3.2.3 Reconstruction and identification of muons

Since muons are used to reconstruct the Higgs boson candidate of the cH process, they represent an important element of the analysis described in this work. Using the available information from the tracker and muon system, three different approaches may be used to intially reconstruct muon tracks.

- **Standalone muon tracks**: A standalone muon track refers to a fit of individual hits present in the muon detector.

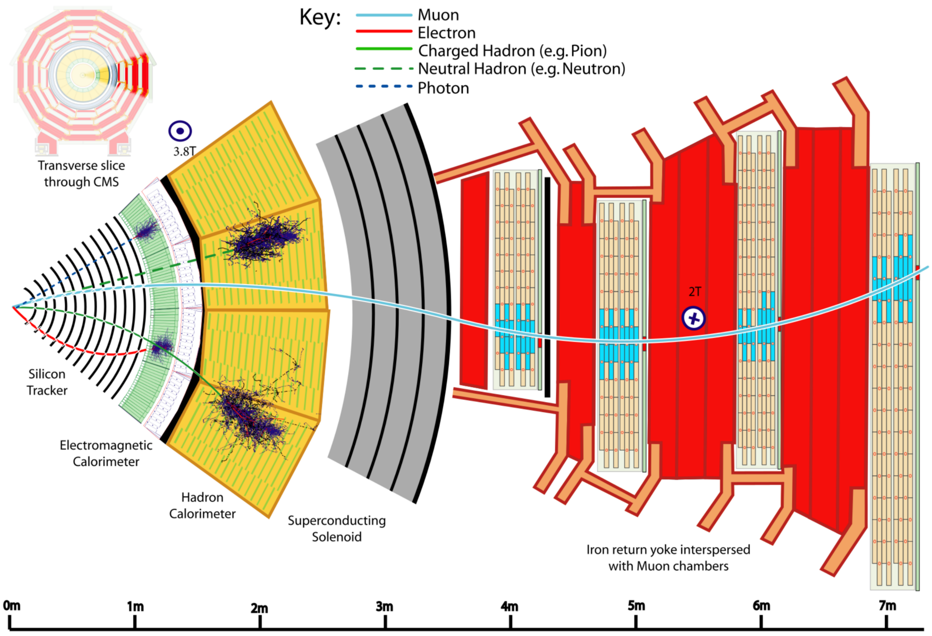


Figure 3.7: An transverse slice of the CMS detector, visualising the signatures that different particles produce in the different detector subsystems. [47].

- **Tracker muon tracks**: Tracker muon tracks are reconstructed by extrapolating tracks from the tracker to the muon detector, referred to as an *inside-out* approach. If a hit in the muon detector can be matched to the extrapolated track, then these matched tracks are identified as a tracker muon track. This reduces the impact from atmospheric muons traversing the detector, which may be falsely interpreted as standalone muon tracks.

- **Global muon tracks**: Global muon tracks are obtained through an *outside-in* appproach, matching standalone muon tracks with tracker muon tracks through a comparison of the respective fitted track parameters. If the tracks are found to match, a combined fit of these tracks is performed. This approach reduces the impact from remnants of hadronic showers that reach the muon chambers, which may be incorrectly reconstructed as a tracker muon track.

Naturally, there is a large overlap between global and tracker muon tracks. If two muon tracks share the same track in the tracker, then they are merged into a single object. The collection of standalone, tracker and global muons is passed to the previously introduced PF alogrithm which, by imposing additional quality requirements (see [47]) produces a set of reconstructed muon candidates.

A useful criterium in identifying muons that originate directly from the proton-proton interaction is the relative isolation $\mathcal{I}^{\mu}_{\text{rel}}$. This is defined as

$$\mathcal{I}^{\mu}_{\rm rel} = \left(\sum p_{\rm T}^{\rm charged} + \max\left(\sum p_{\rm T}^{\rm neutral} + \sum p_{\rm T^{\gamma}} - p_{\rm T}^{\mu,\rm PU} \right) \right) / p_{\rm T}^{\mu}. \tag{3.4}$$

Here, $\sum p_{\rm T}^{\rm charged}$ represents the salar sum of the transverse momenta of charged hadron originating from the primary vertex of the event. The quantities $\sum p_{\rm T}^{\rm neutral}$ and $\sum p_{\rm T}^{\gamma}$ represent the respective transverse momenta sums for neutral hadrons and photons. These sums are calculated by accounting from contributions within a conical volume around the muon direction. The size of a cone between two positions i and j is defiend as $\Delta R(i,j) = \sqrt{\Delta\eta(i,j)^2 + \Delta\phi(i,j)^2}$ and in this case the cone boundary around the muon direction is set at $\Delta R = 0.4$. The contribution to the relative isolation from pileup is estimated by subtracting $p_{\rm T}^{\mu,\rm PU} = 0.5 \sum_k p_{\rm T}^{k,\rm charged}$ in Eq. 3.4, where the sum over k represents charged hadron contributions not originating from the PV. The fator 0.5 corrects for different fractions of charged and neutral particles in the cone [48]. Lastly, $p_{\rm T}^{\mu}$ represents the transverse momentum of the muon. The relative isolation is thus a variable that quantifies the presence of energy deposits in the ECAL and HCAl around the trajectory of the muon, relative to the p_T of the muon. Since muons are expected to produce such deposits only minimally, good muon candidates are expected to be associated with small values of $\mathcal{I}^{\mu}_{\rm rel}$.

Two sets of muon identification criteria are defined for this work:

- **Loose muons**: Loose muons are PF muons reconstructed from either a global or tracker muon track where the perpendicular distance of the extrapolated track to the event's primary vertex is less than 5mm in the z direction and less than 2mm in the r direction.
- **Tight muons**: Tight muons are loose muons which are reconstructed exclusively from a global muon track. A number of additional criteria are applied. This includes that the fit quality of the global muon track must be $\chi^2/\text{ndf} < 10$ as well that the significance of the track's 3D impact parameter $\text{SIP}_{\rm 3D} = \text{IP}/\sigma_{\rm IP}$ satisfies $\text{SIP}_{\rm 3D} < 4$. Here IP is the impact parameter or point of closest approach to the primary vertex and $\sigma_{\rm IP}$ is the associated uncertainty. By taking into account the associated uncertainty, the $\text{SIP}_{\rm 3D}$ parameter is a more robust choice than simply the IP. Additionally, it is required the at least six layers with at least one pixel hit are registered in the tracker in the associated track as well as two segments hit in the muon detector. Lastly, a relative isolation requirement of $\mathcal{I}^{\mu}_{\rm rel} < 0.25$ is imposed.

The tight muon definition is used to select muons for reconstructing Higgs boson candidates while the loose definition is used in the estimation of reducible backgrounds.

3.2.4 Reconstruction and identification of jets

The quarks and gluons that are produce in proton-proton collisions rapidly hadronise, typically producing collimated cones of particle referred to as *jets*. Details on the concept of hadronisation, which results from the nature of the strong interaction, can be found in [49]. Since the c quark of the cH process too will produce a jet, jet objects also represent an important aspect of the analysis presented in this work.

To produce jet objects, the hadrons reconstructed by the PF algorithm must be clustered. To ensure a minimal impact of pileup on this clustering, the contributions of pileup are mitigated through *charged hadron subtraction.* This involves the removal of charged hadron contributions in the HCAL and ECAL if these may be associated with any of the pileup vertices produced in the collision, as described in Subsection 3.2.1. Once this subtraction has been performed, the remaining PF hadrons are passed to the anti-k_{T} algorithm [50]. The anti-k_{T} algorithm is an iterative clustering algorithm that is based on a priniciple of minimal distances between particles. The distance d_{ij} between the particles i and j is defined as well as the distance d_{iB} between particle i and the beam. These are given by

$$d_{ij} = \min(\frac{1}{p_{\mathrm{T},i}}, \frac{1}{p_{\mathrm{T},j}})\frac{\Delta_{ij}^2}{R^2} \tag{3.5}$$

$$d_{iB} = \frac{1}{p_{\mathrm{T}}, i} \tag{3.6}$$

$$\Delta_{ij} = \sqrt{\Delta \mathrm{y}(i,j)^2 + \Delta\phi(i,j)^2}. \tag{3.7}$$

Here, y is the rapidity of a particle and R is a constant parameter that determines the cone size of the clustered jets. The default choice used with the CMS experiment is $R = 0.4$, which is also used in this work. Starting with the highest p_T object in the initial iteration, the distance d_{ij} with the closest PF candidate j is calculated. The two objects are clustered together and this process is repeated until a stopping condition $d_{ih} > d_{iB}$ is met. At this point, the jet is considered fully reconstructed and the PF candidates used in its clustering are removed for the reconstruction of subsequent jets.

Due to the presence of detector noise, unphysical low p_T jets can be erroneously reconstructed. This effect can be mitigated by applying additional criteria on reconstructed jets. This includes requiring that at least two PF candidates are clustered in the jet and that the jet's energy is not solely attributed to neutral hadrons or photons. These requirements remove almost all such unphysical jets while over 99% of physical jets fulfill them [51]. Additionally, a pileup discrimination algorithm is described in [51], of which the loose working point is applied to jets with $p_T < 50$ GeV in this work.

A calibration of jet energies is performed after reconstruction [52] in both simulation and data. This calibration accounts for pileup contributions in the clustering, the non-linearity of the detector response and improper reconstruction of hadrons. A number of methods are used to derive sets of correction factors. An example is the use of events with a Z boson, the p_T of which may be precisely reconstructed via the Z$\rightarrow \mu\mu$ decay, that a single jet recoils against. Additionally, significant discrepancies in the resolution of jets in simulation and data are observed, with the resolution being worse in the latter than the former. This is accounted for by a smearing method, in which the resolution of jets is artifically smeared in simulation so that a better comparison to data is achieved.

3.2.5 Missing transverse momentum

Due to the conservation of momentum, it is expected that the vectorial sum of momenta of all particles produced in a collision adds up to zero. However, this may not be the case when particles such as neutrinos are produced in a collision as these cannot be measured by the detector. As a result, it can be useful to define the missing transverse momentum as

$$p_{\mathrm{T}}^{\mathrm{miss}} = \sum_{i}^{\mathrm{PF}} p_{\mathrm{T}}^{(i)}. \tag{3.8}$$

The presence of significant quantities of $p_{\mathrm{T}}^{\mathrm{miss}}$ may thus be used to identify the presence of neutrinos in an event.

3.3 Identification of charm quark-induced jets

To identify the charm quark-induced jet of the cH process, one must be able to discriminate against both bottom quark as well as light quark or gluon-induced jets. This is a task colloquially referred to as *flavour tagging*, with a jet's *flavour* being determined by the type of particle that inititated it. Modern flavour tagging techniques typically use machine learning to leverage key jet properties that may differentiate jets of different flavours, though this remains a challenging task. To discuss these properties, a definition of jet flavour is necessary. In the context of CMS, a ghost matching procedure [53] is applied to obtain such a definition for simulated events. This involves adding information from the event simulation to the reconstructed event. Specifically, hadrons containing bottom and charm quarks are identified in the simulation and added to the list of reconstructed PF candidates, albeit with negligible momenta. With this addition of so-called *ghost hadrons* the jet clustering is once again performed. Due to the neglible momenta of the ghost hadrons, the clustering procedure itself is unaffected. However, the inclusion of the ghost hadrons can be used for the following definitions:

- **c jets**: If at least one charm (c) ghost hadron and no bottom (b) hadrons are clustered inside the jet, the jet is labelled as a c jet.
- **b jets**: If at least one b ghost hadron is clustered inside the jet, the jet is labelled as a b jet.
- **light jets**: If no ghost hadrons are clustered inside the jet, the jet is labelled as a light jet. Light jets may be initiated by quarks such as the up, down, or strange quark or by gluons. An additional, technical category of *pileupjets* exists depending on whether so-called matching criteria between reconstructed and simulated jets are fulfilled, though they are subsumed into the light jets category for the purpose of this work.

The task of identifying c jets is thus twofold and broken down into two tasks:

1. Discriminating *heavy-flavour* (HF) jets consisting of b jets and c jets against light jets.
2. Discriminating between b jets and c jets.

3.3.1 Properties of heavy-flavour jets

The term heavy-flavour (HF) originates from the mass of the bottom and charm quarks, which is an order of magnitude greater than the next heaviest quark, the strange quark. The c and b hadrons have relatively long lifetimes that allow them to travel an observable distance from the PV before decaying. The typical lifetime of a b hadron of the order of ~ 1.5 ps while that of c quarks ranges down to approximately an order of magnitude less [1]. This typically results in the presence of a secondary vertex (SV) that is measurably displaced from the collision point up to a distance of 1 cm in the case of energetic hadrons and is thus a key signature of HF jets. Tracks originating from the decay of a HF induced jet thus typically originate from a SV.

Another feature of HF jets is the presence of leptons in the jet. This results from the relatively large branching fractions of HF hadrons into states containing leptons. These are typically low-energy and are present in about 20% (10%) of b(c) jets, meaning the identification of a

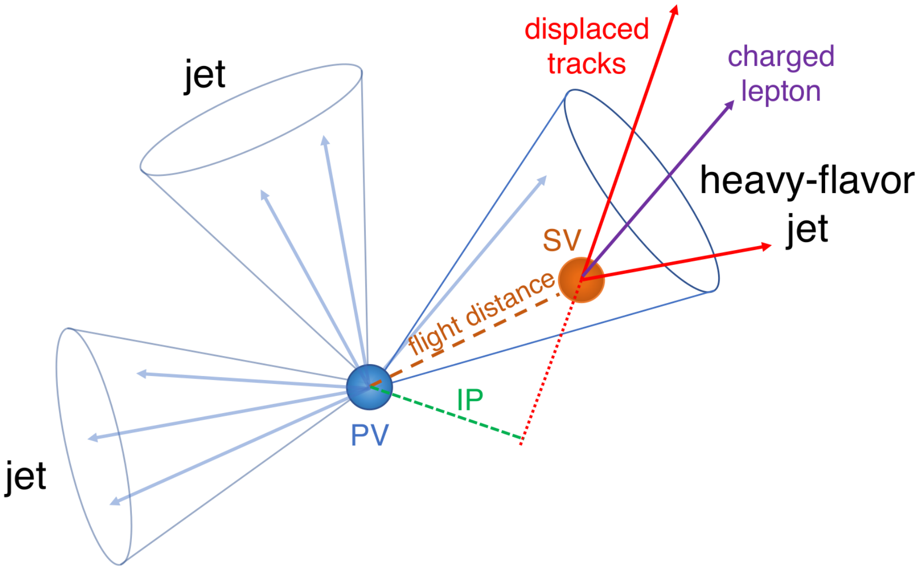


Figure 3.8: An illustration highlighting the properties of HF jets [54]. The presence of a secondary vertex (SV), characterised by the impact parameter (IP) in green, as well as the presence of a lepton is highlighted.

low-energy electron or muon inside a jet serves as a good indicator that a jet originates from a HF hadron. Also of significance are the relatively high masses HF hadrons exhibit in comparison to their lighter counterparts. This results in HF induced jets having a broader energy flux compared to their lighter counterparts, due to higher diffusion of momenta perpendicular to the flight direction as well as a higher hadron multiplicity resulting from the decay of the HF hadron. These features are illustrated in Figure 3.8. Additionally, distributions showing the significance of 2D impact parameters, which are related to the presence of SVs, can be seen for b, c and light jets in Figure 3.9

3.3.2 The DeepJet algorithm

The DeepJet algorithm [55] is a machine learning algorithm used for jet-flavour identification in this work. It improves on previous neural network based algorithms [54] used by CMS in the Run-2 period of the LHC. A notable feature compared to earlier algorithms is its use of lower level information such as use of track, PV and SV information, as well as PF candidate and event kinematics information. An overview of the architecture employed by DeepJet can be see in Figure 3.10. The network is comprised of three branches that individually process neutral and charged hadrons as well as secondary vertices before this information is combined with global variables in a set of fully connected layers. The network ouput consists of six output nodes representing six individual output classes. The output value of the nodes $\mathcal{P}(b/bb/lepb/c/l/g)$ for a given jet are interpreted as the likelihood that a jet belongs to the respective class. These are defined as

- ***b/bb/lepb* (b jets):** These three classes represent subclasses of jets originating from a b hadron. The *b* class represents a jet originating from a single b hadron, the *bb* class originating from two b hadrons and *lepb* representing a jet originating from a b hadron with the presence of a soft lepton.

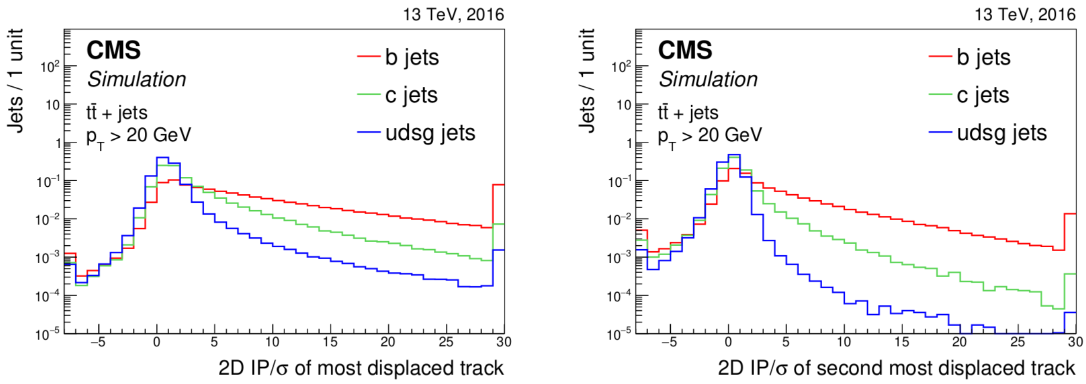


Figure 3.9: Plots showing the significance of the 2D impact parameter of the most and second most displaced tracks in a jet [54]. As can be seen, these variables can differentiate b and c jets from light jets to a significant degree.

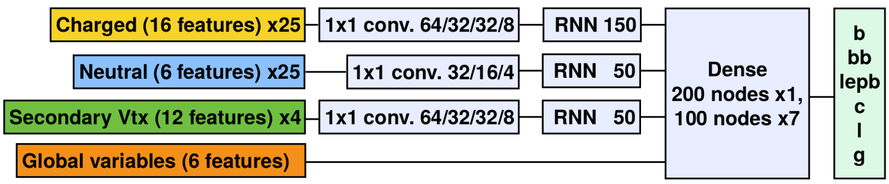


Figure 3.10: An illustration depicting the architecture of the DeepJet neural network [55]. Three individual branches separately process the charged hadrons, neutral hadrons and secondary vertex information before being passed onto a combining, fully connected layer together with global variables.

- ***c* (c jets)**: This class represents a jet originating from a c hadron.
- ***l, g* (light jets)**: These two classes represent light jets originating from a light quark or gluon respectively.

From these output classes, two useful discriminators to identify c jets can be constructed. These are

$$\mathrm{CvsB} = \frac{\mathcal{P}(c)}{\mathcal{P}(c) + \mathcal{P}(b) + \mathcal{P}(bb) + \mathcal{P}(lepb)} \tag{3.9}$$

$$\mathrm{CvsL} = \frac{\mathcal{P}(c)}{\mathcal{P}(c) + \mathcal{P}(l) + \mathcal{P}(g)} \tag{3.10}$$

representing a discrimination of c jets against b jets and light jets respectively. The performance of DeepJet in terms of the CvsL and CvsB discriminators in simulated samples of top quark pair production can be see in Figure 3.11. A comparison to the DeepCSV algorithm (a previous iteration of a jet-flavour identification algorithm) is included, highlighting the performance gain that the DeepJet algorithm achieves. The challenging, two-fold nature of the c-tagging task is demonstrated in Figure 3.12 by showing the 2D distribution of the CvsL and CvsB discriminators for simulated b, c and light (udsg) jets respectively. All three categories exhibit singificant overlap in terms of the discriminator distributions, meaning there is a significant trade off between the efficiency and purity with which c jets may be identified.

Since neural network based algorithms are trained on simulated samples that do not perfectly describe their data counterpart, the neural network output must be calibrated with respect to data. To calibrate the entire shape of the algorithm's output distributions, the approach described in [56] is used. This involves targeting phase spaces enriched in b jets (top quark pair production), c jets (charm associated $W^{\pm}$ production), and light jets (jet associated Drell-Yan production). Using simulation, the fractions of b, c, and light-flavour jets are determined in each phase space and an interative fitting procedure, minimising differences between simulation and data is performed. This allows for the derivation of correction factors which depend on the discriminators CvsL and CvsB as well as the true flavour of a simulated jet.

3.4 Monte Carlo simulation of proton-proton collisions

To produce predictions of the CMS detector response as well as the reconstruction algorithms to pp collisions at the LHC, a simulation of the collision and the CMS detector is required. Since the complexity of pp collisions in a detector cannot realistically be captured by analytic calculations, Monte Carlo methods [58] are used as an approximation. The concept of such a simulation relies on a phenomenoligcal approach by sampling the known distributions of process and detector quanities and properties. In this way, detector and particle properties may be simulated with great detail. The simulation process occurs in discrete steps, each dealing with different aspects of the simulated process. These can be summarised as:

1. **The hard scattering process**: The hard scattering process refers to the immediate, high energy transfer scattering of two protons resulting in the production of additional particles. To calculate this, two main ingredients are required. The first is a calculation of the matrix elements that describe the simulated process in which proton constituents collide to produce additional particles. These matrix elements allow for the calculation of a cross section for the process. However, the proton itself is a complex object consisting not only of

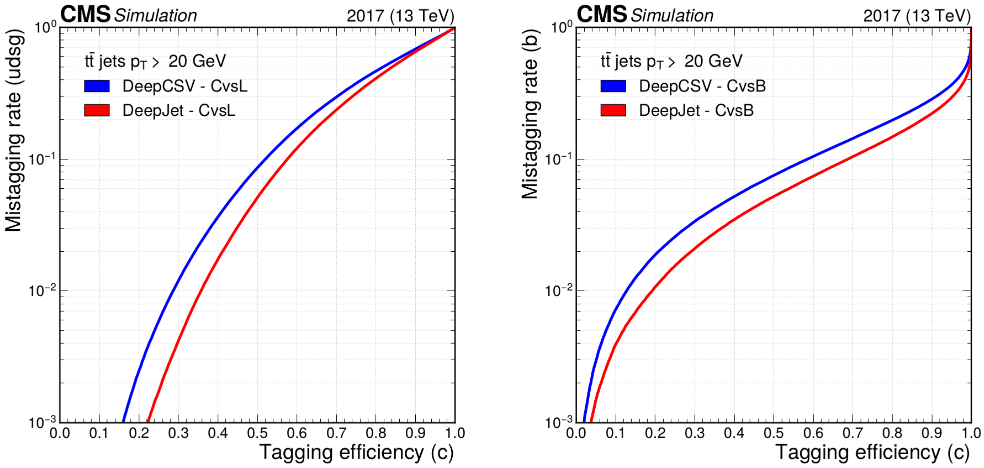


Figure 3.11: Performance of DeepJet algorithm in identifying c jets against b jets and light jets in simulated samples of top quark pair production ($t\bar{t}$), where both top quarks decay hadronically [56]. The x-axis represents the efficiency with which c jets are identified, while the y-axis represents misidentification rate with respect to either b jets or light jets.

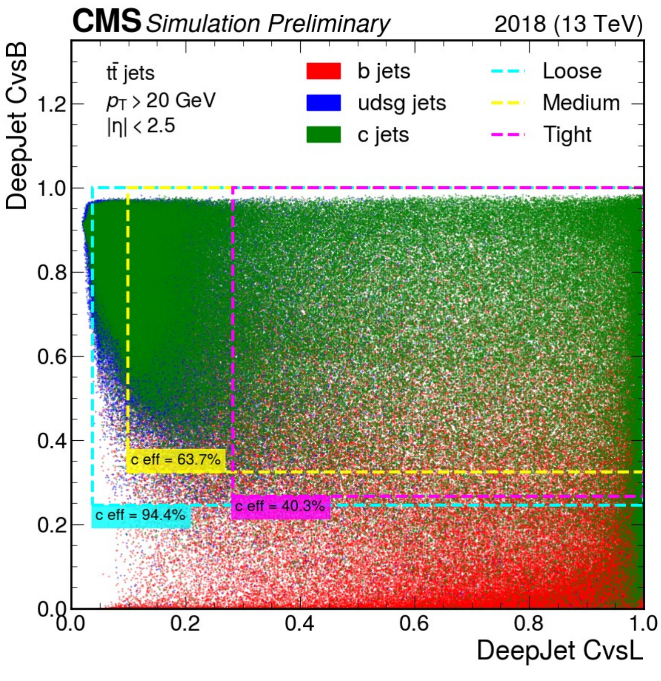


Figure 3.12: A 2D distribution of the CvsL and CvsB discriminators of the DeepJet algorithm, shown for simulated b jets (red), c jets (blue) and light jets (green) in simulated samples of top quark pair production ($t\bar{t}$) [57]. Additionally, three boxes representing different c jet identification efficiencies, each constituting fixed so-called working points, are shown.

its valence quarks (two up-type quarks and one down-type quark) but also of a constantly changing ensemble of additional quarks and gluons that are created and annihilated. This behaviour must thus be captured for an accurate process description and is parametrised via so-called *Parton Distribution Functions*. These describe the likelihood with which a parton, that carries some fraction x of the protons total momentum, may be found in a proton at some energy scale Q^2. The evolution of the PDF with changing Q^2 is described by the DGLAP quations [59]. Software used to simulate the hard scattering are referred to as *event generators*. Commonly used event generators include Madgraph5_aMC@NLO [19] and POWHEG [60].

2. **Parton showering**: Particles such as quarks and gluons that are produced in the hard scattering carry the colour charge of the strong interaction. As a result, these may produce soft radiation or branch into other particles. While a most physically accurate description would be given by including these contributions in the calculation of the hard scattering process, this greatly increases the complexity of the calculation. As such a *parton shower* model, such as in the Pythia software package [61], is used instead to describe the splitting of a single mother particle into two daughter particles. In QCD, this describes to gluon radiation ($q \to qg$) and gluon splitting ($g \to gg$ and $g \to q\bar{q}$) and in QED describes Bremsstrahlung ($f \to f\gamma$) and pair creation ($\gamma \to f\bar{f}$). In case this the parton showering originates from initial state partons it is referred to as initial state radiation (ISR). Accordingly, parton showering originating from final state partons is referred to as final state radiation (FSR). In cases with final states containing multiple partons, there can be some ambiguity in the combination of matrix elements and parton showering since both can describe the same processes. For this merging schemes are applied that resolve potential double counting of events. A prescription used for this work is the FxFx scheme [62].

3. **Hadronisation**: At an energy around the QCD scale Λ_{QCD}, the perturbative parton shower prescription loses its validty as the running coupling of the strong force α_s becomes too strong. Here the individual, colour-charged partons *hadronise* into colour-neutral states. Since this process currently cannot be described from first principles, a phenomenological description must be applied. In Pythia, the *Lund string* model is used [63]. It describes the interaction between two partons as a coloured field, the lines of which pass through a tube that is extended between the partons. The potential energy of the tube (or string) is described by a term linear in the distance between the partons. Thus if the partons are separated at a large enough distance and the potential energy is sufficiently large, the string may 'break' and new colourless quark-antiquark pairs are formed. This procedure may be repeated with these new parton pairs if they posses an invariant mass above some threshold.

4. **The underlying event**: A description of a variety of effects secondary to the hard scattering must be included in the simulation. These can have several origins such as secondary, *soft* interactions of the proton-proton collision or remnants of the collided protons, which will hadronise themselves. These effects are modeled from data [64].

5. **Detector simulation**: Finally, the detector response to the particles emerging from the previously describeds steps must be simulated. This is performed with the GEANT4 package [65], which is configured to model the CMS detector. This includes modelling the curving of particle trajectories due to the detector's magnet, the interaction of particles with the materials of the detector, as well as the digitisation of the signals in the electronic modules of the subdetectors.

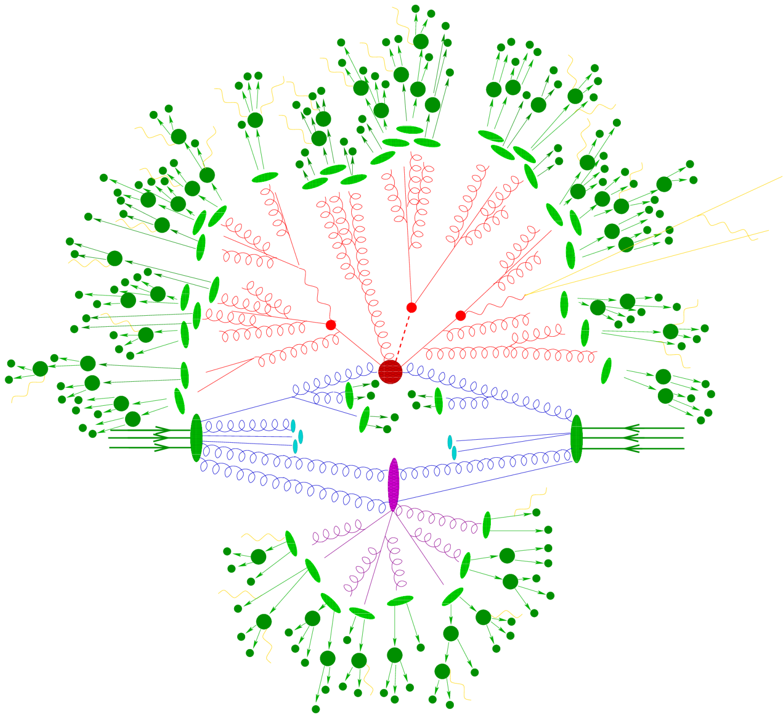


Figure 3.13: An overview of what an event simulation may look like [66]. The initial state particles originating from the protons are shown in blue. The hard scattering process is represented in red. Additional, soft scatterings are represented in purple. The hadronisation to form colourless hadrons (represented by dark green circles) is represented by the light green ellipses. Additional photon radiation is represented in yellow.

A diagrammatic overview of what an event simulation looks like can be found in Figure 3.13. The output of this simulation is passed to the reconstruction algorithms described in this section, thus allowing analysers to produce predictions

Chapter 4

Search for the cH(ZZ$\rightarrow$4μ) process

To probe the charm Yukawa coupling with the cH(ZZ$\rightarrow$ 4μ) process, a methodology must be devised to select and reconstruct cH(ZZ$\rightarrow$ 4μ) candidate events. This candidate selection is described in Section 4.1, with a focus on both the Higgs boson candidate reconstruction as well the selection of the jet associated with the charm quark. Additionally, a model describing the expected contributions from the cH(ZZ$\rightarrow$ 4μ) process as well as a number of background processes for the presented event selection is described in Section 4.2. Here, particularly the simulation of cH(ZZ$\rightarrow$ 4μ) events as well as the estimation of reducible backgrounds are discussed in greater detail. Following this, a statistical evaluation method using discriminators dervived from the reconstructed events is presented in Section 4.4. Finally in Section 4.5, 95% $\mathrm{CL_s}$ CL upper limits on κ_c are set by applying this method for a variety of scenarios, including statistical extrapolations to full Run-2 and High Luminosity-LHC scenarios.

4.1 cH(ZZ$\rightarrow$ 4μ) event selection

To reconstruct a cH(ZZ$\rightarrow$ 4μ) candidate event, a Higgs boson candidate needs to be reconstructed and a corresponding jet candidate needs to be identified. These two procedures are described in this section. Distributions of cH(ZZ$\rightarrow$ 4μ) candidate events are shown using a simulation of the cH(ZZ$\rightarrow$ 4μ) process.

To reconstruct a Higgs (jet) candidate, an initial selection of muon (jet) objects must be made. These are summarised in Table 4.1, along with the HLT trigger path requirement used in this analysis. The objective of this selection is to identify events with well-reconstructed, isolated muons as well as a least on well-reconstructed jet. Following this initial selection, the corresponding objects are passed onto respective algorithms to select a final Higgs and jet candidate.

Table 4.1: Muon, jet object and HLT path selection requirements.

Object	**Selection criteria**
Muons	$p_T > 5$ GeV $\lvert\eta\rvert < 2.4$ Tight muon identification criteria
Jets	$p_T > 25$ GeV $\lvert\eta\rvert < 2.5$ Jet ID Pile-up ID, loose working point ΔR(jet, selected muons) > 0.4 Jets in veto regions of detector are excluded
HLT	HLT_IsoMu24 is triggered

4.1.1 Higgs candidate selection

A Higgs boson reconstruction algorithm (and muon object selection) very similar to those presented and validated in [67] is implemented. This reconstruction is performed for events in which exactly four selected muons are present to avoid introducing a potential bias in the mass of the Higgs boson candidate. This is relevant due to the potential contributions of background processes in which no Higgs boson is produced. For these events, the following reconstruction steps are applied:

1. Of the four selected muons, the p_T-leading muon is required to satisfy p_T >20 GeV and the sub-leading muon is required to satisfy p_T >10 GeV. Additionally, the HLT_IsoMu24 trigger requirement must be met. Lastly, to ensure two muons are not spuriously reconstructed from shared tracks, it is required that each muon candidate is separated from the others by ΔR >0.02.

2. Opposite-sign muon pairs are merged into Z boson candidates. At least two Z boson candidates must be reconstructed to proceed. Additionally, the invariant mass of any combination of opposite-sign muons must satisfy $m(\mu\mu)> 4$ GeV, to remove any contributions from low mass resonances such as J/ψ [1].

3. The Z candidate with a mass closest to the known Z boson mass of $m(Z) = 91.19$ GeV [1] is interpreted as an on-shell Z_1 candidate. The Z_1 candidate should satisfy 40 GeV < $m(Z_1) < 120$ GeV. The other candidate is taken as the Z_2 candidate, which is typically more off-shell and thus the invariant di-muon mass requirement is relaxed to 12 GeV < $m(Z_1) < 120$ GeV.

4. The Z_1 and Z_2 candidates are combined to form a Higgs boson candidate. The four-muon invariant mass of the Higgs boson candidate must satisfy $m(H) > 70$ GeV.

The reconstructed Higgs boson candidate mass distribution in simulated cH(ZZ→ 4μ) events can be seen in Figure 4.1. As expected, a peak around the known Higgs mass $m(H) = 125$ GeV [1] can be oberserved, with an elongated tail towards lower masses that originate from increasingly off-shell Z candidate contributions.

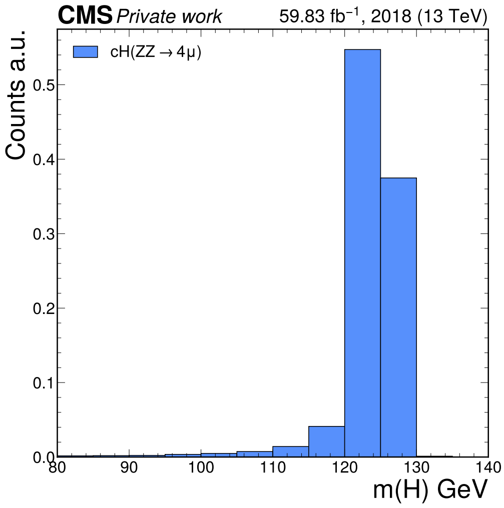


Figure 4.1: The distribution of the reconstructed Higgs boson candidate mass $m(H)$ in simulated cH(ZZ→ 4μ) events. A clear peak around the nominal Higgs boson mass of 125 GeV is observed.

4.1.2 Jet candidate selection

Once a Higgs boson candidate is reconstructed, a likelihood ratio algorithm is applied to best identify and select the jet that is associated with (i.e. recoils off) the reconstructed Higgs boson. This algorithm does not use jet-flavour identification methods and is based solely on kinematic properties of the jets, so as to minimise the introduction of any flavour bias in the selection. Specifically, two variables related to momentum conservation in the transverse plane are exploited:

1. The difference in azimuthal angle $\Delta\phi$(H, jet) between the Higgs boson candidate H and the jet is used. Due to an initial zero net momentum in the direction of the azimuthal angle, the Higgs boson and associated jet are expected to recoil off eachother *back-to-back* and thus $\Delta\phi$(H, jet) is expected to be $\sim \pm\pi$.

2. Since the Higgs boson and associated jet recoil off eachother, their p_T is expected to be approximately balanced. This information can be captured by transverse momentum ratio p_T(H)/p_T(jet).

To derive the relevant distributions to be used in a likelihood ratio, a parton-to-jet matching is performed in simulated cH(ZZ→ 4μ) events. This is achieved by, in each simulated event, taking the directional information of the simulated parton and matching it to a reconstructed jet with the matching requirement $\Delta R(\text{jet}, \text{parton})<0.3$. All jets which match the initial jet selection are considered for this process and a matching efficiency of $\sim$80% is achieved for events in which the parton is a charm quark, while an efficiency of $\sim$75% is achieved for events in which the parton is a gluon. A jet which is matched in this way is labelled as the associated jet, while the remaining non-matched jets are labelled as non-associated jets. Once this labelling is performed, the distributions of both kinematic variables are extracted as templates for both associated and non-associated jets and treated as probability density functions. To capture kinematic differences associated with higher and lower p_T Higgs candidates, this procedure is repeated in different bins of $p_T(H)$ listed in Table 4.2. The templates that are extracted in this way can be seen in Appendix Section B.

Using the extracted templates, a per-jet likelihood evaluation can be made in each event. For this, the per-variable likelihood ratio

$$\mathcal{L}(x) = \frac{\mathcal{L}_{\mathrm{associated}}(x)}{\mathcal{L}_{\mathrm{non-associated}}(x)}, \text{ with } x \in \left\{\Delta\phi(\mathrm{H}, \mathrm{jet}), \frac{p_T(H)}{p_T(\mathrm{jet})}\right\} \tag{4.1}$$

is defined. From this follows the per-jet likelihood

$$\mathcal{L}(\mathrm{jet}) = \mathcal{L}\Big(\Delta\phi\big(\mathrm{H}, \mathrm{jet}\big)\Big)\mathcal{L}\left(\frac{p_T(H)}{p_T(\mathrm{jet})}\right). \tag{4.2}$$

The jet with the highest associated likelihood in an event is selected as the jet candidate. The distribution of $\mathcal{L}(\mathrm{jet})$ for both associated jets and non-associated jets can be seen in Figure 4.2 for simulated cH(ZZ→ 4μ) events in which an associated jet can be identified via a parton-to-jet matching. The distribution of the transverse momentum of the selected jet can be seen in Figure 4.3.

Table 4.2: The p_T(H) bins in which the jet selection procedure is performed.

Bin number	**p_T(H) range**
1	0 - 15 GeV
2	15 - 30 GeV
3	30 - 50 GeV
4	50 - 100 GeV
5	100 - 200 GeV
6	>200 GeV

With this, all components of cH(ZZ→ 4μ) candidate events can be reconstructed. Events satisfying the described requirements are thus selected for further evaluation.

4.1.3 Reconstruction efficiency

A total selection efficiency of ∼10% is achieved on the simulated cH(ZZ→ 4μ) sample using the described methods. The losses in efficiency can be attributed to the following:

- Approximately 50% of generated events fall outside of the geometrical detector acceptance.
- Of the 50% of generated events within the geometrical detector acceptance, approximately 50% fulfill the described muon selection requirements.
- Approximately 91% of these events pass the Higgs reconstruction, leading to an additional event yield loss of 9%
- Of the remaining events, the event yield is reduced by close to 60% by the jet selection requirements that are imposed.

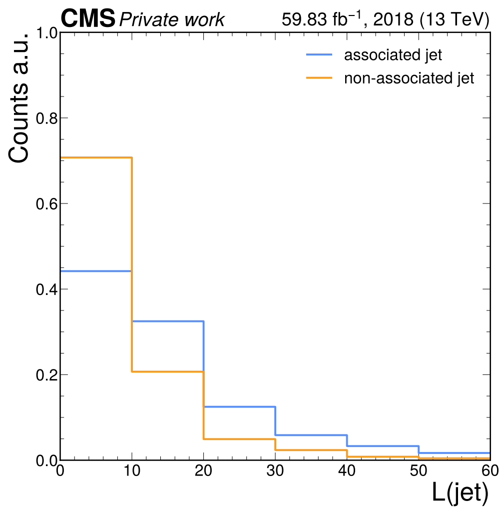


Figure 4.2: The distribution of the jet selection likelihood $\mathcal{L}$(jet), in simulated cH(ZZ→ 4μ) events in which an associated jet can be identified via a parton-to-jet matching, for associated and non-associated jets. It can be seen that on average, applying the jet selection algorithm on the associated jet produces a higher likelihood value than on non-associated jets, as expected.

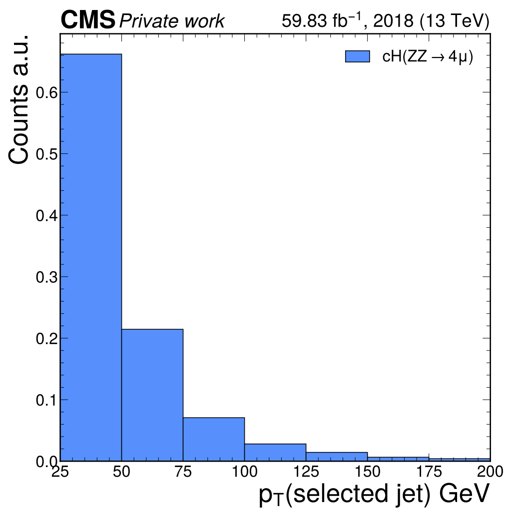


Figure 4.3: The distribution of the selected jet's transverse momentum in simulated cH(ZZ→ 4μ) events. As expected, the majority of jets come at relatively low transverse momenta.

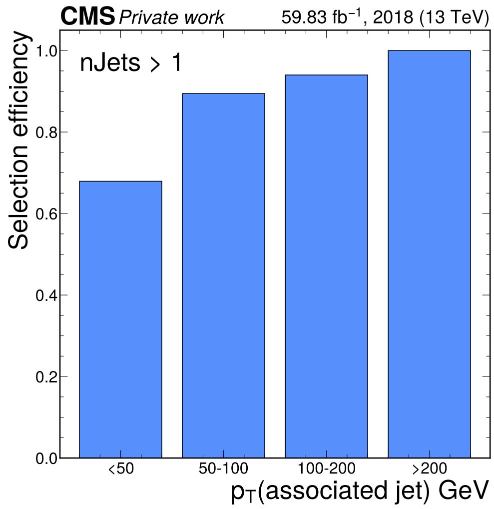


Figure 4.4: The efficiency with which the jet associated with the Higgs boson is selected via the described algorithm in simulated cH(ZZ→ 4μ) events. This efficiency is shown in bins of the associated jet's transverse momentum for events in which more than one jet passes the jet selection requirements and an associated jet can be identified via a parton-to-jet matching.

Also of interest is the efficiency with which the jet associated with the Higgs boson is selected with the described jet selection method. For events in which more than one jet passes the jet selection criteria, an efficiency of 68% is achieved for associated jet transverse momenta below 50 GeV, with the efficiency exceeding 90% for associated jet transverse momenta above 50 GeV, as seen in Figure 4.4. It should be noted that this calculation can only account for events in which an associated parton is generated at the matrix element level (see Subsection 4.2.1 for more details) and in which the parton can be matched to a jet.

4.2 Signal and background estimation

The cH(ZZ→ 4μ) process, as well as background processes which may mimic its signature, must be estimated. This includes estimating the underlying physics processes as well as the interactions that they produce with the detector. This is done primarily using the Monte Carlo methods described in Section 3.4, with the simulation of the cH(ZZ→ 4μ) process being specifically discussed in Subsection 4.2.1. Overall two primary types of background to the cH(ZZ→ 4μ) process exist. These are the *irreducible* backgrounds, which refer to processes in which four muons are produced in the hard scattering, and the *reducible* backgrounds, which refers to processes in which less than four muons are produced in the hard scattering. These are discussed in Subsection 4.2.2 and Subsection 4.2.3, respectively. From these estimations, a comprehensive model of the expected yields and distributions that would result from applying the described selection to a given CMS detector dataset can be constructed.

4.2.1 Estimation of cH(ZZ→ 4μ) process

The cH(ZZ→ 4μ) process is estimated using a simulation generated by MadGraph5_aMC@NLO. The following MadGraph5_aMC@NLO syntax is used, which illustrates some important concepts related to the simulation of the cH process:

```
import model loop_sm_MSbar_yb_yc-yc4FS
define p = g u u~ d d~ s s~ c c~
define j = g u u~ d d~ s s~ c c~
generate p p >h [QCD] @ 0
generate p p >h j [QCD] @ 1
```

In the first line, it can be read off that the loop_sm model is used, a model allowing NLO in QCD calculations of SM processes. Only the Yukawa couplings of the bottom and charm quarks are included to ensure orthogonality of the cH simulation to simulations of other Higgs production processes such as gluon fusion. Additionally, a so-called *four flavour scheme* (4FS) version of the model is used. The flavour scheme denotes which quarks are included as constituents of the proton, in which they are approximated as massless. The 4FS includes the up, down, strange and charm quarks as proton constituents. In contrast to the 4FS, a three flavour scheme 3FS could also be used. Here, the charm quark is not included in the proton and initial state charm quarks must thus instead be produced via gluon splitting, i.e. $g \to c\bar{c}$.

In the following two lines, the proton and jet constituents are defined. Finally in the last two lines, the processes included in the simulation are defined. These are, calculated at next to leading order in QCD, the $pp \to H$ and $pp \to H + j$ processes. Both are included to give the most accurate possible kinematic description of the cH process. The reasoning for this is related to the modelling of final state partons and can be better understood by considering what is included in the leading order (LO) and next-to-leading (NLO) contributions to $pp \to H$ and $pp \to H + j$ respectively. At leading order, an additional jet in $pp \to H$ can only be generated via the parton shower. Thus, this contribution is expected to best model the lower momentum behaviour of final state partons. The NLO contributions to $pp \to H$, which correspond to LO contritubtions of $pp \to H + j$, in turn are expected to better model higher momentum behaviour of the final state parton. The same logic is applied to the LO contributions of $pp \to H + j$ and the NLO contributions of $pp \to H + j$, where two final state partons explicitly appear in the calculation of the latter. However, this approach introduces double counting of processes. These are accounted for along with double counting between parton shower and matrix element contributions using the FxFx merging scheme.

The types of event topologies generated by these commands can be best understood by categorising them according to the partons that inititate them. In this way, four categories can be identified:

- $c\bar{c}$: At leading order, this topology arises from an initial state c and anti-c quark that interact to produce a Higgs boson. Final state partons must thus be produced via next-to-leading order contributions to the matrix element (i.e. gluon radiation) or via parton showering. Notably, the charm quarks associated with the Higgs boson are not present in the final state of this type of topology. This means that for the described event selection strategy, this analysis is only be sensitive to these topologies via an 'incorrect' jet association. A visualisation of this type of event topology can be seen in Figure 4.6a. It is the most prominent topology in the generated sample.

- ***cg***: At leading order, this topology arises from an initial state (anti-)c quark that emits a Higgs boson and absorbs a gluon. This topology corresponds to the leading order cH process that is discussed in Subsection 2.3.1. Additional final state partons may be produced via next-to-leading order contributions to the matrix element or via parton showering. A visualisation of this type of event topology can be seen in Figure 4.6b. It is the second most prominent in the generated sample.

- ***gg***: This topology arises from two initial state gluon which split to form $c\bar{c}$ pairs that interact to produce a Higgs boson. Since this is already considered a next-to-leading order contribution to the matrix element by the generator, additional final state partons can only be generated via parton showering. A visualisation of this type of event topology can be seen in Figure 4.6c. It contributes to only a small fraction of events in the generated sample.

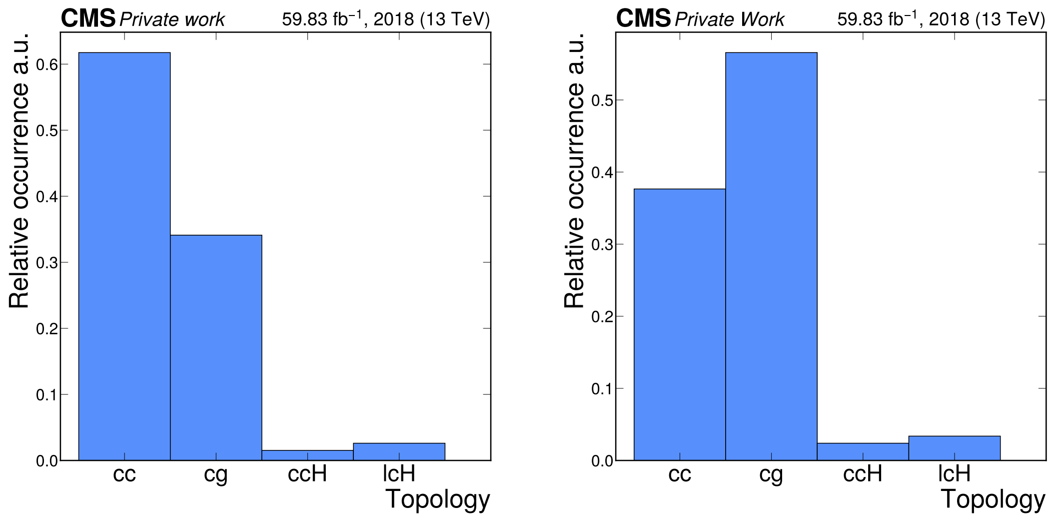
- ***cl***: This topology arises from an initial state (anti-)c quark that interacts with an additional initial state light quark via gluon exchange, which proces a Higgs boson. Since this is already considered a next-to-leading order contribution to the matrix element by the generator, additional final state partons can only be generated via parton showering. A visualisation of this type of event topology can be seen in Figure 4.6d. Like the *gg* topology, it contributes to only a small fraction of events in the generated sample.

An overview of the relative occurrence of these topologies in the generated sample can be seen in Figure 4.5a. Clearly, the $c\bar{c}$ topology dominates with a relative proportion $> 60\%$, while the *cg* topology only comprises $\sim 35\%$ of events. Since the jet identification plays no role in the former topology, it is worth considering how the selection algorithm described in Section 4.1 affects the relative occurrence of each topology. This can be seen in Figure 4.5b. Clearly, the selection algorithm (specifically the jet selection) introduces a significant bias towards *cg* topologies as expected and desired. However, a relative porportion of close to 40% of $c\bar{c}$ topologies remain, which may still contribute towards the sensitivity of the analysis. These are typically associated with light flavour jets. This motivates the use of an inclusive statistical evaluation strategy, exemplied by for example the use of the full CvsB and CvsL DeepJet distributions of the selected jet.

To capture the effects of uncertainties associated with the choice of a particular flavour scheme in the simulation of cH(ZZ→ 4μ) , two additional signal samples are used. These specifically simulate the cH(ZZ→ 4μ) process in the 3FS and 4FS, without the use of FxFx merging, effectively exclusively describing cases where a charm *must* originate from gluon splitting or originate directly from the proton, respectively. From these samples, an uncertainty envelope is constructed for the discriminators distributions used in the statistical evaluation presented in Section 4.4. This envelope is determined by taking the relative discrepancy between the 3FS and 4FS samples and interpreting it as an uncertainty band around the nominal 4FS FxFx sample distributions. An overview of all used signal samples can be seen in Table 4.3.

4.2.2 Estimation of irreducible backgrounds

Irreducible background processes are background processes that produce the same final state particles as the signal process in question. Thus, processes which produce four final state muons along with the presence of a, or several, jet(s) constitute the irreducible background. These again fall into two categories, namely those where the four muons originate from a Higgs boson and those in which they do not. When analysing e.g. the mass spectrum of Higgs candidates,



(a) Before applying selection algorithm.

(b) After applying selection algorithm.

Figure 4.5: The relative occurrence of the $c\bar{c}$, cg, gg and cl event topologies in the simulated cH(ZZ$\rightarrow 4\mu$) sample before and after applying the selection algorithm.

the processes of the former category is thus clearly resonant around the Higgs mass of ∼125 GeV, while those of the latter category may take on a more continuous shape. The irreducible backgrounds considered in this work are:

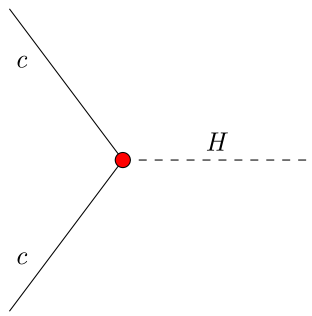
- **Gluon fusion (ggH)**: In these processes, a pair of gluons produce an intermediary (top) quark pair loop that produces a Higgs boson. The Higgs boson decay may produce four muons while parton radiation may produce additional jets. An example Feynman diagram of the ggH process can be seen in Figure 4.7a.

- **Top quark pair assocaited Higgs production(ttH)**: In ttH processes, a pair of top quarks is produced alongside a Higgs boson. The Higgs boson decay may produce four muons while the decay of the top quark pairs produces additional b jets. An example Feynman diagram of the ttH process can be seen in Figure 4.7b.

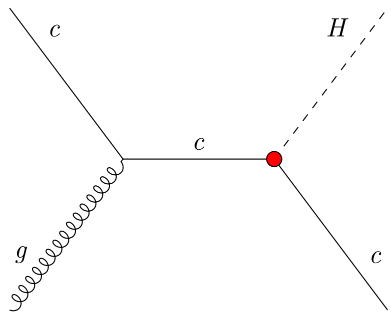
- **Vector boson associated Higgs production (VH)**: The vector boson produced alongside a Higgs boson may refer to W$^{\pm}$ and Z bosons. Thus, three different process types contribute to this background, namely W^{+}H, W^{+}H and ZH. The Higgs boson decay may produce four muons while a W$^{\pm}$ boson decay as well as a Z boson decay may produce additional jets. An example Feynman diagram of these processes can be seen in Figure 4.7c.

- **Vector boson fusion (qqH)**: In vector boson fusion processes, two vector bosons emitted by initial state quarks fuse to produce a Higgs boson. The Higgs boson decay may produce four muons while additional final state quarks that produce jets are also present. An example Feynman diagram of the qqH process can be seen in Figure 4.7d.

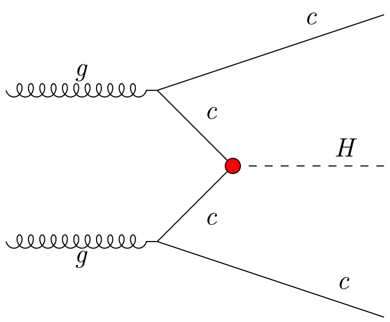
- **Top quark plus quark associated Higgs production (tqH)**: In tqH processes, a top quark as well as a quark of a different flavour are produced alongside a Higgs boson. The



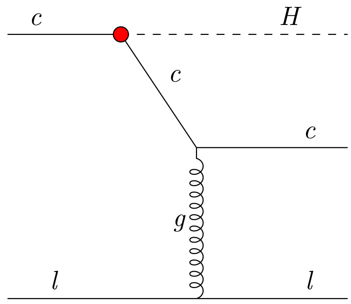
(a) The $c\bar{c}$ topology at lowest order.



(b) The cg topology at lowest order.



(c) The gg topology at lowest order.



(d) The cl topology at lowest order.

Figure 4.6: Examples of feynman diagrams that illustrate the four different event topologies generated in the cH(ZZ→ 4μ) sample at the lowest order. The red dots represent the y_{c} coupling.

Higgs boson decay may produce four muons while the additional (top) quark will produce a (b) jet. An example Feynman diagram of the tqH process can be seen in Figure 4.7e.

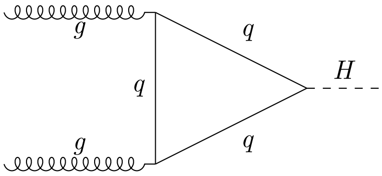
- **ZZ production from gluons (gg→ZZ)**: A pair of Z bosons can be produced from a pair of initial state gluons via an intermediate quark loop. The Z boson decays may produce four muons while additional parton radiation may produce jets. An example Feynman diagram of the gg→ ZZ process can be seen in Figure 4.7f

- **ZZ production from quarks (qq→ZZ)**: A pair of Z bosons can be produced from two initial state quarks. The Z boson decays may produce four muons while additional parton radiation may produce jets. An example Feynman diagram of the qq→ZZ process can be seen in Figure 4.7g.

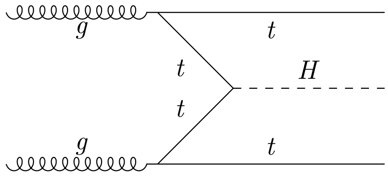
The irreducible backgrounds are estimated using simulation at next-to-leading order in QCD. The exception to this is the gg→ZZ process, for which only a sample in leading order in QCD is available. To correct for this, a k-factor of 1.7 is applied to the cross section to extrapolate the cross section to next-to-leading (NLO) order in QCD [68]. Similarly, a k-factor of 1.1 is applied to the NLO cross section of the gg→ZZ process, extrapolating the cross section to next-to-next-to-leading (NNLO) order in QCD [69]. This is done due to the significant, non-neglible contributions to the qq→ZZ process at NNLO in QCD. An overview of the samples that are used can be found in Table 4.4. Like with the cH(ZZ→ 4μ) process, additional samples are used for the bH($ZZ \to 4\mu$) background to account for an uncertainty related to the choice of flavour scheme. However, here the five flavour scheme (5FS) is now the nominal FS to include the bottom quark in the proton and the 4FS is used to estimate the bH contribution where the initial state bottom quark originates from gluon splitting.

Table 4.3: cH(ZZ→ 4μ) samples used in this work

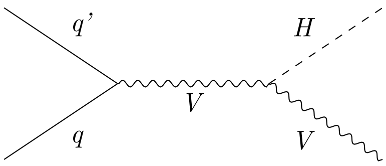
Process	$\sigma\cdot$BR	**# of simulated events**
cH(ZZ→ 4μ) 4FS FxFx	0.025 fb	4,854,537
cH(ZZ→ 4μ) 3FS	0.014 fb	1,941,000
cH(ZZ→ 4μ) 4FS	0.022 fb	1,628,000



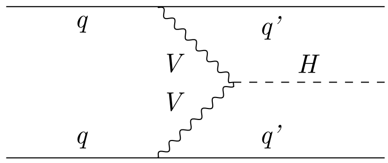
(a) The ggH process.



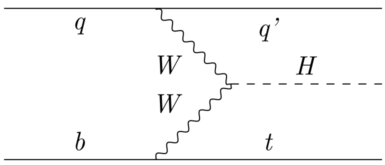
(b) The ttH process.



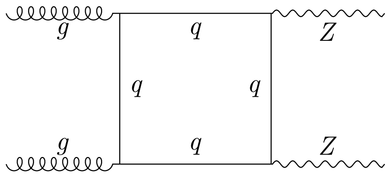
(c) The VH process.



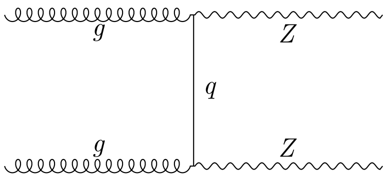
(d) The qqH process.



(e) The tqH process.



(f) The gg→ZZ process.



(g) The qq→ZZ process.

Figure 4.7: Representative feynman diagrams of the irreducible background processes relevant to the cH(ZZ→ 4μ) analysis. This includes the ggH, ttH, VH, qqH, tqH, gg→ZZ and qq→ZZ processes. When no final state parton is explicitly present in the diagram it is implied via, for example, parton radiation.

Table 4.4: Simulated processes used for background estimation in this work. The listed processes represent the irreducible backgrounds to the cH(ZZ→ 4μ) analysis with the exeception of the WZ→ $3\ell\nu$ process, which is used solely in the estimation of the irreducible backgrounds.

Process	σ·BR	**# of simulated events**
ggH(ZZ→4L)	12.18 fb	940,000
ttH(ZZ→4L)	0.393 fb	490,043
W^-H(ZZ→4L)	0.147 fb	193,893
W^+H(ZZ→4L)	0.232 fb	298,647
ZH(ZZ→4L)	0.668 fb	544,550
qqH(ZZ→4L)	1.044 fb	477,000
tqH(ZZ→4L)	0.0213 fb	969,000
gg→ZZ(4μ)	1.59 fb	786,136
gg→ZZ(4τ)	1.59 fb	493,998
gg→ZZ($2\mu2\tau$)	3.19 fb	434,000
qq→ZZ/Zγ^* →4L	1.256 pb	84,366,000
bH 5FS FxFx	0.192 fb	3,869,701
bH 4FS	0.099 fb	2,000,000
bH 5FS	0.153 fb	1,944,000
WZ→ $3\ell\nu$	4.43 pb	9,821,283

4.2.3 Estimation of reducible backgrounds

Reducible background processes are background processes that do not produce the same final state particles as the signal process but where mis-identification of physics objects can still falsify the sought-after signature. Since jets are typically abundant in most collisions, this amounts to the misidentification of additional muons for this analysis. Major contributions to this background are expected to come from the Drell-Yan process while other processes which may produce at least two leptons, such as $t\bar{t}$, also contribute [70]. Example Feynman diagrams of these two processes can be seen in Figure 4.8. Since the simulation of misidentified muons is subject to significant modelling uncertainties, a data-driven approach may be used. This involves determining the misidentification rate of muons in data and applying it to a sideband region from which the contributions of reducible backgrounds are extrapolated into the signal region. The methodology used to this end is presented in this section and follows that of [67].

Determination of muon misidentification rate

To determine the misidentification of muons with respect to the tight muon requirement outlined in Subsection 3.2.3, a three-muon selection is applied to data. Specifically, events with a $Z \to \mu^+\mu^-$ decay that also contain a third reconstructed muon are targeted. Since hard-scattering processes that produce a Z boson are not expected to produce any additional muons, the third reconstructed muon (mmreferred to as the *probe muon*) is assumed to be one that is misidentified as such. To determine a misidentification rate, the ratio of probe muons that pass the tight muon requirement with respect to those that pass the loose muon requirement is calculated. This procedure is performed in bins of the probe muon $p_T$for the barrel ($|\eta| \leq 1.2$) and endcap ($|\eta| > 1.2$) regions respectively. The exact reconstruction algorithm that is applied is the following:

1. Events that contain at least two muons that pass the tight identification requirement and where the third passes at least the loose identification requirement are chosen. The p_T-

(a) The Drell-Yan process. (b) The $t\bar{t}$ process.

Figure 4.8: Representative feynman diagrams of the Drell-Yan and $t\bar{t}$ processes, which are expected to contribute significantly to the reducible background of the cH(ZZ→ 4μ) analysis. Jets are produced via for example parton radiation for the Drell-Yan process, while the decay of the top quarks in the $t\bar{t}$ process automatically produces b jets from the hadronisation of b quarks.

leading muon is required to satisfy p_T >20 GeV and the sub-leading muon is required to satisfy p_T >10 GeV. Additionally, the HLT_IsoMu24 trigger requirement must also be met. Lastly, to ensure two muons are not spuriously reconstructed from shared tracks, it is required that each muon candidate is separated from the others by ΔR >0.02.

2. Opposite-sign muon pairs are merged into Z boson candidates and the candidate closest to the nominal Z mass is taken as the final Z candidate. Additionally, the invariant mass of any combination of oppsite-sign muons must satisfy $m(\mu\mu)>$ 4 GeV, to remove any contributions from low mass resonances such as J/ψ.

3. The remaining, third muon not selected as part of the Z candidate is taken as the probe muon.

The misidentification rate of the probe muon that is determined in this way in bins of the probe muon p_T, can be seen in Figure 4.9. However, contributions from processes that indeed produce three muons in the hard-scattering must be subtracted. This consists primarily of WZ→ $3\ell\nu$ processes that artifically inflate the calculated misidentification rate at higher probe muon p_T. This contribution is subtracted using simulation. It is this corrected version of the misidentification rate that is used in the following section.

Application of muon misidentification rate

The muon misidentification rate is applied to a control region to estimate the contribution of reducible backgrounds to the previously described cH(ZZ→ 4μ) selection. It is useful to introduce some of the related terminology at this point. The four pass (4P) region henceforth refers to the inclusive signal region that is defined via the cH(ZZ→ 4μ) selection. The three-pass-one-fail (3P1F) and two-pass-two-fail (2P2F) regions respectively refer to regions in which the cH(ZZ→ 4μ) reconstruction is performed as previously described but where only three (two) of the muons satisfy the tight identification criteria and the remaining one (two) muon(s) satisfy only the loose identification criteria. The 3P1F and 2P2P are collectively referred to as the application region (AR).

The extrapolation of the AR to the 4P is performed using the previously determined misidentification rate. The prescription for this application can be obtained from the misidentification

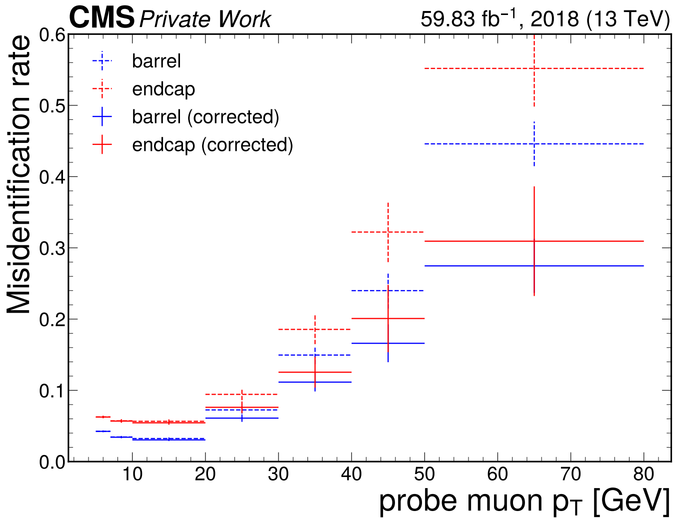


Figure 4.9: The muon misidentification rate in bins of the probe muon transverse mometum in both the barrel and endcap regions. The doted lines represent the uncorrected misidentification rate while the solid lines repsent the misidentification rate where contributions from WZ$\rightarrow 3\ell\nu$ processes are subtracted.

rate f_i which is defined as

$$f = \frac{N_{\text{tight}}}{N_{\text{loose}}}. \tag{4.3}$$

Here N_{loose} and N_{tight} are the number of probe muons in a given bin that pass the loose and tight identification criteria respectively. From this, the relation

$$N_{\text{tight}} = N_{\text{loose}} f \tag{4.4}$$

follows. Since one is interested in the contributions of muons which pass the loose but not the tight identification requirement in the AR, Eq. 4.4 can be reinterpereted as

$$N_{\text{loose}-\text{not}-\text{tight}} = N_{\text{loose}}(1-f). \tag{4.5}$$

By substituting this back into Eq. 4.4, the desired prescription is found:

$$N_{\text{tight}} = N_{\text{loose}-\text{not}-\text{tight}} \frac{f}{(1-f)}. \tag{4.6}$$

Thus, for each muon that fails the tight identification requirements but passes the loose ones in the 3P1F and 2P2F regions, the weight $f/(1-f)$ is applied, where f is the p_T and η dependant muon misidentification rate. This leads to the following expressions for the individual contributions of the AR to the 4P region:

1. **2P2F**: Since this region contains two muons that pass the loose identification criteria but not the tight, the weight $f/(1-f)$ must be applied twice. The total contribution of this region in the 4P region can thus be written as

$$N_{\mathrm{4P}}^{(\mathrm{2P2F})} = \sum_{k}^{N_{\mathrm{2P2F}}} \frac{f_k^{(3)}}{(1-f_k^{(3)})} \frac{f_k^{(4)}}{(1-f_k^{(4)})}, \tag{4.7}$$

where the $f_k^{(3/4)}$ is the misidentification rate associated with each of the non-passing muon for the k-th event. Major contributors to the 2P2F region the Drell-Yan and $t\bar{t}$ processes, which produce only two prompt muons.

2. **3P1F**: Since this region contains only one muon that passes the loose identification criteria but not the tight, the weight $f/(1-f)$ is only applied once. The total contribution of this region in the 4P region can thus be written as

$$N_{\mathrm{4P}}^{(\mathrm{3P1F})} = \sum_{k}^{N_{\mathrm{3P1F}}} \frac{f_k^{(4)}}{(1-f_k^{(4)})}, \tag{4.8}$$

where the $f_k^{(4)}$ is the misidentification rate associated with the non-passing muon for the k-th event. The major contributor to the 3P1F region consists of WZ→ $3\ell\nu$ due to the presence of three prompt leptons.

To obtain the total contribution of the AR to the 4P region, potential contaminations of the 2P2F and 3P1F regions must be accounted for. The first source of contamination is the potential overlap of the 3P1F region with contributions from the 2P2F region, where an additional muon has been misidentified in the former. This may lead to an overestimation of the 3P1F region. Such contributions can be estimated via the term

$$N_{\mathrm{exp.}}^{(\mathrm{3P1F})} = \sum_{k}^{N_{\mathrm{2P2F}}} \left(\frac{f_k^{(3)}}{(1-f_k^{(3)})} + \frac{f_k^{(4)}}{(1-f_k^{(4)})}\right). \tag{4.9}$$

Effectively, contributions from the 2P2F region are weighed with the misidentification rate for either muons that may fail the tight identification criteria. To then extrapolate this contribution to the 4P region, the fake rate must once again be applied, this time to the respective complementary muon. This produces the final term

$$N_{\mathrm{4Pexp.}}^{(\mathrm{3P1F})} = \sum_{k}^{N_{\mathrm{2P2F}}} \left(\frac{f_k^{(3)}}{(1-f_k^{(3)})} \frac{f_k^{(4)}}{(1-f_k^{(4)})} + \frac{f_k^{(4)}}{(1-f_k^{(4)})} \frac{f_k^{(3)}}{(1-f_k^{(3)})}\right) \tag{4.10}$$

$$= 2 \sum_{k}^{N_{\mathrm{2P2F}}} \left(\frac{f_k^{(3)}}{(1-f_k^{(3)})} \frac{f_k^{(4)}}{(1-f_k^{(4)})}\right) \tag{4.11}$$

An additional source of potential contamination is the contribution of four muon processes in which muons either fail the tight identification criteria or fall outside the detector acceptance. This is primarily relevant in the 3P1F region with contributions from qq→ ZZ/Zγ^* →4ℓ processes, which again lead to an overestimation of the 3P1F region. Contributions to the 4P region from this, denoted with $N_{\mathrm{4P}}^{(\mathrm{ZZ,3P1F})}$ are estimated via simulation with the same prescription as data events in the 3P1F region:

$$N_{4\mathrm{P}}^{(\mathrm{ZZ,3P1F})} = \sum_{k}^{N_{3\mathrm{P1F}}^{ZZ}} \frac{f_k^{(4)}}{(1-f_k^{(4)})}. \tag{4.12}$$

The total contribution to the 4P region $N_{4\mathrm{P}}$ may finally be estimated by taking the sum of the 2P2F and 3P1F contributions and subtracting the discussed contamination terms. This leads to

$$\begin{aligned} N_{4\mathrm{P}} =& N_{4\mathrm{P}}^{(\mathrm{2P2F})} + N_{4\mathrm{P}}^{(\mathrm{3P1F})} - N_{4\mathrm{Pexp.}}^{(\mathrm{3P1F})} - N_{4\mathrm{P}}^{(\mathrm{ZZ,3P1F})} & (4.13) \\ =& \sum_{k}^{N_{3\mathrm{P1F}}} \frac{f_k^{(4)}}{(1-f_k^{(4)})} - \sum_{k}^{N_{3\mathrm{P1F}}^{ZZ}} \frac{f_k^{(4)}}{(1-f_k^{(4)})} - \sum_{k}^{N_{2\mathrm{P2F}}} (\frac{f_k^{(3)}}{(1-f_k^{(3)})} \frac{f_k^{(4)}}{(1-f_k^{(4)})}) & (4.14) \end{aligned}$$

By applying this strategy, an inclusive yield of 6.59 events for the reducible background in the 4P region is observed. In the signal region around the Higgs mass peak of 120 GeV $< m(H) <$ 130 GeV, this reduces to a yield of 0.42 events. However, the derived observable distributions are significiantly discontinuous. This arises from the strong influence of statistical fluctuations on the small yields of the individual control regions. The effects of these fluctuations are compounded by the subtractions performed in Eq. 4.13, which introduce a significant number of bins with negative, unphysical yields. Given the small contribution of this background, a simplifcation to the estimation is made to resolve this problem. Specifically, the shapes of the distributions for observables in the 2P2F region, which is the most populated control region in the reducible background estimation, are taken and scaled to the yield of the full estimation. This approach is motivated by a comparison of the 2P2F and 3P1F flavour tagging disciminator shapes, which show reasonable similarity as seen in Appendix. Section C. The contribution of the reducible background estimation to distributions of observables that is derived in this way can be seen in Section 4.3.

A number of systematic uncertainty sources can influence the reducible background estimation. This includes the effects of statistical uncertainties on each bin of the muon misidentification rate, effects related to differences in the background composition between different control regions as well as the discussed simplication introduced in the estimation strategy. However, due to this background's small contribution in the phase space sensitive to the signal and the technical limitations resulting from the limited event yield, it is difficult to estimate the effect of these uncertainty sources. Instead a conservative rate uncertainty of 100% is introduced, allowing the yield of this background to effectively be determined by a best fit value in the statistical evaluation described in Section 4.4.

4.3 Validation of cH(ZZ→ 4μ) event selection in sideband regions

Here, a validation of the described event selection is presented. Specifically, the 2018 dataset of the CMS detector is compared to simulation in sideband regions of the cH(ZZ→ 4μ) signal for a variety of observables. This sideband region isdefined as all selected events outside of the 100 GeV $<m$(H) $<$130 GeV region, which is where the cH(ZZ→ 4μ) contribution is expected to be found. The only exception to this is the distribution of the mass of the Higgs boson candidate,

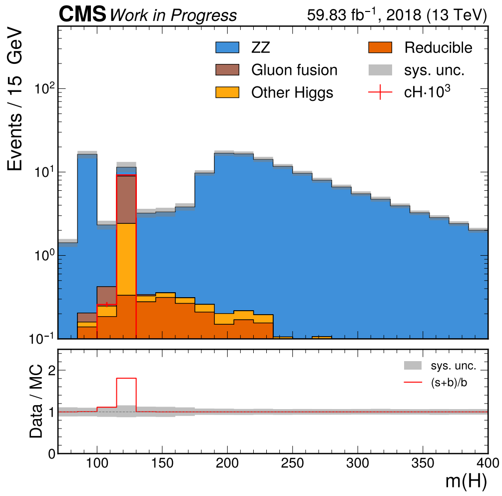


Figure 4.10: The mass spectrum of the Higgs boson candidate, showing the agreement between the 2018 dataset and simulation in the side band regions of the spectrum. The contribution from a cH(ZZ→ 4μ) signal, scaled by a factor 10^3 for better visibility, is also shown. The signal contribution is clearly concentrated around the Higgs boson mass of $m(H) = 125$ GeV, as expected.

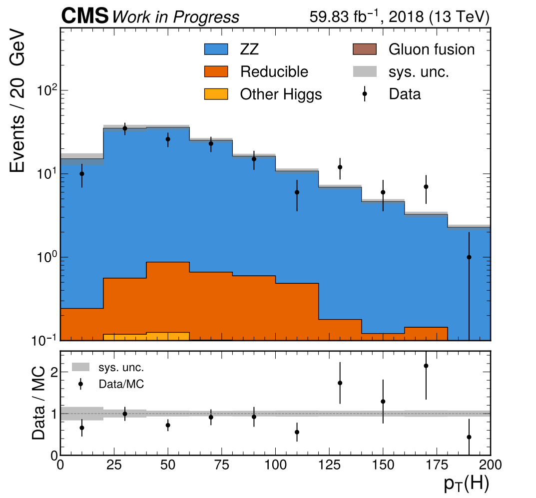
shown in Figure 4.10, for which the entire mass spectrum is shown for simulation to illustrate the presence of the cH(ZZ→ 4μ) signal around the Higgs boson candidate mass peak at $m(H) = 125$ GeV. The spectra in the sideband region of the Higgs boson candidate transverse momentum, the mass of the Z_1 and Z_2 candidate, as well as the transverse momentum, and the CvsL and CvsB discriminators of the selected jet are shown in Figure 4.11a – Figure 4.11f, respectively. As seen in these distributions, a reasonable agreement between data and simulation is observed when accounting for the small event yields.

4.4 Statistical inference

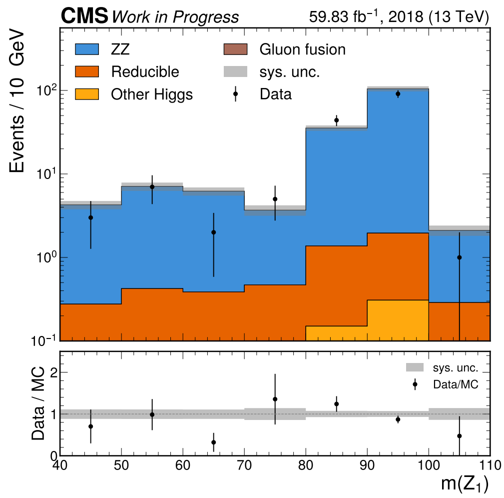
For the statistical inference process, the Combine tool is used. Specficially, a fit is performed to derive expected 95% CL CL$_s$ upper limits on σBR(cH(ZZ→ 4μ)) using distributions obtained from the selected, simulated cH(ZZ→ 4μ) candidates for some chosen discriminating variable. To perform such a fit, a statistical model describing the discriminator distributions of the signal and background processes must be constructed. This is described in Subsection 4.4.1. Additionally, an uncertainty model describing the various systematic uncertainty sources that affect the analysis is needed. This is described in Subsection 4.4.2

4.4.1 Statistical model

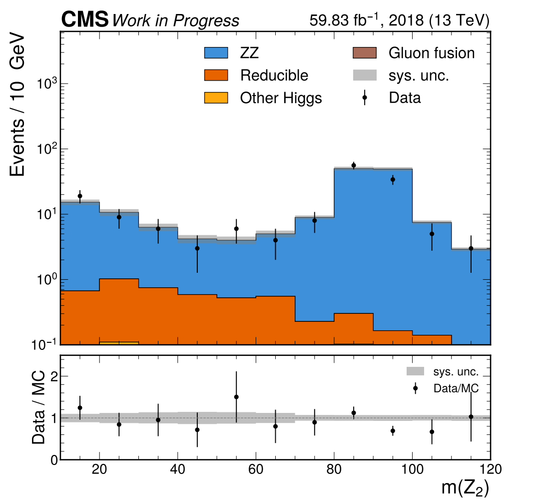
Assuming the absence of a measurable signal, given the very small signal cross section, an expected 95% CL CL$_s$ upper limit on σBR(cH(ZZ→ 4μ)) may be set. This in turn may be



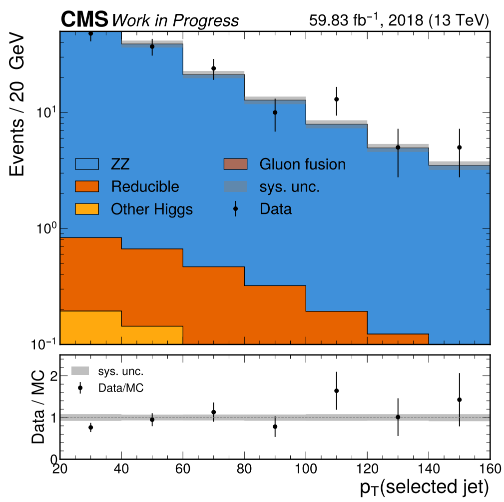
(a) Higgs boson candidate transverse momentum.



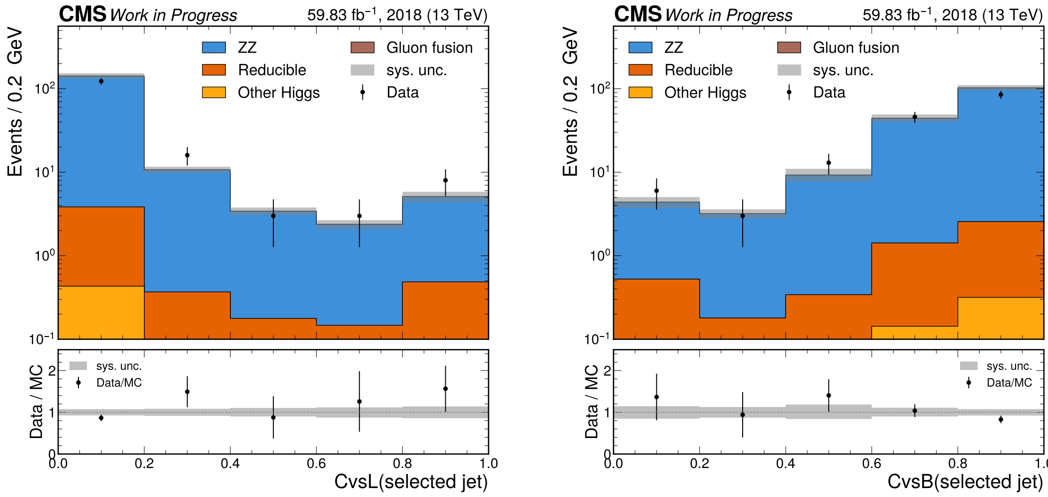
(b) Mass of the Z_1 candidate.



(c) Mass of the Z_2 candidate.



(d) Transverse momentum of the selected jet.



(e) CvsL discriminator of the selected jet.

(f) CvsB discriminator of the selected jet.

Figure 4.11: Spectra of cH(ZZ→ 4μ) candidate observables, showing the agreement between the 2018 dataset and simulation. Variables related to the Higgs boson reconstruction and the selected jet in sideband regions of the Higgs boson candidate mass are shown.

interpreted as an expected limit on κ_c using the prescription derived in Subsection 2.3.2. To do this, a statistical model is required to predict the distribution of the number entries in each bin of the discriminator histogram according to a given hypothesis. Since each of the bins are filled in a discrete couting process, a Poissonian Ansatz with

$$\mathcal{P}(k;\lambda) = \frac{\lambda^k e^{-\lambda}}{k!} \tag{4.15}$$

is a fitting choice. Here, λ denotes the expectation value with k observed events. In the i-th bin of the discriminator distribution, the expectation value λ_i is calculated using the models of the signal and background processes discussed in Section 4.2. This amounts to $\lambda_i = \mu s_i + b_i$ where s_i and b_i are the yields of the signal and background estimations respectively. The signal strength modifier μ allows for an arbitrary scaling of the signal contribution and is used as a floating parameter in the inference process. From this, a combined binned likelihood function

$$\mathcal{L}(d\,|\,\mu \cdot s(\boldsymbol{\theta}) + b(\boldsymbol{\theta})) = \prod_{i \in \text{bins}} \mathcal{P}(d_i\,|\,\mu \cdot s(\boldsymbol{\theta}) + b(\boldsymbol{\theta})) \text{ x } \prod_{j \in \text{nuis.}} \mathcal{C}(\hat{\theta}_j\,|\,\theta_j). \tag{4.16}$$

may be derived for some set of measured data d_i. Here, an uncertainty model is introduced via the nuisance parameters $\boldsymbol{\theta}$ that account for uncertainties related to the signal and background estimation. These follow a distribution $\mathcal{C}$ and may alter the scaling of the signal and background contributions as well as their histogram shape with respect to the discriminator that is used. The estimate of $\boldsymbol{\theta}$ that is used to obtain estimations of s_i and b_i in the inference process is denoted by $\hat{\boldsymbol{\theta}}$ and is found by finding the global maximum of the likelihood. The specifics of the uncertainty model employed in this analysis, which is parametrised by the nuisance parameters, are discussed in Subsection 4.4.2. To characterise hypotheses, the test statistic that is used is the *profile likelihood ratio*

$$q_\mu = -2\ln\left(\frac{\mathcal{L}(d\,|\,\mu\cdot s(\hat{\boldsymbol{\theta}}_{\boldsymbol{\mu}})+b(\hat{\boldsymbol{\theta}}_{\boldsymbol{\mu}}))}{\mathcal{L}(d\,|\,\hat{\mu}\cdot s(\hat{\boldsymbol{\theta}})+b(\hat{\boldsymbol{\theta}}))}\right), \tag{4.17}$$

where $(\hat{\mu}, \hat{\boldsymbol{\theta}})$ are the parameter values that globally maximise the likelihood while $\hat{\boldsymbol{\theta}}_{\boldsymbol{\mu}}$ maximises the likelihood for a given μ. A significant advantage of this test statistic is that in the large sample limit, the distribution $f(q\,|\mu)$ approaches the χ^2_k distribution with $k = 1$ degrees of freedom, according to Wilk's theorem [71]. This is very useful as it allows for a simple calculcation of a p-value

$$p_\mu = \int_{q_{\mathrm{obs.}}}^{\infty} f(q\,|\,\mu)\mathrm{d}q. \tag{4.18}$$

With this p-value, the $\mathrm{CL_s}$ method may be applied [72]. This involves calculating the $\mathrm{CL_s}$ value

$$\mathrm{CL_s} = \frac{p_\mu}{1-p_0} = \frac{\int_{q_{\mathrm{obs.}}}^{\infty} f(q\,|\,\mu)\,\mathrm{d}q}{\int_{q_{\mathrm{obs.}}}^{\infty} f(q\,|\,0)\,\mathrm{d}q}. \tag{4.19}$$

A feature of this method is that large overlaps of test statistic distributions between the H_μ and H_0 hypotheses, due to the very small signal cross section associated with H_μ, can be accounted for. This is achieved by weighting p_μ with $(1-p_0)^{-1}$, thus decreasing the $\mathrm{CL_s}$ value for larger overlaps.

4.4.2 Uncertainty model

The sources of uncertainty in the model of the signal and background processes are described in this section. These consist of two types. The first are referred to as shape uncertainties and may change the normalisation as well as the shape of the distributions in question. The effect of these uncertainties are captured by creating variations of these distributions, which are interpreted as $\pm1\sigma$ variations of the underlying parameter describing the uncertainty source. During the statistical evaluation process, an interpolation is performed between the nominal distributions as well the respective variations to characterise their effect. The second uncertainty type are referred to as normalisation uncertainties. These only affect the normalisation of distributions without affecting the shape and can thus be described with a single, real paarameter. Both types of uncertainties are associated with a respective nuisance parameter in the statistical model introduced in Subsection 4.4.1.

Theoretical uncertainties common to simulation

A number of theoretical uncertainties that are common to all simulated sample are included in the uncertainty model. This includes:

- Shape uncertainties related to the choice of the normalisation and factorisation scales μ_R and μ_F in the simulated Monte Carlo samples. These are varied independently by factors of 2 and 0.5 using a reweighting technique that is applied to the simulation, thus introducing two shape-changing nuisance parameters per process. Yield changes associated with the uncertainty on μ_R range from approximately 1% – 40%, with the largest change being observed for the ggH process. The yield change of the cH(ZZ→ 4μ) process associated with this uncertainty source is approximately 23%. Yield changes associated with the uncertainty on μ_F are comparatively smaller, ranging from 1% – 15%. The the bH process accounts for the largest change while the cH(ZZ→ 4μ) process sees a yield change of approximately 9%.

- A shape uncertainty related to the modelling of the Parton Distribution Function (PDF). This uncertainty is obtained from the NNPDF3.1 PDF set that is used in simulation [73] and includes 100 PDF variations associated with variations of individual parameters. These are combined into a single nuisance parameter with the prescription

$$\sigma^+ = \sigma^- = \sqrt{\sum_{i=1}^{N_{\mathrm{par}}} (F_i - F_0)^2}\,, \tag{4.20}$$

  where F_0 is the nominal value of the observable in question and F_i is the varied value associated with one of the N_{par} parameter variations [74]. A yield uncertainty of approximately 6.5% on the total yield of the background processes is associated with this uncertainty. The yield uncertainty on the cH(ZZ→ 4μ) process is somewhat larger, at approximately 12%.

- An uncertainty related to the choice of the strong coupling constant α_s in the Monte Carlo simulation. This uncertainty is associated with a total yield change of approximately 2%.

- Shape uncertainties related to the tuning of the parton showering used in the Monte Carlo generator. These are implemented by varying parameters related to initial-state radiation and final-state radiation independently, introducing two nuisance parameters. Both uncertainties are associated with yield changes of less than 1% for the background processes. These uncertainties are associated with slightly larger yield changes for the signal process at 2–3%.

- A shape uncertainty related to the application of a pileup reweighting procedure, with which the pileup profile of simulation is matched to that of data. This uncertainty is described with a single nuisance parameter. The uncertainty on process yields that is introduced by this uncertainty is almost negligible for both background and signal processes.

Experimental uncertainties common to simulation

The following experimental uncertainties are common to all simulated sample:

- Shape uncertainties related to the jet energy scale. For this, a simplified schema is used in which closely correlated uncertainty sources are grouped and described through 11 individual nuisance parameters. Sources of uncertainty include for example the limited set of data available for the calibration procedure or differences in calibration response to different jet flavours. The yield uncertainty introduced due to these uncertainties ranges between <1% - 7% for background and signal processes.

- A shape uncertainty related to the smearing method used to correct the jet energy resolution. This is described with a single nuisance parameter and is associated with small yield changes of approximately 1.5%.

- Shape uncertainties related to the calibration of the jet flavour-tagging algorithm that is used in the jet-driven approach described in Section 4.5. The effect of these are captured via 13 nuisance parameters and include, for example, uncertainty sources related to the individual phase spaces targeted in the calibration or the limited set of available data. No yield uncertainties are associated with these nuisance parameters. This is by design, as no flavour tagging discriminants are used for the event selection criteria. A number of jet flavour-tagging uncertainty sources such as the jet energy resolution, parton shower tuning and pile-up reweighting uncertainties also affect the calibration of the flavour-tagging algorithm. These are fully correlated to the corresponding, nominal uncertainties considered in the analysis.

- A shape uncertainty related to the pileup identification method that is used. This uncertainty is described with a single nuisance parameter and is associated with yield changes of less than 1%.
- Shape uncertainties related to the muon calibration used in the analysis. The response of the muon identification criteria, the isolation as well as the muon trigger path in simulation is calibrated to match data. Uncertainties associated with this calibration are combined into a single nuisance parameter. The yield uncertainty associated with this uncertainty source is approximately 4.5%. This is increased for the cH(ZZ$\rightarrow 4\mu$) process at approximately 9%.
- A normalisation uncertainty related to the luminosity, which is know with a limited precision, and with which the simulation is scaled. This amounts to an uncertainty of 2.5% on the yields for the 2018 dataset [75] and is described with a single nuisance parameter.

Uncertainties related to cH(ZZ$\rightarrow 4\mu$) and bH(ZZ$\rightarrow 4\mu$) modelling

There is, alongside the previously described uncertainties, a systematic uncertainty that is unique to cH(ZZ$\rightarrow 4\mu$) and bH(ZZ$\rightarrow 4\mu$) simulation and is associated with the choice of a specific flavour scheme. As discussed in Subsection 4.2.1, an envelope representing a $\pm 1\sigma$ variation is extracted from the respective 3FS (4FS) and 4FS (5FS) samples and applied to the nominal cH(ZZ$\rightarrow 4\mu$) 4FS FxFx (bH(ZZ$\rightarrow 4\mu$) 5FS FxFx) sample. A shape changing nuisance parameter reflecting the flavour scheme uncertainty is thus introduced for both the cH(ZZ$\rightarrow 4\mu$) and bH(ZZ$\rightarrow 4\mu$) processes, respectively. For the former, this corresponds to an uncertainty on the yield of approximately 59% and 42% for the latter.

Uncertainty related to reducible background modelling

Due to the minor nature of the reducible background contribution, a simple normalisation uncertainty is introduced to account for uncertainties associated with the reducible background estimation procedure. The uncertainty is set as 100% of the yield, giving significant freedom in the fitting procedure to find a best-fit yield.

Uncertainties related to statistical precision of background estimation

The employed simulation estimation methods themselves include a statistical uncertainty due to the limited number of events that are generated to estimate each process. This results in the introduction of nuisance parameter per process per bin of the final discriminator. This can be simplified to introducing only a single, per bin nuisance parameter by using the Barlow-Beeston approach [76][77].

Additional uncertainties

The following additional unceratinties are also included:

- Two normalisation uncertainties are introduced for the theoretical uncertainty associated with the application of the k-factors for the qq→ZZ and gg→ZZ processes. These amount to an uncertainty of 4% and 1.5% on the yields, respectively.

- A normalisation uncertainty originating from the theoretical uncertainty on the branching ratio $\mathcal{B}(H \to ZZ \to 4\ell)$ [20]. This is applied to simulated samples in which a Higgs boson is produced and corresponds to an uncertainty of 2% on the yields.

4.5 Results

In this section, the results obtained by performing the statistical evaluation described in Section 4.4 on a simulated Asimov dataset [71] are discussed. A number of different scenarios are considered and expected 95% CL $\mathrm{CL_s}$ upper limit (henceforth simply referred to as expected upper limits) on $\sigma\mathcal{B}$(cH(ZZ→ 4μ)) and κ_c are presented for each scenario. Additionally, the impacts of the individual uncertaintie sources on the fitting procedure performed during the statistical evaluation are estimated for the nominal result. Additionally, extrapolations of the results to full Run-2 and High Luminosity LHC scenarios are presented. Finally, a number of potential improvements to the analysis are discussed.

4.5.1 Expected upper limits on $\sigma\mathcal{B}$(cH(ZZ→ 4μ))

As a baseline scenario, the mass of the Higgs boson candidate is used as a discriminator for the statistical evaluation, which is applied to the simulated 2018 dataset of the CMS detector. The distribution used for this corresponds to the one previously shown in Figure 4.10, in **??**. For this scenario, an expected upper limit of $\sigma\mathcal{B}$(cH(ZZ→ 4μ)) = 1088 is set. This corresponds to an expected upper limit of $\kappa_c = 191$. Given the resonant nature of the peak around the Higgs boson candidate mass, which is shared by all Higgs production processes, this approach to setting upper limits on κ_c can be interpreted as a measurement of an excess in jet-associated Higgs boson production.

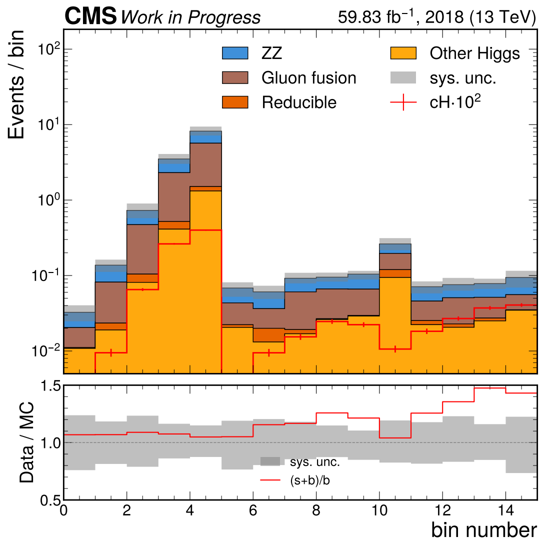
To improve upon this baseline scenario, it is investigated whether using the information of the selected jet can improve upon this result. This henceforth referred to as the *jet-driven approach* To this end, the DeepJet CvsB and CvsL discriminators are used in a pseudo two-dimensional fit with the simulated 2018 dataset of the CMS detector. Specifically, an *unrolling* process is applied to project the two-dimensional information provided by the CvsB and CvsL discriminators into a single, final discriminator. This involves constructing a histogram of the CvsB discriminator in bins of the corresponding CvsL value. The chosen bins for the CvsL discriminator are [0, 0.4, 0.7, 1.0]. The CvsB dicriminator is split into five equidistant bins. This choice is made to find a good compromise between the separation of signal enhanced and background dominanted regions, and the smoothness of the distribution. Additionally, it is required that the Higgs boson mass be in the [100, 130] GeV window to reduce the presence of non-Higgs boson backgrounds. The resulting histogram can be seen in Figure 4.12a. In this scenario, an expected upper limit of $\sigma\mathcal{B}$(cH(ZZ→ 4μ)) = 999 times the SM expectation is set. This corresponds to an expected upper limit of $\kappa_c = 176$. While this is an improvement with respect to the baseline scenario, the gain in analysis sensitivity that is achieved is comparatively small. This behaviour can be understood by the fact that a large portion of the jets of the cH(ZZ→ 4μ) signal are associated with a light jet-like discriminator (bins 0 − 4). For a large portion of the signal, this results in a significant similarity with the discriminator distributions associated with background processes such as the ggH process, which are also typically associated with a light jet-like discriminator. As a result, though there is a signal-like region enriched with the cH(ZZ→ 4μ) process (for example bins 11 – 14), the impact of this appears to be subdominant given its much smaller overall proportion. Motivated by this, it is investigated whether an additional cut on the CvL discriminator yields an improvement in the expected upper limit that is obtained. However, it is found that this is not the case, likely due to the fact that a significant portion of the signal is also removed with this cut.

Overall, a significant drawback to the described jet-driven approach is that the predicted number of events in a majority of bins of the discriminator is around one tenth of an event for the 2018 dataset, with only two bins predicting a yield greater than one. This impairs the interpretability of this approach when applying it to 2018 data, as the shape of the discriminator distributions predicted by simulation cannot be reflected in the data, due to the fractional predicted yields. This problem is expected to remain for this approach if it were expanded to the full Run-2 dataset of the CMS detector, corresponding to 138 fb^{-1} of data. Only with significantly larger datasets such as at the HL-LHC, assuming an expected total of 3000 fb^{-1} [11], would this issue be resolved. A possible solution to reduce the impact of this statistical limitation would be to reduce the number of bins used in constructing the discriminator. However, given the significant reduction in bin numbers that would be required to achieve this, the information gained from the shape of the discriminator distributions is effectively lost. As such, such a version of the measurement would have significant similarity to interpretation of the baseline scenario, albeit with reduced control over the non-Higgs backgrounds that dominante in the sideband regions of the Higgs boson candidate mass distribution. Since the described statistical limitation originates from the criteria imposed on the Higgs boson candidate mass, an additional variation on the jet-based approach is trialed, in which the criteria are not applied. The resulting discriminator distribution can be seen in Figure 4.12b. However, the expected upper limit of $\sigma\mathcal{B}$(cH(ZZ→ 4μ)) = 3244 that is derived is significantly worse than in the baseline scenario. This can be again explained by the large overlap in the distributions of the signal and background processes in the light jet-like region of the discriminator, the impact of which is more significant than previously, due to the large contributions of ZZ production processes in this region.

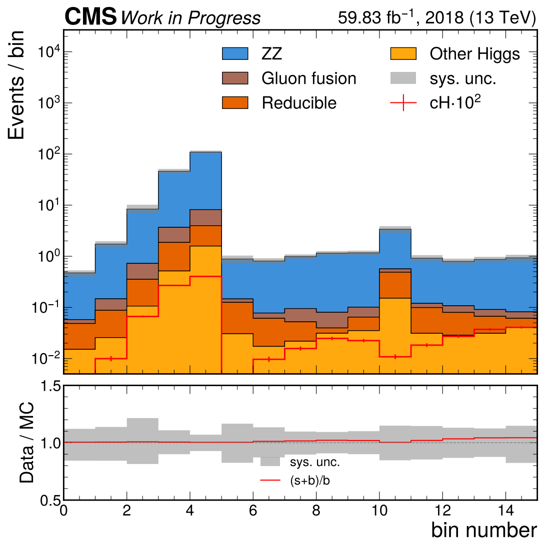
Given the described challenges associated with the jet-driven approach, the result of the baseline approach is taken as the nominal result.

4.5.2 Impact of nuisance parameters

For the nominal result, the impact of the nuisane parameters on the derived expected upper limit on σBR(cH(ZZ→ 4μ)) are investigated. This can be achieved by performing a series of fits to the Asimov dataset used to derive the expected upper limits, for which the Asimov dataset is injected with a cH(ZZ→ 4μ) signal at a signal strength r corresponding to the nominal result. In these fits, the profile likelihood ratio is maximised to find a best fit value of the cH(ZZ→ 4μ) signal strength $\hat{r}$. Naturally, in a nominal fit on the Asimov dataset, $\hat{r}$ corresponds to the signal-strength with which the cH(ZZ→ 4μ) signal is injected into the dataset. To now gauge the impact of the nuisance parameters on the fit, a series of fits where the individual nuisance parameters θ_i are varied by $\pm\sigma$ to find the best fit values $\hat{r}_{\pm\theta_i}$ are performed. The size of the differences $\Delta\hat{r}_i = \hat{r} - \hat{r}_{\pm\theta_i}$ thus gives an indication of the uncertainty on $\hat{r}$ that each nuisance parameter introduces. With this technique, the leading uncertaintiy sources that limit the sensitivity the of the nominal result can thus be identified. These can be seen in Figure 5.3, which shows the $\Delta\hat{r}_i$ values of the leading 30 (of a total of 70) uncertainty sources. As can be



(a) Higgs boson candidate mass limited to [100,130] GeV.



(b) Inclusive Higgs boson candidate selection.

Figure 4.12: The unrolled discriminator distributions for the jet-driven approach, representing the CvsB discriminator in three bins of the CvsL discriminator for the selected jet of the reconstructed cH(ZZ→ 4μ) candidates. It is shown once inclusively in the Higgs boson candidate mass spectrum and once with Higgs boson candidate mass criteria applied. As can be seen in the lower panel of both figures, the relative contribution of a cH(ZZ→ 4μ) signal with respect to the background generally increases with increasing bin number for both distributions. The region most enriched with signal are the last bins on the right side of both distributions, which as expected, corresponds to selected jets with both a high CvsL and CvsB value. The cH(ZZ→ 4μ) signal contribution is scaled by a factor of 10^2 to improve its visibility.

seen, the flavour scheme uncertainty from the cH(ZZ→ 4μ) simulation is clearly the leading uncertainty source. Given the relatively large yield changing effect associated with this uncertainty and the very small expected signal strength of the cH(ZZ→ 4μ) process, this is expected. Other leading uncertainty sources are also of theoretical origin, introducing significant yield changes to the primary background processes. These include a yield effect of the renormalisation scale both on the ggH process as well as the signal process. Also playing a role are PDF uncertainties and the uncertainty associated with the factorisation scale in the signal modelling. The leading experimental uncertainty is the uncertainty associated with the muon calibration. Uncertainties associated with the jet energy scale and jet energy resolution are associated with comparitively sub-dominant effects, which may be expected given that the used discriminator is not directly related to the jet. The distributions associated with the 30 leading uncertainty sources can be seen in Appendix Section D. Only the shape changing uncertainties are shown since normalisation uncertainties only modifies the scaling of distributions. To ensure that the statistical model does not behave significantly differently during the fitting procedure in the absance of signal, the impacts of the nuisance parameters in a fit scenario where no signal is injected into the Asimov dataset is also investigated. The leading 30 uncertainty sources in this scenario can be seen in Appendix Subsection D.1. No unexpected changes are observed, demonstrating the robustness of the model. The uncertainties associated with the signal model no longer play a noticeable role, as expected in the absence of a signal in the Asimov dataset.

4.5.3 Fitting nuisance parameters in sideband regions

By performing a fit of the estimated background contributions to data in sideband regions, which are defined as the region outside of [100,130] GeV window in the Higgs boson candidate mass spectrum, it can be demonstrated that a number of nuisance parameters can already be fixed to best-fit values in this way. This produces some improvement in agreement between the background estimation and data in these sideband regions, as can be seen in Figure 4.13. Here, *pre-fit* refers to the unmodified, nominal background distributions, while *post-fit* refers to the background distributions that have been modified by fixing the associated nuisance parameters to their best-fit values. The observed difference between pre-fit and post-fit distributions is small. This shows that already the pre-fit background estimation model describes the data reasonably well, and indicates that the effect of the considered systematic uncertainty sources is subdominant to the significant statistical fluctuations observed in data. It should be noted that, looking at the lower panel of Figure 4.13, a slight trend may be interpreted in the ratio of the post-fit distribution to the data. However, it is unclear whether this is an effect of the background process estimation in the sideband region or also a result of statistical fluctuations. This behaviour could be reevaluated in future analyses of this phase space with higher yields.

The best-fit values of the individual nuisance parameters can be seen in Appendix Subsection D.2. These fall into three categories. The first contains nuisance parameters associated with processes where a Higgs boson is produced. Since contributions from these are primarily in the [100,130] GeV Higgs boson candidate mass region, no information can be gained about the respective nuisance parameters from a fit in the sideband region. The second category contains nuisance parameters associated with non-Higgs processes, such as ZZ production. These are indeed fixed to best-fit values that best reflect the data and are distributed within the $\pm1\sigma$ band around the respective nominal values. This reflectss the indication that no unexpectedly large deviations between the nominal background process estimation and data are observed. Additionally, for some nuisance parameters, the relative uncertainty on the nuisance parameters value can be con-

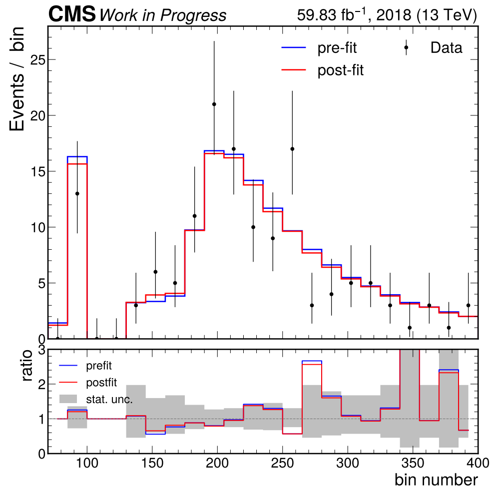


Figure 4.13: Figure showing the agreement between the pre- and post-fit distributions of the background estimation model and data, in the sideband region of the Higgs boson candidate mass spectrum. A reasonable agreement is observed already in the pre-fit distribution, supporting the indication that the model is able to describe the data well within the statistical fluctuations observed in data.

strained. This is for, for example, the case for the nuisance parameter describing renormalisation scale of the qq→ZZ process. However, the uncertainties for the majority of nuisance parameters remain unconstrained. The last category contains the nuisance parameters assigned to the per-bin, statistical uncertainty associated with the estimation of background processes. These best-fit values of these are again reasonably distributed around their respective nominal values.

4.5.4 Extrapolation to larger datasets

To make a comparison to published cH(WW) and cH($\gamma\gamma$) results of the CMS collaboration, a statistical extrapolation of the baseline scenario to the full Run-2 dataset of the CMS detector is performed. This consists of scaling the prediction of the 2018 dataset to the full Run-2 dataset. However, it should be noted that this extrapolation neglects any changes in detector operating conditions and assumes no changes in any of the uncertainty sources, meaning this result should only be taken as an indication of the analysis sensitivity that my be expected. In this scenario, an expected upper limit of $\sigma\mathcal{B}(\mathrm{cH(ZZ\rightarrow 4\mu)}) = 725$ ($\kappa_c = 128$) is derived. Given this, the presented cH(ZZ→ 4μ) analysis is expected to be comparatively less sensitive to κ_c, with an expected upper limit of $\kappa_c = 95$ being reported in the cH(WW) analysis and $\kappa_c = 72.5$ in the cH($\gamma\gamma$) analysis. Given that the sensitivity of the presented analysis appears to be strongly constrained by signal selection efficiency, possible explanations for this could be the larger Higgs boson decay branching ratios in the H(WW) and H($\gamma\gamma$) channels compared to H(ZZ→ 4μ). However, a direct comparison is difficult due to significantly different background process compositions in all three analyses.

Similarly, the result may be extrapolated to the expected 3000 fb−1 dataset of the High-Luminosity LHC (HL-LHC). Again, it should be noted that this neglects changes to the CMS detector and operating conditions, which are expected to differ significantly during HL-LHC operation [78]. A result of an expected upper limit of $\sigma\mathcal{B}(\mathrm{cH(ZZ}\rightarrow 4\mu)) = 318$ ($\kappa_c = 58$) is derived in this case. Additionally, an extrapolation to the expected HL-LHC for the jet-driven approach is made, since the described statistical limitations become less significant in this scenario. In this scenario, an expected upper limit of $\sigma\mathcal{B}(\mathrm{cH(ZZ}\rightarrow 4\mu)) = 154$ ($\kappa_c = 29$) is achieved. Clearly, this represents a significant improvement with respect to the baseline approach in a HL-LHC scenario and thus motivates a jet-driven approach in analysis scenarios with significantly larger datasets, where information from differential distributions can be leveraged more effectively as the limited signal efficiency becomes less influential.

4.5.5 Potential analysis improvements

A number of potential analysis improvements may be considered for future implementation, besides using a larger targeting a larger dataset. One such possibility would be to loosen the muon selection criteria that are applied. This would allow for a more reliable estimation of the reducible background component as well as increase the signal selection efficiency. Additionally, this could enhance analysis sensitivity as signal efficiency is increased and potentially allow for better constraints of nuisance parameters. The motivation to pursue this would be especially clear in the jet-driven analysis approach, as it could alleviate the discussed statistical constraints. Another method through which an increase in total yields can also be achieved, would be the inclusion of electrons in the analysis. This would introduce $H(4e)$ as well as $H(4\mu)$ final states, clearly broadening the targeted phase space and thus increasing the expected signal yield.

One factor that clearly the signal acceptance is also the minimum jet transverse momentum crierium of 25 GeV that is applied, as the majority of the cH signal is associated with jets at lower momenta. Lowering this value could significantly increase the yield, though it would also require a dedicated calibration of the jet-flavour tagging algorithm that is used. As such, the development of dedicated low transverse momentum jet flavour-tagging algorithms could be very interesting to pursue, given such an algorithm would also be naturally calibrated in the low transverse momentum region.

Lastly, since theoretical uncertainties associated with with the modelling of the background, and especially the signal process, appear to have a significant impact on the derived expected upper limits, improvements in this domain could greatly improve the sensitivity of the presented analysis, as well as other analyses of the cH process.

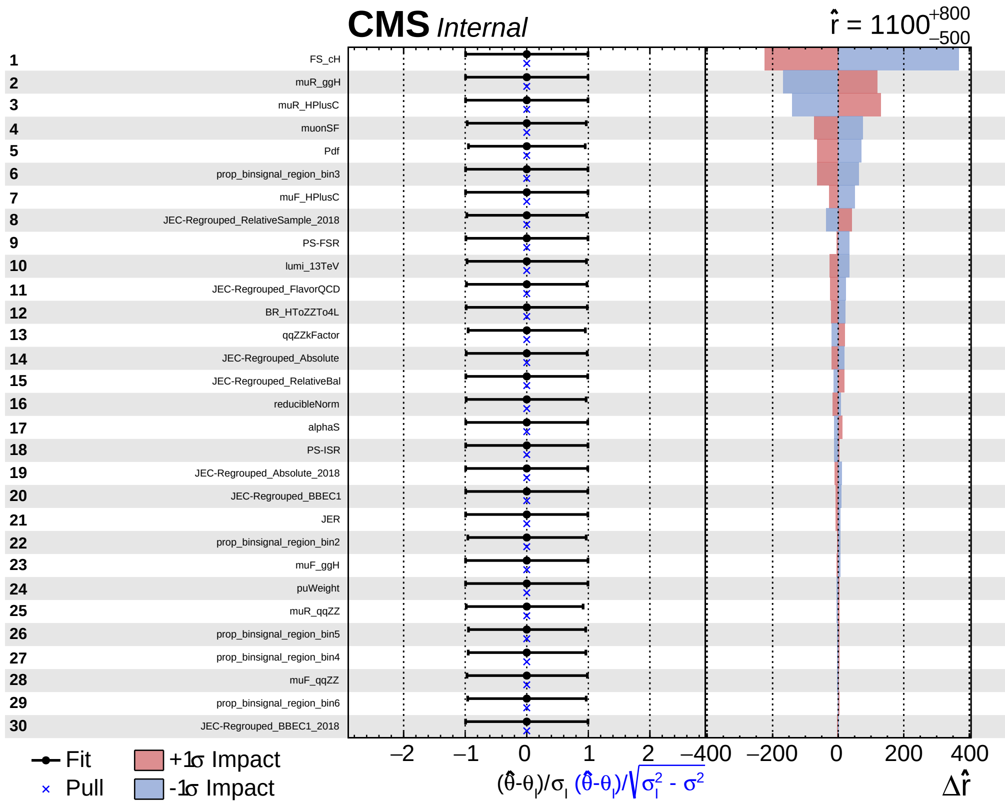


Figure 4.14: Figure showing the 30 leading uncertainty sources in a fit of $\sigma\mathcal{B}(\mathrm{cH(ZZ}\to 4\mu))$, orderd by their relative impact on the fit. In the left panel the nuisance parameter names can be seen. In the central panel, the fitted nuisance parameter values as well as their so-called pulls are shown. On the right, the $\Delta\hat{r}_i$ values can be seen for the $\pm\sigma$ variations associated with each nuisance parameter.

Chapter 5

An EFT interpretation of the cH(ZZ$\rightarrow$4μ) process

The methods developed in Chapter 4 may also be used to investigate EFT interpretations of the cH signal. In this chapter, a demonstration of this idea is presented by targeting a modified cH process in which the CMD SMEFT operator, as discussed in Subsection 2.4.1, is introduced. This follows the ideas presented in [**PhenoPaper**]. Since the vast majority of analysis components remain unchanged from the SM interpretation of the cH(ZZ$\rightarrow$ 4μ) analysis, only the relevant changes to this variant of the analysis are explicitly discussed. This includes a brief discussion in Section 5.1 on the simulation of the signal associated with the CMD operator, here referred to as cH(ZZ$\rightarrow$ 4μ)$_{cG}$, and how this motivates a change in the jet selection strategy. Subsequently, a limit is set on the signal strength of this signal in Section 5.2, which is interpreted as 95% CL$_{\mathrm{s}}$ CL upper limit on the Wilson coefficient C_{cG}.

5.1 The cH(ZZ$\rightarrow$ 4μ)$_{cG}$ process

5.1.1 Simulation of the cH(ZZ$\rightarrow$ 4μ)$_{cG}$ process

Like in the SM case, the signal associated with the $\hat{O}_{cG}$ operator is simulated using MadGraph5_aMC@NLO. To specifically simulate SMEFT, the SMEFTSim model [79][80] is used, which provides an implementation of dimension six SMEFT operators in the Warsaw basis. For this analysis, the contributions of all SMEFT operators except for the $\hat{O}_{cG}$ operator are turned off. While the MadGraph5_aMC@NLO syntax used for the event generation is very similar to that presented in Subsection 4.2.1, a key difference is that unlike in the SM, SMEFT models such as this one are curently limited to providing simulations at LO in QCD. A consequence of this is that in the simulation of the cH(ZZ$\rightarrow$ 4μ)$_{cG}$ process, the ambiguity associated with a choice of flavour scheme in the simulation falls away. This is as a 3FS version of the cH(ZZ$\rightarrow$ 4μ)$_{cG}$ process cannot be generated due to the process only being defined at NLO in QCD in the 3FS. Accordingly, the 4FS becomes the default choice.

The simulation of the cH(ZZ$\rightarrow$ 4μ)$_{cG}$ process consists of three components, the first of which is the purely SM component. This consists of the SM cH(ZZ$\rightarrow$ 4μ) process, simulated with a cross section that is fixed to the SM expectation. Second is the so-called quadratic component, which consists exlusively of SMEFT contributions introduced by the $\hat{O}_{cG}$ operator. The last component is the so-called interference term, consisting of a mix of SM and SMEFT contributions. The effect of this behaviour behaviour on the cH(ZZ$\rightarrow$ 4μ)$_{cG}$ cross section σ_{total} can be parametrised as

$$\sigma_{\text{total}} = A\sigma_{\text{SM}} + B\sigma_{\text{int.}} + C\sigma_{\text{EFT}}, \tag{5.1}$$

where A, B and C are coefficients describing the relative strength of each contribution. This expression can be normalised to the SM cH cross section and expressed in terms of C_{cG} and the new physics scale Λ as

$$\sigma_{\text{EFT}} = \sigma_{\text{SM}}(1 + \frac{C_{cG}}{\Lambda^2}\mu_{\text{int.}} + \frac{C_{cG}}{\Lambda^2}\mu_{\text{EFT}}), \tag{5.2}$$

introducing the signal strength modifiers $\mu_{\text{int.}}$ and μ_{EFT} for the interference and quadratic terms respectively. By simulating the cH(ZZ$\rightarrow$ 4μ)$_{cG}$ for a range of Wilson coefficient C_{cG} values, the scaling behaviour of Eq. 5.2 with C_{cG} can be obtained through a polynominal fit. Using the SM, LO in QCD cross section of $\sigma_{\text{SM}} = 0.058$ pb and $\Lambda = 1$ TeV, it is found that $\mu_{\text{int.}} = 2.2$ and $\mu_{\text{EFT}} = 55.8$ [**PhenoPaper**].

A technical complication in setting upper limits on C_{cG} arises from the fact that the interference and quadratic terms are, for some chosen discriminator, associated with different distribution shapes and their relationship is determined by the value of C_{cG}. This produces a non-trivial effect when scaling the signal with C_{cG}, such that the scaling produces not only cross section changes but also changes in the shape of the discriminator distribution. To address this, a simplification is made here. Given the relatively small size of the interference signal strength in comparison to the quadratic signal strength, the interference term is neglected and the total contribution sensitive to C_{cG} is approximated using just the quadratic term. This allows for a simple scaling of the cross section of the signal dsitribution during the limit setting procedure, to which the SM component now becomes a background process. It should be noted that this only holds true for $C_{cG} > 1$ and as such, this approximation is only valid when probing this part of the parameter space in the limit setting procedure.

5.1.2 Modifications to the analysis strategy

The event selection used in this analysis variant is kept almost identical to the selection presented in Section 4.1. The only change that is made is that a simplification with respect to the jet selection is made. Instead of applying the previously introduced jet selection alogrithm, the jet with the highest transverse momentum a candidate event is selected instead. This choice is motivated by the $\hat{O}_{cG}$ operator introducing a significant kinematic modification to the cH process in which the associated charm quarks can be produced at much higher transverse momenta than in the SM case. This behaviour can be seen in Figure 5.1 in a simulation of the cH(ZZ$\rightarrow$ 4μ)$_{cG}$ process. The transverse momentum of the jet that is selected in this way for cH(ZZ$\rightarrow$ 4μ)$_{cG}$ candidates can be seen in Figure 5.2, along with the full background estimation. Clearly, the cH(ZZ$\rightarrow$ 4μ)$_{cG}$ produces a strong enhancement in the high transverse momentum regime of the selected jet, as expected. This quality makes the transverse momentum of the selected jet an excellent candidate as the final discriminator in the statistical evaluation. The background processes considered here are the same as in Chapter 4, with the exception being that the SM component of the cH(ZZ$\rightarrow$ 4μ) process is now also considered a background process.

The same statistical evaluation strategy as presented in is used Chapter 4. Again, a range of systematic uncertainty sources are considered in the uncertainty model. This includes all

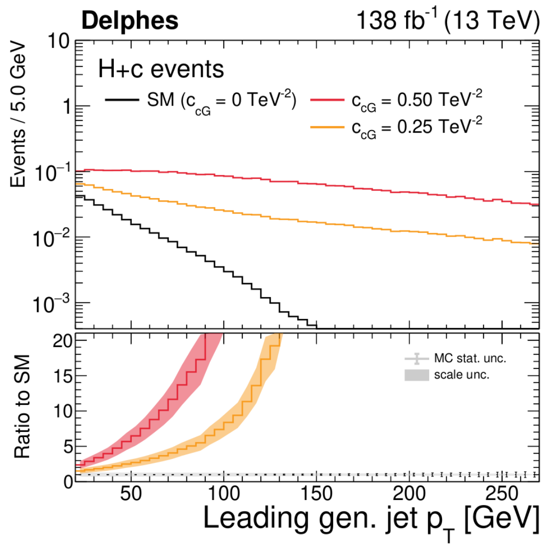


Figure 5.1: The transverse momentum of the charm quark associated with the Higgs boson for a cH process modified by the CMD operator $\hat{O}_{cG}$[**PhenoPaper**]. The enhancement in the high p_T region associated with $\hat{O}_{cG}$ can be seen for a number of different C_{cG} values.

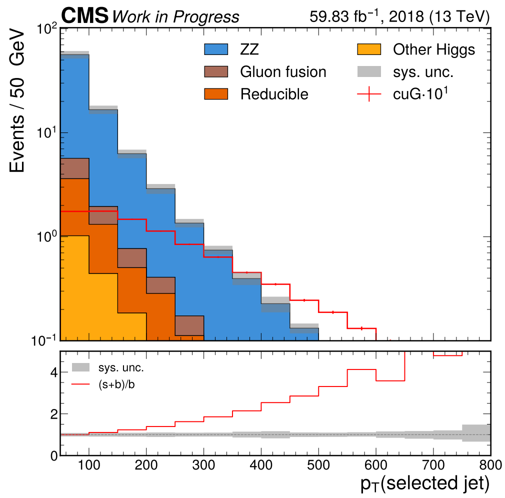


Figure 5.2: The spectrum of the transverse momentum of the jet selected in a SMEFT variant of the cH(ZZ→ 4μ) analysis. Contributions from the c_{cG} operator, simulated for a nominal value of c_{cG}=0.5, are scaled by a factor 10 to improve visibility and can be clearly see to produce jets at high transverse momenta.

the previously introduced uncertainty sources, with the exception of those associated with jet flavour tagging, since not jet flavour tagging is performed. Since only the jet selection method has changed, the relative magnitude of the uncertainties remains unchanged with respect to the numbers quoted in Chapter 4. For the cH(ZZ→ $4\mu)_{cG}$ signal simulation, the same uncertainty sources as for the cH(ZZ→ 4μ) process are considered aisde from the flavour scheme uncertainty, which is not defined for the cH(ZZ→ $4\mu)_{cG}$ signal simulation. The magnitude of the theoretical uncertainties on the cH(ZZ→ $4\mu)_{cG}$ signal are, like for the cH(ZZ→ 4μ) process, significant. This encompasses especially the uncertainty associated with the PDF set, which produces a yield change of approximately 20%, as well as the renormalisation and factorisation scales that are associated with yield changes of approximately 18% and 10%, respectively.

5.2 Results

The tranverse momentum of the selected jet is used as a final discriminator, which corresponds to the distribution shown in Figure 5.2, to set expected upper limits on C_{cG}. An additional cut of p_T(jet)> 50 GeV is implemented, since this cuts away a significant portion of background process contributions. To ensure compatibility with the perturbativity requirements discussed in Subsection 2.4.2, a $M_{cut} = 1$ TeV criterium is also applied to candidate events. Here M_{cut} is estimated via the invariant mass of the Higgs boson candidate and jet system. At a luminosity equivalent to the 2018 dataset of the CMS detector, an expected upper limit of $C_{cG} < 1.99$ is set.

Like for the SM oriented analysis, a decomposition of the effects of the individual uncertainty sources on this result can be performed. Again, the 30 leading uncertainty sources are considered for a signal strength corresponding to the upper limit on C_{cG} that is set, which are listed in Figure 5.3. Leading theoretical uncertainties include the uncertainty associated with the PDF used in simulation, the renormalisation and factorisation scale of the cH(ZZ→ $4\mu)_{cG}$ signal, as well the uncertainties associated with $\mathcal{B}(H \to ZZ \to 4\ell)$ and the qq→ZZ k-factor. Leading experimental uncertainty include the normalisation of the reducible background estimation, the uncertainty on the luminosity as well as the calibration of the muons. A common factor to all these uncertaintiy sources is that they are associated with larger relative yield changes in the background and signal processes, which can potentially mask the presence of the signal. Also noticeable is that due to the sparse population of bins in the discriminator bins at very high transverse momenta, the statistical modelling uncertainties associated with these bins is also significant.

A statistical extrapolation to full Run-2 and and HL-LHC scenario is again performed. In a Run-2 scenario, an expected upper limit of $C_{cG} < 1.61$ is derived. For a HL-LHC scenario, this improves to $C_{cG} < 0.76$. It should be noted that in this regime, the contribution from the linear term associated with $\hat{O}_{cG}$ becomes increasingly significant and should be accounted for. These values are approximately in line with those reported in [**PhenoPaper**], indicating that the expected results presented there are expected to translate well into an experimental setting.

Given that the presented analysis serves as a basic demonstration, it is clear that a number of potential improvements can be made in future analysis iterations. Like in the SM case, methods with which to increase the overall signal selection efficiency would be interesting to explore. While the low transverse momentum region of the jet is not relvant in the way it was previously, the loosening of the muon selection criteria used in the Higgs boson candidate reconstruction is again of interest. Additionally, other selection optimisations could be explored, such as multi-

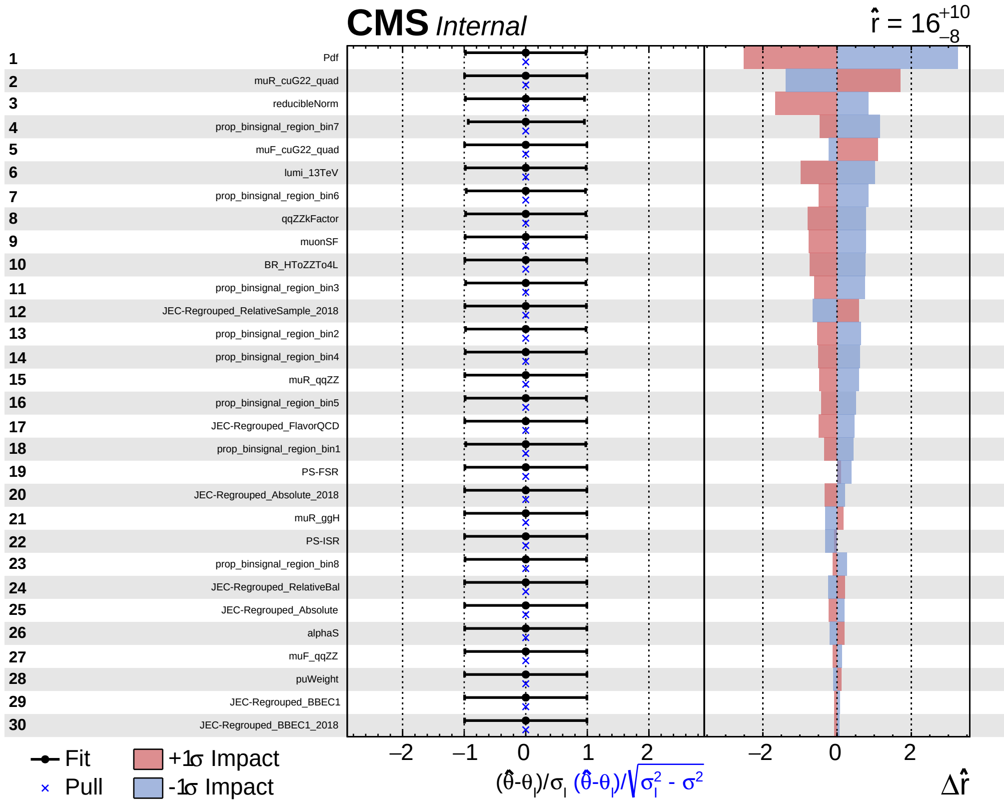


Figure 5.3: Figure showing the 30 leading uncertainty sources in a fit of C_{cG}, orderd by their relative impact on the fit. In the left panel the nuisance parameter names can be seen. In the central panel, the fitted nuisance parameter values as well as their so-called pulls are shown. On the right, the $\Delta\hat{r}_i$ values can be seen for the $\pm\sigma$ variations associated with each nuisance parameter.

variate methods. Since theoretical uncertainties associated with the modelling of the background and signal processes again constrains the analysis sensitivity significantly, improvements in this domain are also of significant interest here.

Conclusion

In this thesis, a first-time analysis of the cH process in the $H \to ZZ \to 4\mu$ channel is presented, using the 2018 dataset of the CMS detector, with the ultimate goal of placing constraints on the charm quark Yukawa interaction modifier κ_c. In Chapter 2, the Standard Model of particle physics is discussed with a particular focus on the Higgs boson, as it is the Higgs boson which is responsible for the generation of particle masses in the SM. This is also true for the charm quark, which gains its mass through a Yukawa interaction with the Higgs boson. Since this interaction ist yet to be verified experimentally like it has been for other heavier particles in the SM, the cH process is identified as a process that is of experimental interest to this end. This process is discussed in a SM context and the method by which a signal strength measurement of the cH process can be translated to a contraint on κ_c. Subsequently, the interpretation of the cH process in a beyond-the-SM physics scenario, parametrised through SMEFT, is discussed. Specifically, the chronomagnetic dipole operator is identified as a SMEFT operator that may be uniquely contrained with the cH process, motivating the extension of a cH process analysis to additionally target such a SMEFT interpretation.

In Chapter 3, the CMS detector, which can reconstruct pp collisions at the LHC to identify cH candidate events, is introduced. This includes a discussion of the various subdetector systems, as well as the reconstruction methods relevant to the analysis. Additionally, the jet flavour tagging algorithm DeepJet is introduced, along with the Monte Carlo simulation methods used to estimate the physics processes that occur at the LHC. The full analysis strategy, targetting specifically the cH(ZZ$\to$ 4μ) channel in the SM, is detailed in Chapter 4. This includes the reconstruction of Higgs boson candidates and the selection of a jet candidate to reconstruct cH(ZZ$\to$ 4μ) candidate events, with a particular focus on the selection of the latter. Subsequently, the Monte Carlo estimation of the cH(ZZ$\to$ 4μ) signal as well as a number of background processes is discussed. A data-driven estimation of background processes not estimated via simulation is also presented. This is followed by a discussion of the statistical methods used to derive expected 95% CL $\mathrm{CL_s}$ upper limit on the signal strength of the cH(ZZ$\to$ 4μ) process, as well as the model of systematic uncertainty sources that is employed. Finally, upper limits on the cH(ZZ$\to$ 4μ) signal strength and $\kappa_c$are set for a number of scenarios. In the nominal scenario, this corresponds to an expected upper limit of $\sigma\mathcal{B}(\mathrm{cH(ZZ}\to 4\mu)) = 1088$ ($\kappa_c = 191$). Additional scenarios using different discriminators are considered and for the nominal scenario, a statistical extrapolation to a full Run-2 data and the anticipated High Luminosity-LHC dataset is performed. The corresponding expected upper limits are set at $\sigma\mathcal{B}(\mathrm{cH(ZZ}\to 4\mu)) = 725$ ($\kappa_c = 128$) and $\sigma\mathcal{B}(\mathrm{cH(ZZ}\to 4\mu)) = 318$ ($\kappa_c = 58$), respectively. Additionally, the impact of leading systematic uncertainty sources are discussed, along with a fit in sidebands region of the analysis and potential analysis improvements. Finally, in Chapter 5, the potential of the presented analysis to constrain the Wilson coefficient of the chronomagnetic dipole operator in a SMEFT comtext is demonstrated and discussed. An expected constraint of $C_{cG} < 1.94$ is set at 95% CL using the $\mathrm{CL_s}$ method. Similarly, a statistical extrapolation to full Run-2 and High Luminosity-LHC scenarios is performed, with the expected constraints being set at $C_{cG} < 1.58$ and $C_{cG} < 0.75$. Potential improvements to this interpretation of the analysis are also discussed.

While the derived, expected analysis sensitivity to $\kappa_{\mathrm{c}}$is lower than in other analyses of the cH(WW), cH($\gamma\gamma$) processes as well as analyses targetting H$\rightarrow$ c$\bar{\mathrm{c}}$ decays, it remains to be an important, complimentary contribution in the quest to determine the nature of the charm quark Yukawa coupling. Given the strong statistical limitations encoutered in all of the named analyses, a combination of many different analysis channels is expected to be an integral part in an ultimate measurement of this coupling, as we look towards future, larger datasets such as that of the HL-LHC to do so. As a result, contributions from less sensitive analysis channels remain valuable. Additionally, the SMEFT interpretation of the cH process presented here may provide exciting hints at deviations from the expected SM behaviour and represents a unique method with which to probe SMEFT operators.

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Appendix

A Columnar analysis framework

For this work, an analysis framework built upon the Coffea [81] software package and based on columnar analysis methods was concipated and written. To better understand the idea of columnar analysis, it helps to consider the computations are required of a typical physics analysis. On a per-event basis, an analyser must repeatedly perform a set of operations on some input data (event data delivered for example via the CMS NanoAOD [82] format) to produce a desired set of outputs. This may consist of applying object selection criteria or producing new, higher level variables from the input. In the context of an entire dataset, the subset of data associated with a single event can thus be thought of as a single row in a larger matrix. With this matrix picture in mind, two approaches to producing the desired outputs may be implemented. The first is a row based approach where the necessary computations are performed serially, once per event, with one event following another. This represents a 'traditional' event loop approach to analysis. The second is the column based approach where operations are performed on entire columns of event data, thus lending the *columnar* analysis name. This means a single computational step is effectively performed simultaneously across a set of events, allowing for an improved optimisation of computations as relevant data may be stored in contiguous memory. Both of these approaches are visualised in Figure 4. The implementation of the columnar approach is facilitated by the use of the Awkward array [83] python package, which generalises numpy [84] arrays to data with non-regular (or *jagged*) shapes.

To perform the analysis presented in Chapter 4 it is necessary to process approximately $5.6 \cdot 10^{10}$ events, equating to approximately 8.4 TB worth of data. Given the scope of these processing requirements, the motivation to optimise the associated computing effort is clear. Using the described columnar approach, the analysis framework is able to achieve a turn around time of approximately one working day for the complete analysis(including the handling of systematic uncertainties) using the distributed grid computing resources available at the IIHE. This represents an almost tenfold speed increase to a previous framework iteration using a traditional CMS software C++ event loop structure. Along with the implementation of the algorithms described in Chapter 4 and Chapter 5, the framework design includes an automatic, central handling of samples and relevant metadata, job submission to distributed computing resources as well as plotting tasks and production of histogram templates for the CMS Combine tool. The development of the analysis framework thus represents a central element of the presented analysis. Additionally, a variant of this framework was also used for the analysis presented in [**PhenoPaper**].

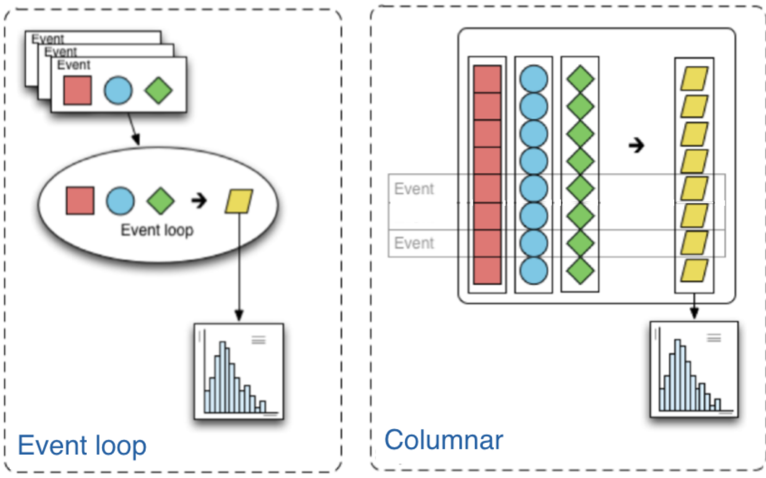


Figure 4: A representation of event loop based and columnar based analysis approaches can be seen on the left and right respectively [81]. The squares, circles and diamonds represent event data that is transformed via an algorithm into output that is histogrammed, represented by the trapezoids.

B Likelihood templates derived for jet selection algorithm

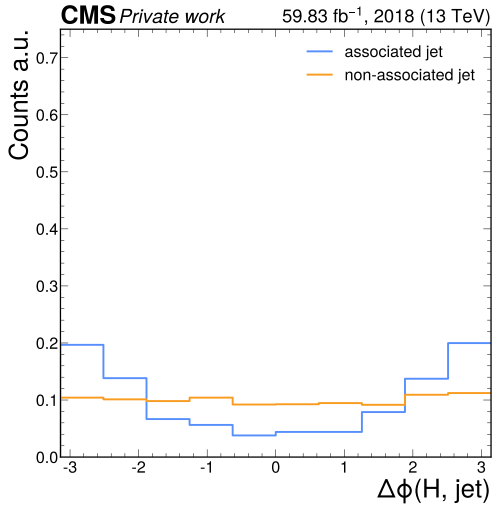
A number of liklihood templates are derived for the jet selection algorithm using the $\Delta\phi(H, jet)$ and $p_T(\text{jet})/p_T(\text{H})$ variables, as decribed in Subsection 4.1.2. These can be seen in Figure 5 to Figure 10.

C Comparison of discriminators in 3P1F and 2P2F regions

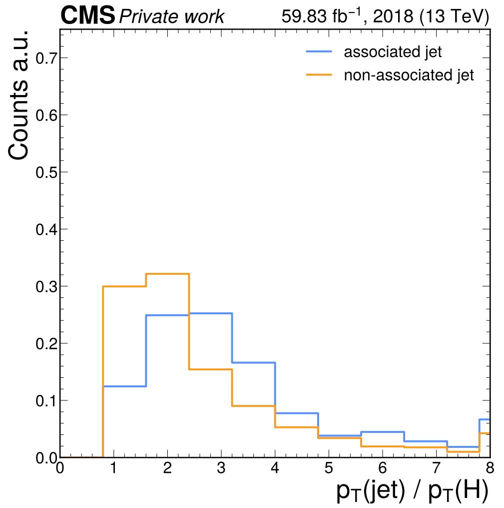
A comparison between the 2P2F and 3P1F regions of the reducible background estimation for the DeepJet c-tagging discriminators, as well as the Higgs mass and selected jet transverse momentum is shown is shown in Figure 11 and Figure 12. Overall, there exists a reasonable agreement between the c-tagging discriminator shapes discriminator shapes, with significant differences only being observed in sparsely poopulated bins. Though there are significant diferences when considering the Higgs boson candidate mass spectrum, there is a reasonable agreement in the signal enriched region around $m(H) = 125$ GeV. There is also an acceptable agreement observed for the transverse momentum of the selected jet, which is only relevant for the EFT analysis that is presented. Given this reasonable agreement of the shapes of the discriminator used for statistical evaluation in this work, and considering the small relative total contribution of the reducible background, the 2P2F discriminator shapes are used to estimate the shape of the entire reducible background distribution in Subsection 4.2.3, which otherwise cannot be well estimated due to statistical limitations.

D Uncertainty distributions

Here, the distributions associated with dominant uncertainty sources used in the statistical evaluation presented in Section 4.5, are shown in Figure 13, Figure 14 and Figure 15. Unless explicitly

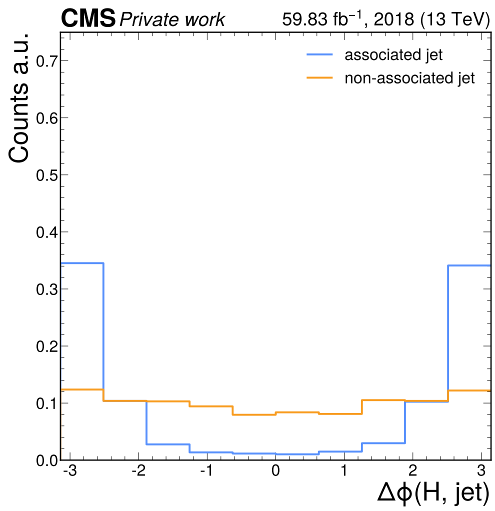


(a) The $\Delta\phi(H, jet)$ template.

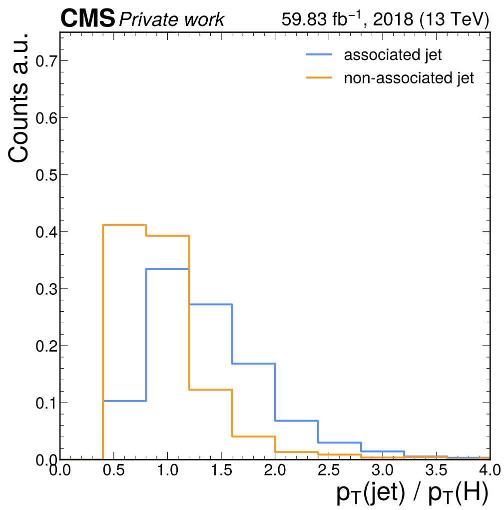


(b) The p_T(jet)/p_T(H) template.

Figure 5: The $\Delta\phi(H, jet)$ and p_T(jet)/p_T(H) templates in the 0-15 GeV bin of the Higgs candidate mass.

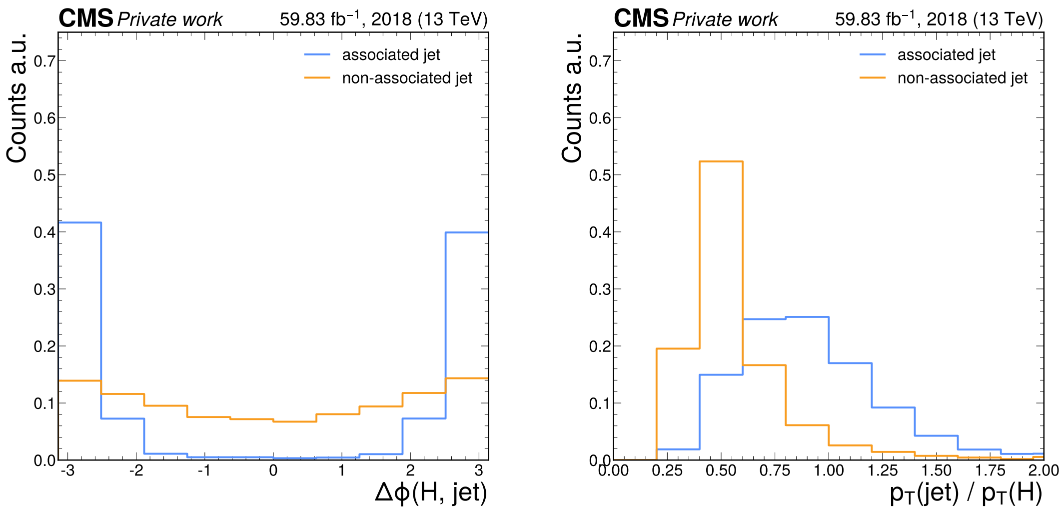


(a) The $\Delta\phi(H, jet)$ template.



(b) The p_T(jet)/p_T(H) template.

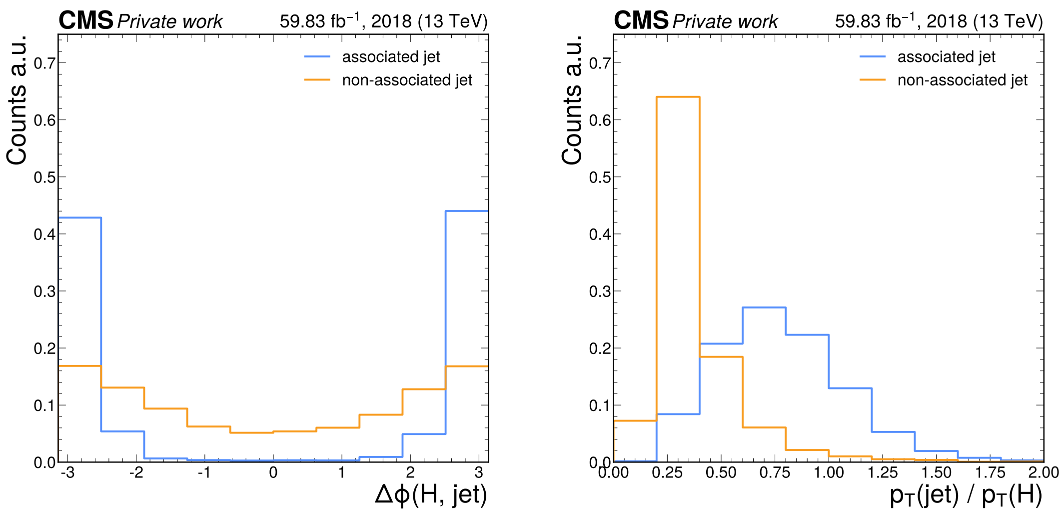
Figure 6: The $\Delta\phi(H, jet)$ and p_T(jet)/p_T(H) templates in the 15-30 GeV bin of the Higgs candidate mass.



(a) The $\Delta\phi(H, jet)$ template.

(b) The $p_T(\text{jet})/p_T(\text{H})$ template.

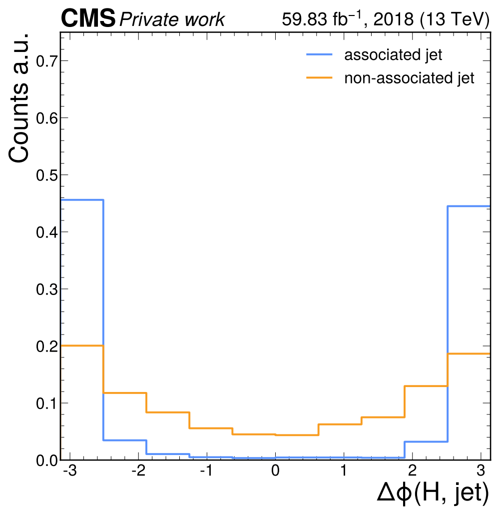
Figure 7: The $\Delta\phi(H, jet)$ and $p_T(\text{jet})/p_T(\text{H})$ templates in the 30-50 GeV bin of the Higgs candidate mass.



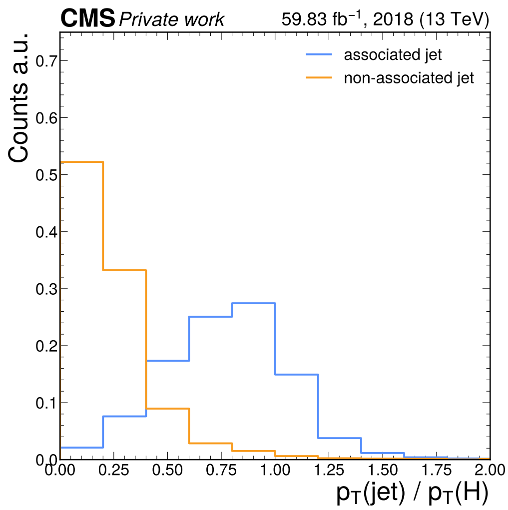
(a) The $\Delta\phi(H, jet)$ template.

(b) The $p_T(\text{jet})/p_T(\text{H})$ template.

Figure 8: The $\Delta\phi(H, jet)$ and $p_T(\text{jet})/p_T(\text{H})$ templates in the 50-100 GeV bin of the Higgs candidate mass.

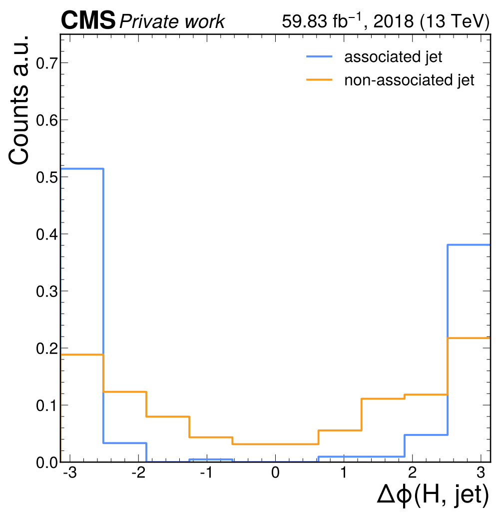


(a) The $\Delta\phi(H, jet)$ template.

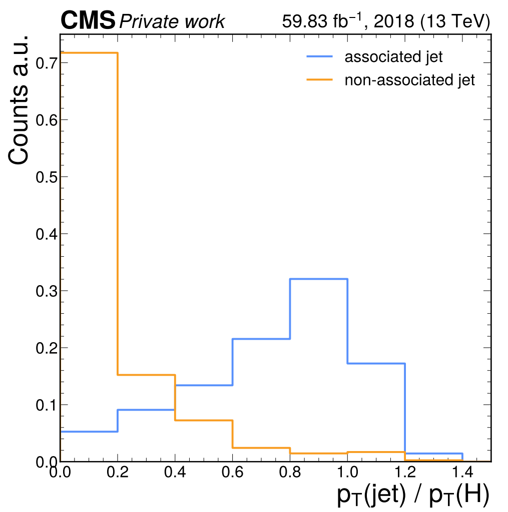


(b) The p_T(jet)/p_T(H) template.

Figure 9: The $\Delta\phi(H, jet)$ and p_T(jet)/p_T(H) templates in the 100-200 GeV bin of the Higgs candidate mass.

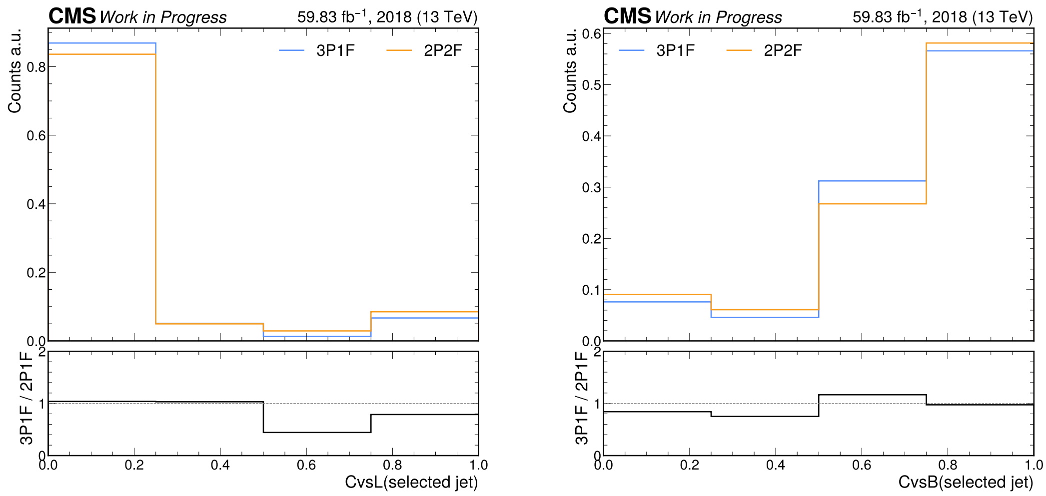


(a) The $\Delta\phi(H, jet)$ template.



(b) The p_T(jet)/p_T(H) template.

Figure 10: The $\Delta\phi(H, jet)$ and p_T(jet)/p_T(H) templates in the >200 GeV bin of the Higgs candidate mass.



(a) The CvsL discriminator.

(b) The CvsB discriminator.

Figure 11: A comparison between the 2P2F and 3P1F regions of the reducible background estimation, looking at the shape of the DeepJet c-tagging discriminator distributions. A reasonable agreement is observed, with the most significant differences being observed in more sparsely populated bins.

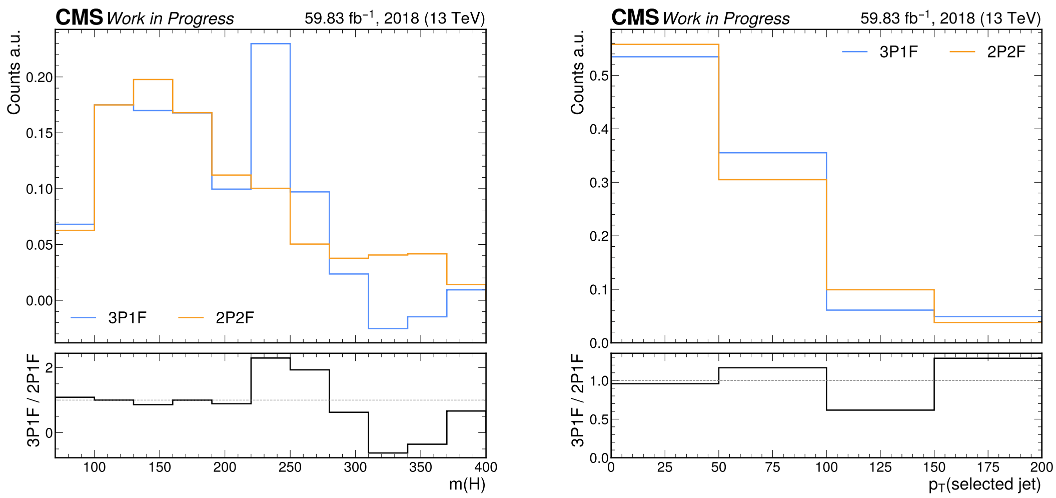


Figure 12: A comparison between the 2P2F and corrected 3P1F regions of the reducible background estimation, looking at the Higgs boson candidate mass spectrum and transverse momentum of the selected jet. For Higgs boson candidate mass, a reasonable agreement is observed in the region around $m(H) = 125$ GeV, with the most significant differences being observed regions not sensitive to the signal. An acceptable agreement for the transverse momentum of the selected jet (here for the EFT analysis interpretation) is also observed.

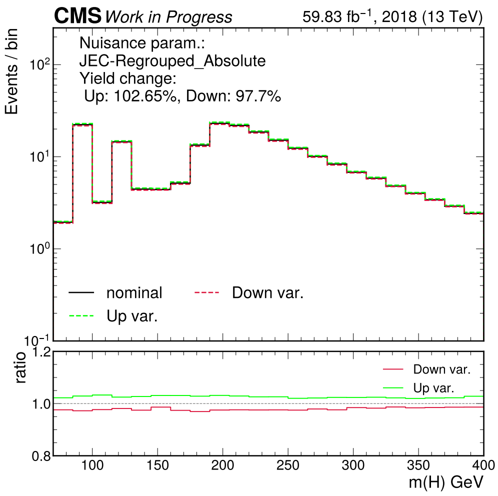
stated otherwise, the distributions shown are the sum of the background processes. Distributions for individual processes (such as the cH(ZZ→ 4μ) signal) are shown separately, per uncertainty, in case significant differences in the distribution shapes associated with the uncertainty are observed for that process with respect to the sum of background processes.

D.1 Nuisance parameter impacts for $r = 0$ fit of cH(ZZ→ 4μ)

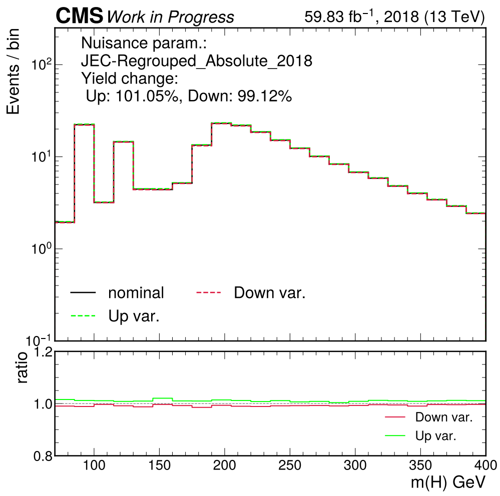
In Figure 16, the impacts of the nuisance parameters on a fit with the baseline scenario can be seen for an expected signal strength of zero. This can be understood as a test of the robustness of the uncertainty model. As expected, the uncertainties associated with the cH(ZZ→ 4μ) signal modelling no longer have a significant impact on the fit, while the remaining nuisance parameters otherwise behaves similarly to when a signal is injected into the Asimov dataset. This indicates that the uncertainty model is working as expected.

D.2 Evaluation of nuisance parameters in sideband region

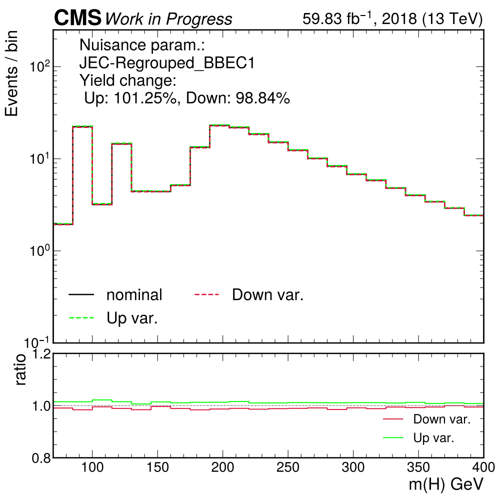
Figure 17 shows the best-fit nuisance parameters of the background estimation model in a fit to data in the sideband region of the Higgs boson candidate mass spectrum. These are distributed within the $\pm 1\sigma$ uncertainty band of the pre-fit uncertainies. The post-fit uncertainties of the nuisance parameters are also indicated. Given the absence of a signal contribution in this region, the background-only (B-only) and signal plus background (S+B) fits are effectively identical.



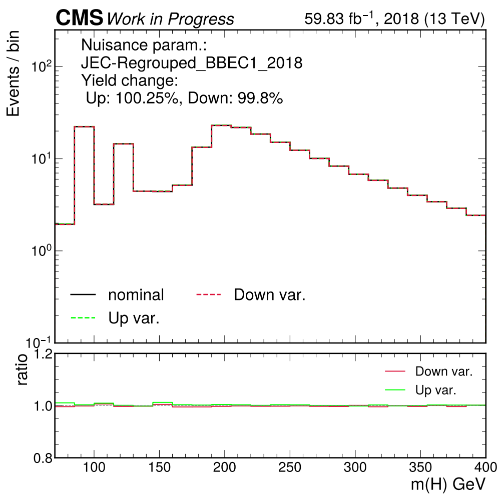
(a) JES: Regrouped_Absolute



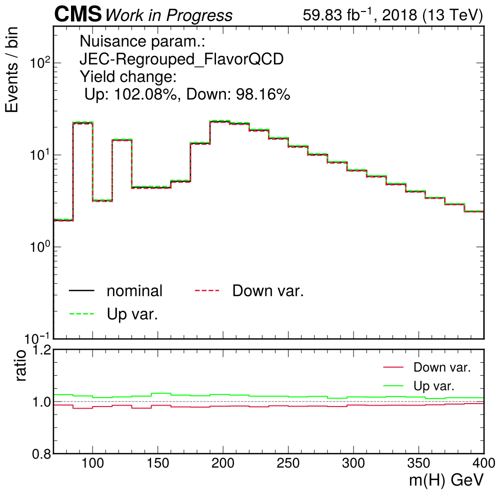
(b) JES: Regrouped_Absolute_2018



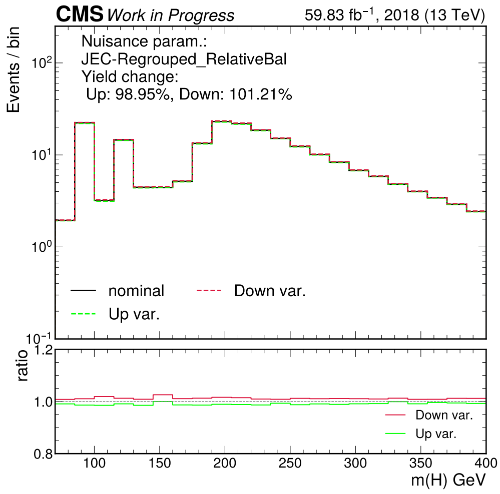
(c) JES energy scale: Regrouped_BBEC1



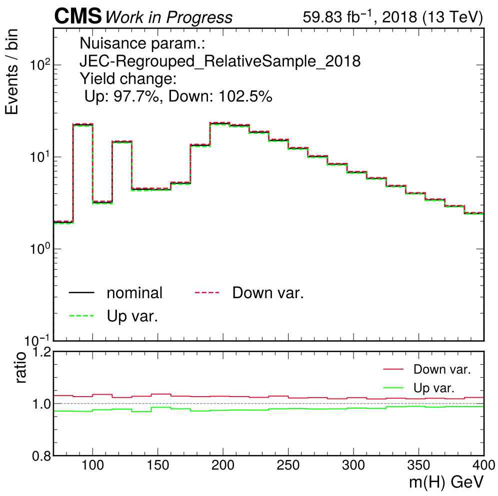
(d) JES: Regrouped_BBEC1_2018



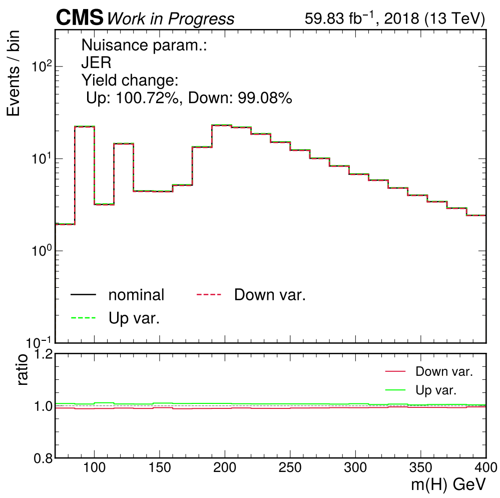
(e) JES: Regrouped_FlavorQCD



(f) JES: Regrouped_RelativeBal

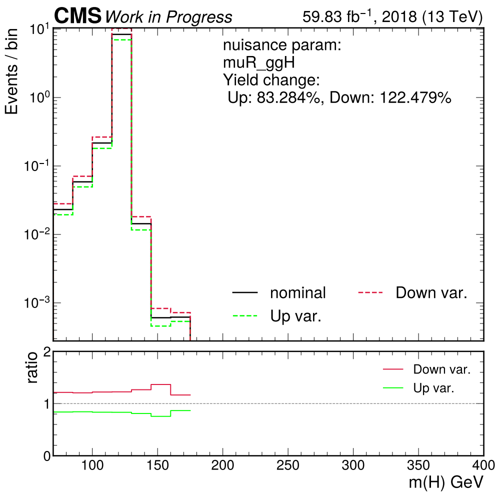


(g) JES: Regrouped_RelativeSample_2018

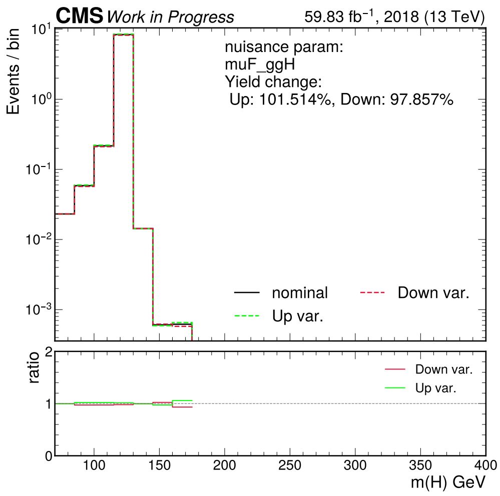


(h) JER

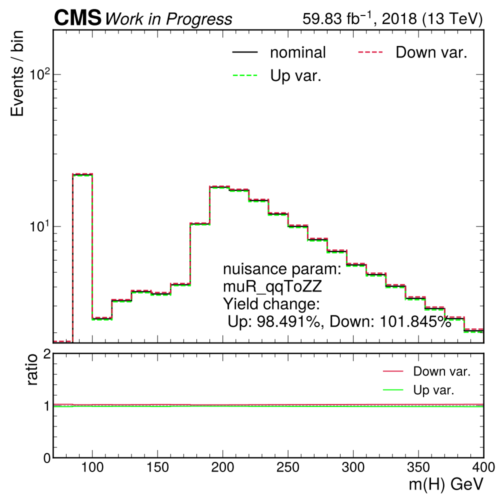
Figure 13: Leading uncertainty sources associated with the jet energy scale (JES) and jet energy resolution (JER) calibration calibration.



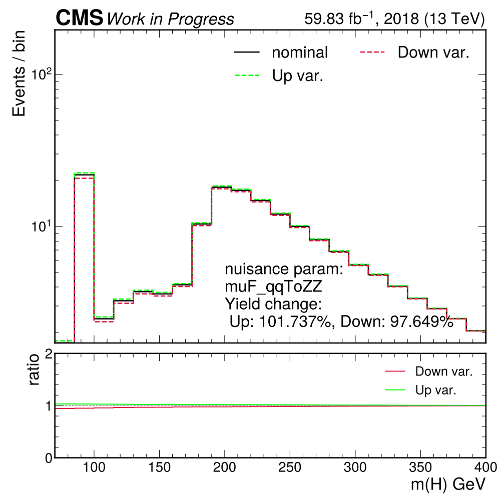
(a) Renormalisation scale: ggH



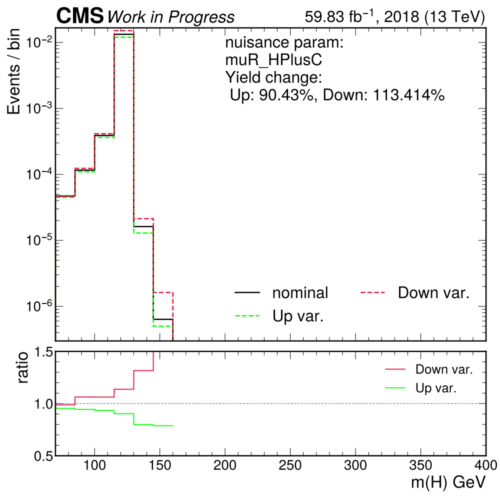
(b) Factorisation scale: ggH



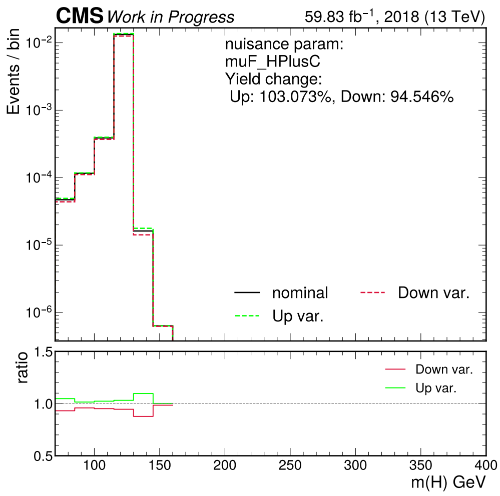
(c) Renormalisation scale: qq→ZZ



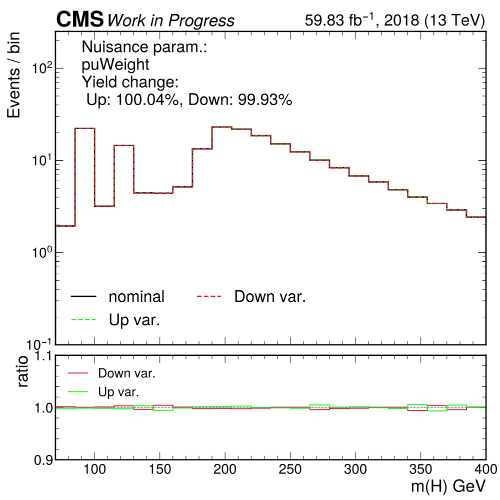
(d) Factorisation scale: qq→ZZ



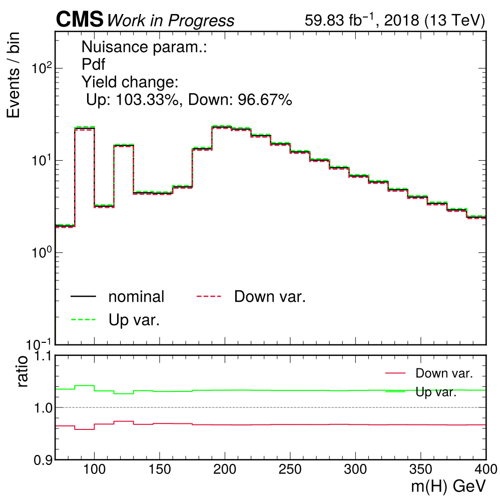
(e) Renormalisation scale: cH(ZZ→ 4μ)



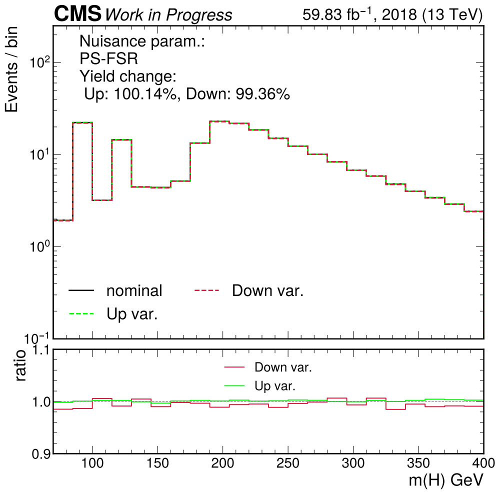
(f) Factorisation scale: cH(ZZ→ 4μ)



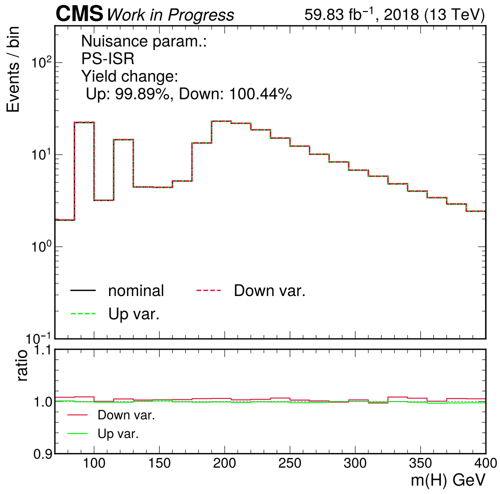
(g) Pileup reweighting



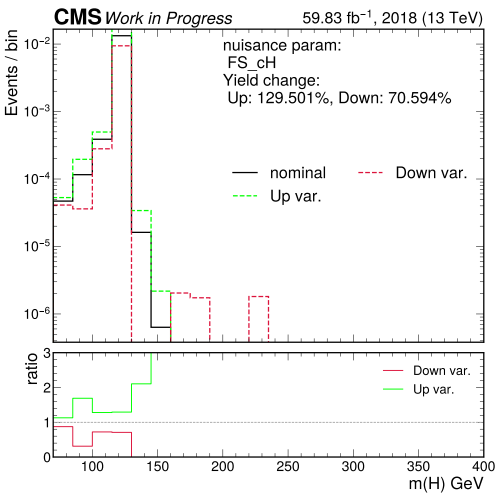
(h) Parton distribution function



(i) Parton showering: FSR



(j) Parton showering: ISR



(k) cH(ZZ→ 4μ) flavour scheme

Figure 14: Leading theoretical uncertainty sources related to the modelling of the signal and background processes.

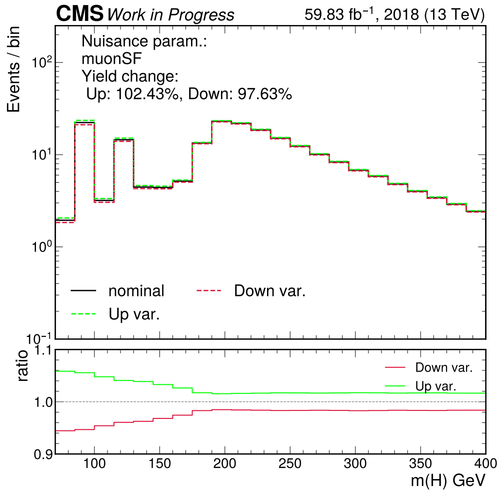


Figure 15: The uncertainty associated with the calibration of muons.

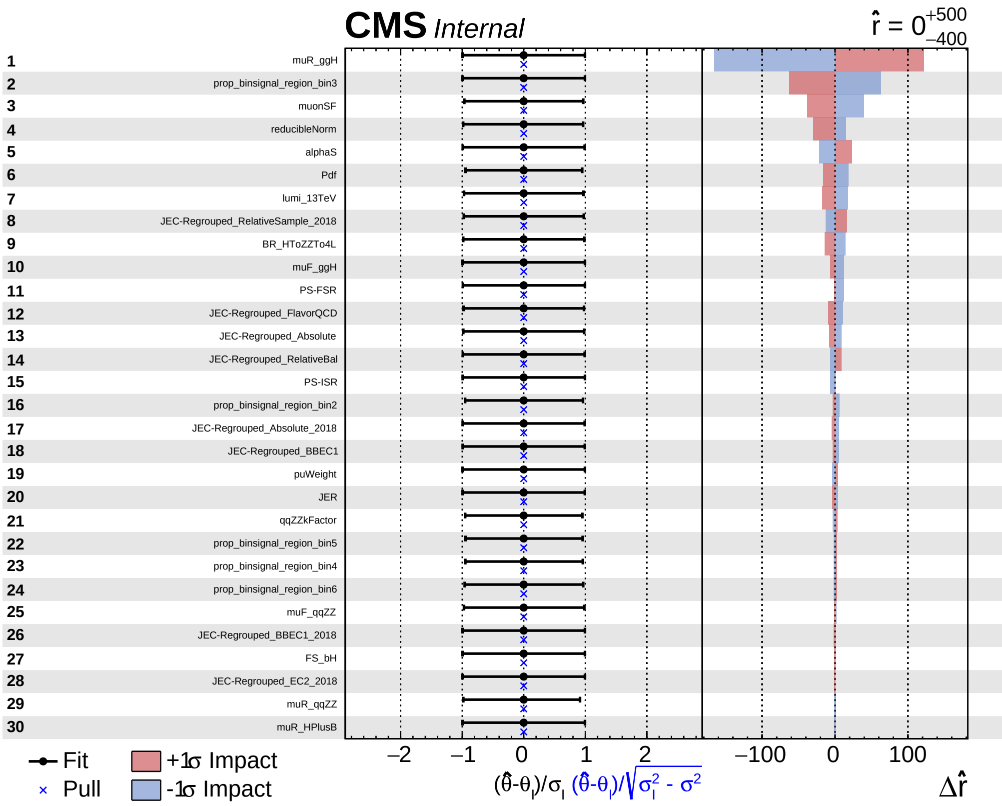


Figure 16: The 30 leading uncertainty sources in a fit of $\sigma\mathcal{B}(\mathrm{cH}(\mathrm{ZZ}\to 4\mu))$ with $r = 0$, orderd by their relative impact on the fit.

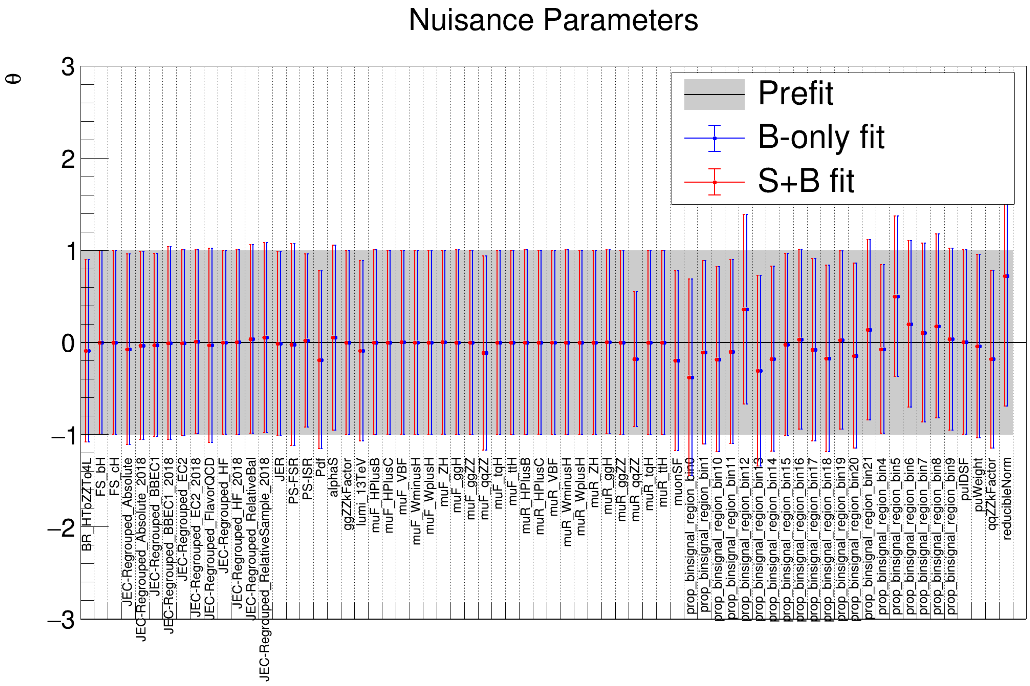


Figure 17: The best-fit values of the nuisance parameters that describe the uncertainty model of the background estimation model in the baseline scenario.